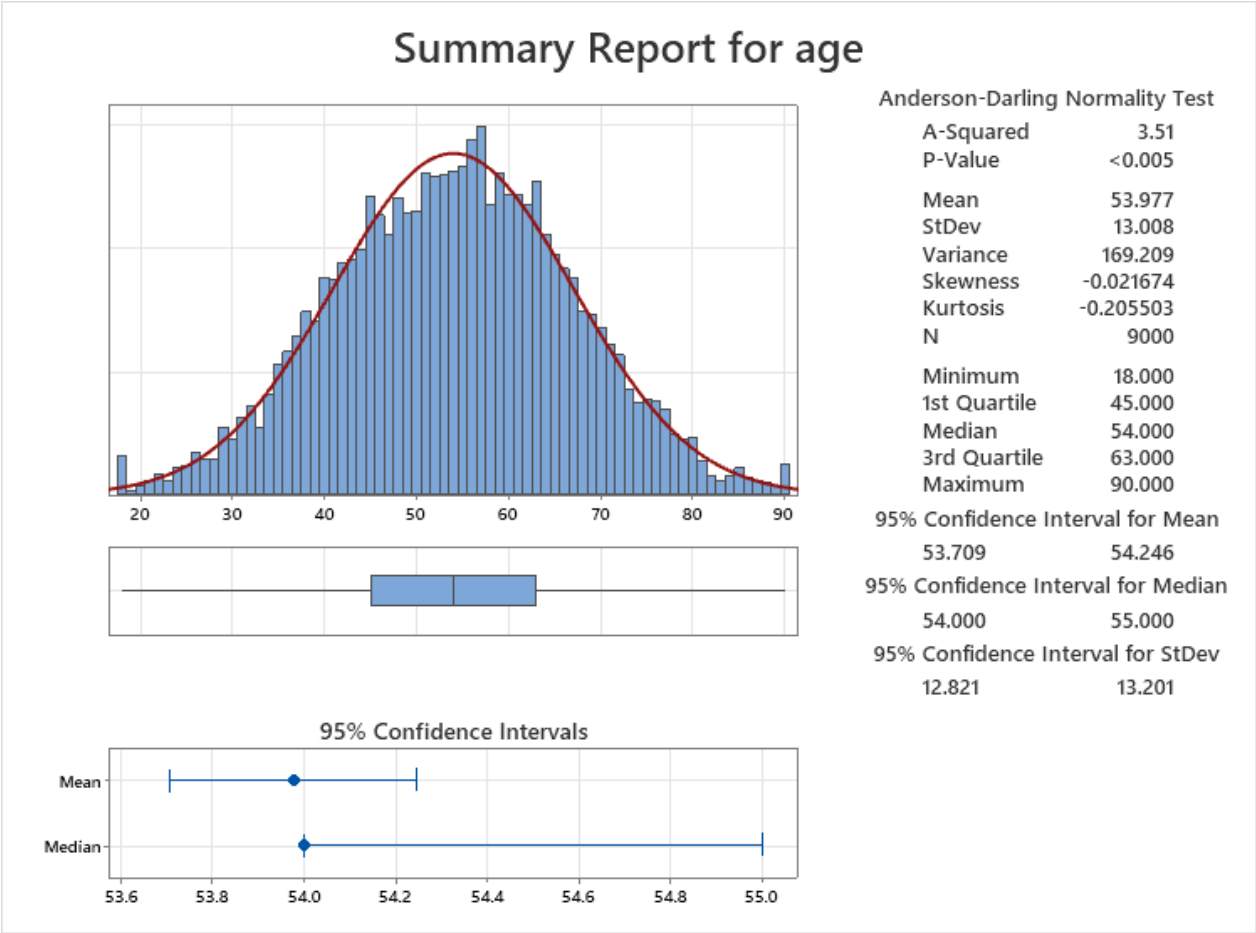
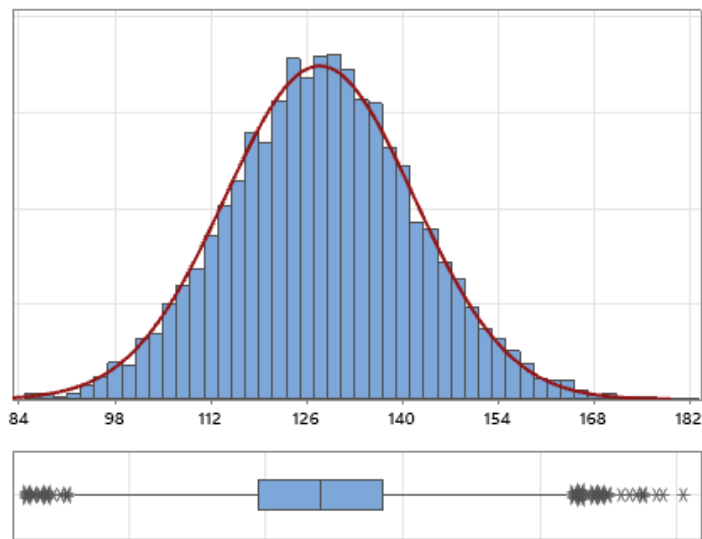


Summary Report :



Summary Report for resting_bp_systolic



Anderson-Darling Normality Test

A-Squared	2.57
P-Value	<0.005
Mean	127.85
StDev	13.77
Variance	189.54
Skewness	0.0676858
Kurtosis	0.0290895
N	9000
Minimum	85.00
1st Quartile	119.00
Median	128.00
3rd Quartile	137.00
Maximum	181.00

95% Confidence Interval for Mean

127.57 128.13

95% Confidence Interval for Median

127.00 128.00

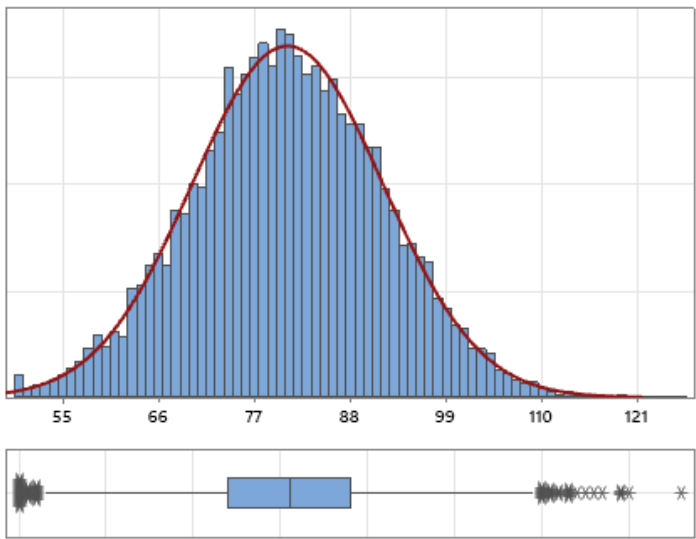
95% Confidence Interval for StDev

13.57 13.97

95% Confidence Intervals



Summary Report for resting_bp_diastolic



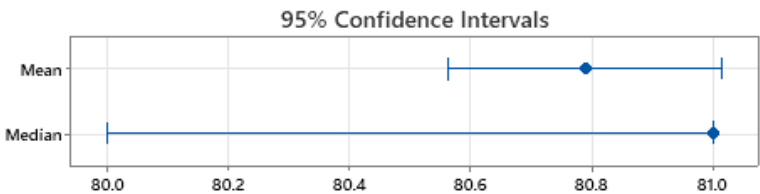
Anderson-Darling Normality Test

A-Squared	3.30
P-Value	<0.005
Mean	80.789
StDev	10.940
Variance	119.682
Skewness	0.0086138
Kurtosis	-0.0797391
N	9000
Minimum	50.000
1st Quartile	74.000
Median	81.000
3rd Quartile	88.000
Maximum	126.000

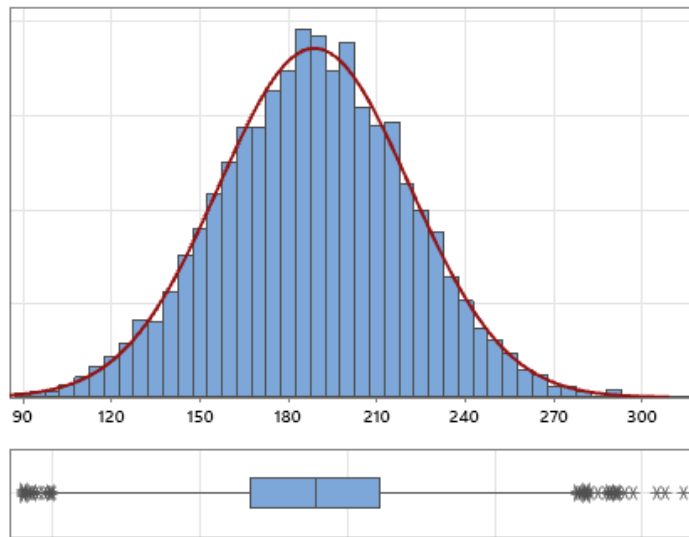
95% Confidence Interval for Mean
80.563 81.015

95% Confidence Interval for Median
80.000 81.000

95% Confidence Interval for StDev
10.782 11.102



Summary Report for cholesterol_total



Anderson-Darling Normality Test

A-Squared	0.74
P-Value	0.053

Mean	188.75
StDev	32.27
Variance	1041.46
Skewness	0.0038504
Kurtosis	-0.0280421
N	9000

Minimum	90.00
1st Quartile	167.00
Median	189.00
3rd Quartile	211.00
Maximum	314.00

95% Confidence Interval for Mean

188.09	189.42
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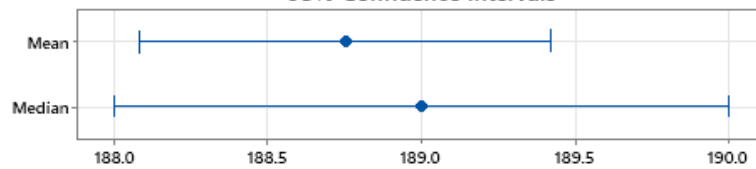
95% Confidence Interval for Median

188.00	190.00
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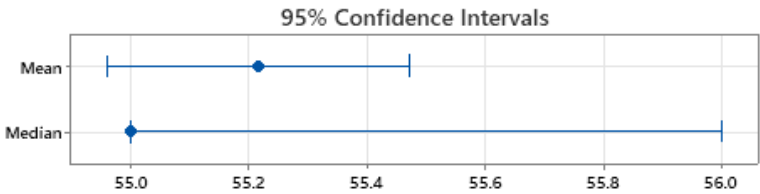
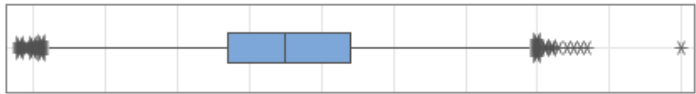
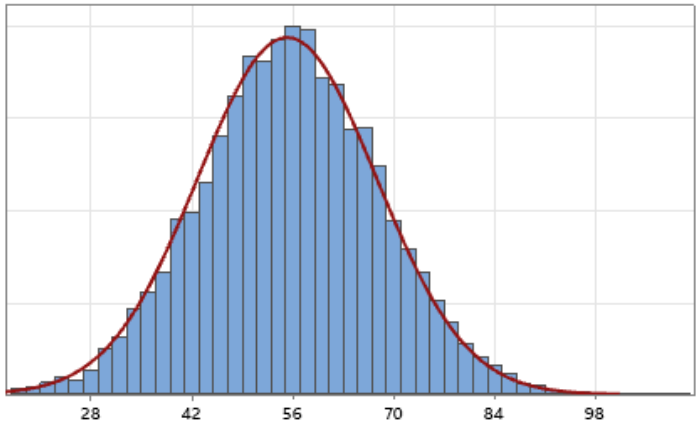
95% Confidence Interval for StDev

31.81	32.75
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95% Confidence Intervals



Summary Report for hdl



Anderson-Darling Normality Test

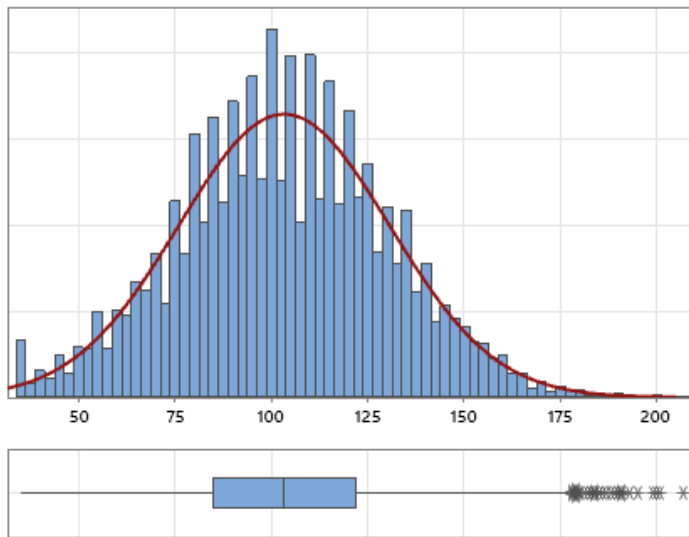
A-Squared	2.54
P-Value	<0.005
Mean	55.215
StDev	12.394
Variance	153.621
Skewness	0.0204590
Kurtosis	-0.0315272
N	9000
Minimum	18.000
1st Quartile	47.000
Median	55.000
3rd Quartile	64.000
Maximum	110.000

95% Confidence Interval for Mean
54.959 55.471

95% Confidence Interval for Median
55.000 56.000

95% Confidence Interval for StDev
12.216 12.578

Summary Report for Idl



Anderson-Darling Normality Test

A-Squared	1.04
P-Value	0.010
Mean	103.28
StDev	27.34
Variance	747.60
Skewness	0.0449627
Kurtosis	-0.0985205
N	9000
Minimum	35.00
1st Quartile	85.00
Median	103.00
3rd Quartile	122.00
Maximum	207.00

95% Confidence Interval for Mean

102.71 103.84

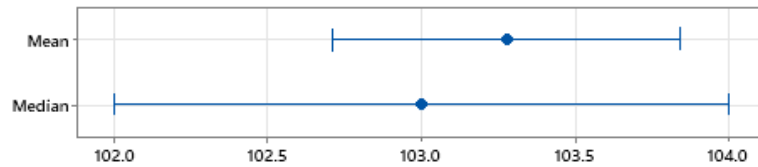
95% Confidence Interval for Median

102.00 104.00

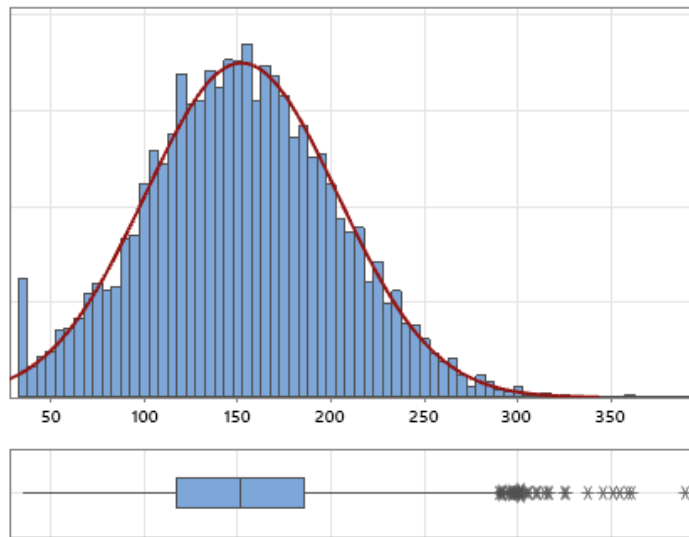
95% Confidence Interval for StDev

26.95 27.75

95% Confidence Intervals



Summary Report for triglycerides



Anderson-Darling Normality Test

A-Squared 1.59
P-Value <0.005

Mean 152.04
StDev 51.45
Variance 2646.94
Skewness 0.125859
Kurtosis -0.024352
N 9000

Minimum 35.00
1st Quartile 117.00
Median 151.00
3rd Quartile 186.00
Maximum 390.00

95% Confidence Interval for Mean

150.98 153.11

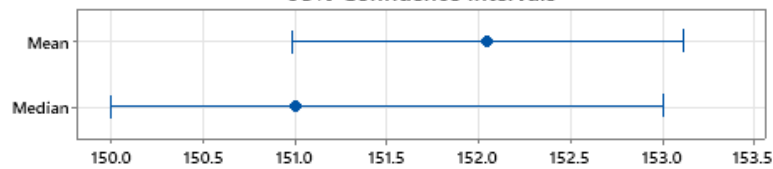
95% Confidence Interval for Median

150.00 153.00

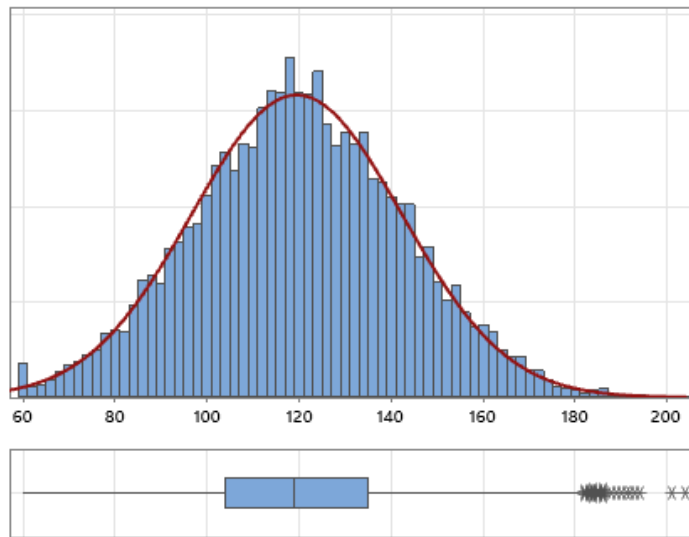
95% Confidence Interval for StDev

50.71 52.21

95% Confidence Intervals



Summary Report for fasting_blood_sugar



Anderson-Darling Normality Test

A-Squared 1.27
P-Value <0.005

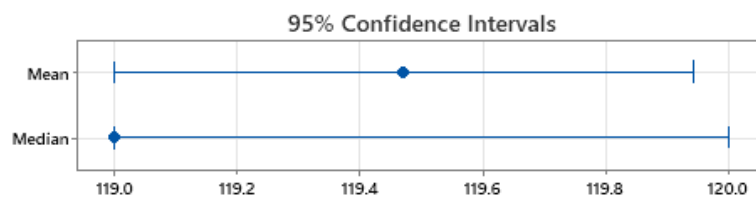
Mean 119.47
StDev 22.78
Variance 518.84
Skewness 0.040870
Kurtosis -0.140283
N 9000

Minimum 60.00
1st Quartile 104.00
Median 119.00
3rd Quartile 135.00
Maximum 204.00

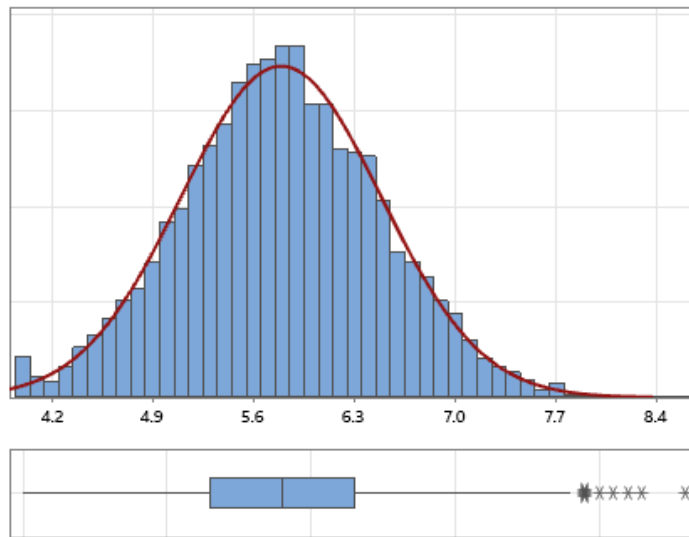
95% Confidence Interval for Mean
119.00 119.94

95% Confidence Interval for Median
119.00 120.00

95% Confidence Interval for StDev
22.45 23.12



Summary Report for hba1c



Anderson-Darling Normality Test

A-Squared	8.26
P-Value	<0.005
Mean	5.7916
StDev	0.6926
Variance	0.4798
Skewness	0.0291825
Kurtosis	-0.0760069
N	9000
Minimum	4.0000
1st Quartile	5.3000
Median	5.8000
3rd Quartile	6.3000
Maximum	8.6000

95% Confidence Interval for Mean

5.7773	5.8059
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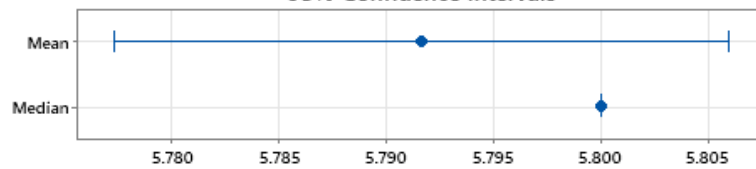
95% Confidence Interval for Median

5.8000	5.8000
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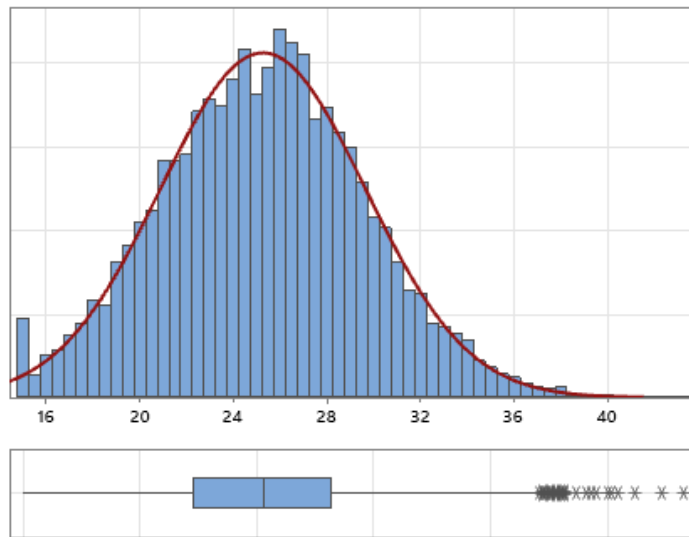
95% Confidence Interval for StDev

0.6827	0.7029
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95% Confidence Intervals



Summary Report for bmi



Anderson-Darling Normality Test

A-Squared 1.36
P-Value <0.005

Mean 25.265
StDev 4.372
Variance 19.110
Skewness 0.091722
Kurtosis -0.122467
N 9000

Minimum 15.000
1st Quartile 22.300
Median 25.300
3rd Quartile 28.200
Maximum 43.300

95% Confidence Interval for Mean

25.175 25.355

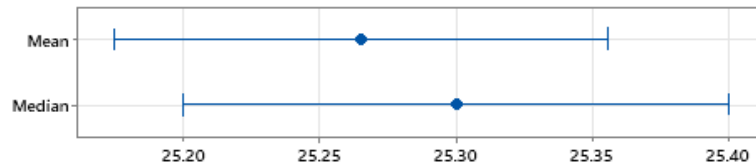
95% Confidence Interval for Median

25.200 25.400

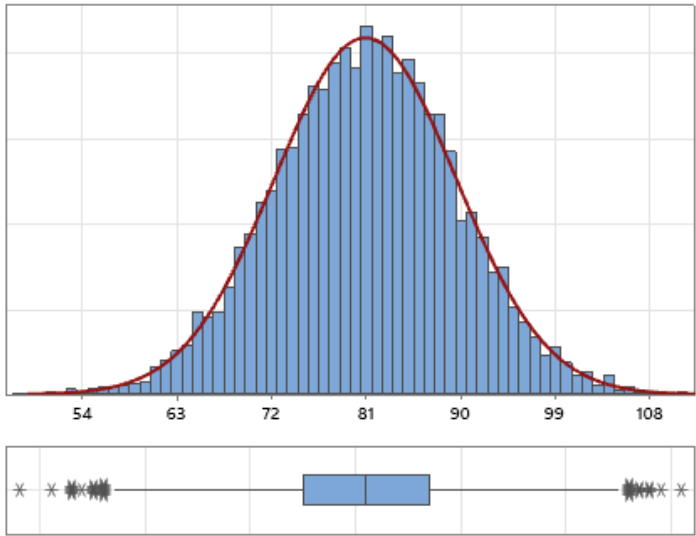
95% Confidence Interval for StDev

4.309 4.436

95% Confidence Intervals



Summary Report for resting_heart_rate



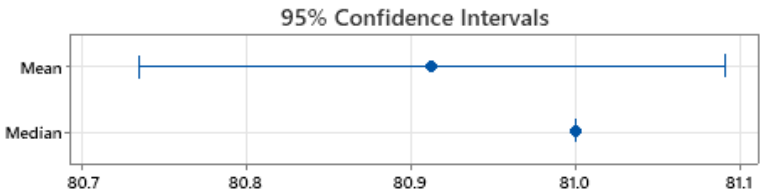
Anderson-Darling Normality Test

A-Squared	5.18
P-Value	<0.005
Mean	80.913
StDev	8.605
Variance	74.041
Skewness	-0.0189122
Kurtosis	-0.0164839
N	9000
Minimum	48.000
1st Quartile	75.000
Median	81.000
3rd Quartile	87.000
Maximum	111.000

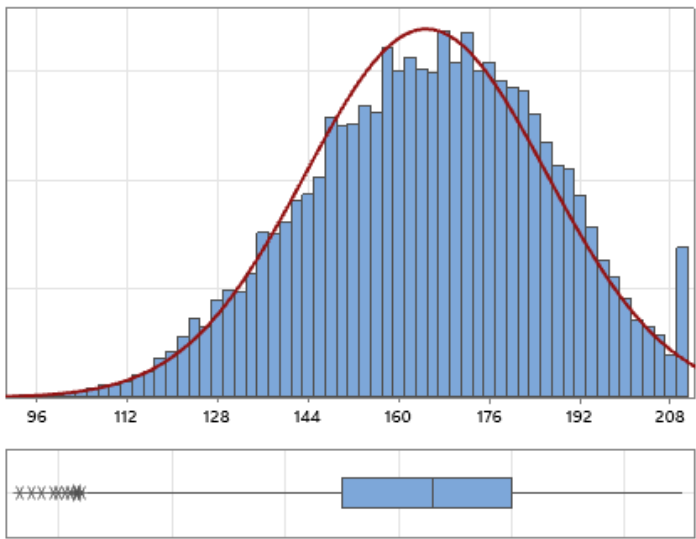
95% Confidence Interval for Mean
80.735 81.090

95% Confidence Interval for Median
81.000 81.000

95% Confidence Interval for StDev
8.481 8.732



Summary Report for max_heart_rate_achieved



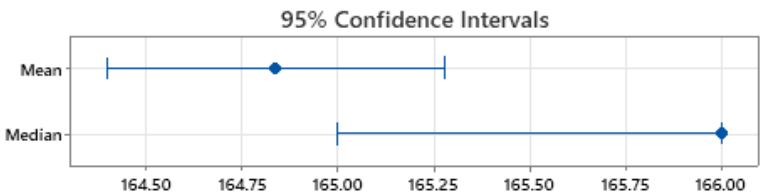
Anderson-Darling Normality Test

A-Squared	9.33
P-Value	<0.005
Mean	164.84
StDev	21.29
Variance	453.30
Skewness	-0.184662
Kurtosis	-0.368012
N	9000
Minimum	93.00
1st Quartile	150.00
Median	166.00
3rd Quartile	180.00
Maximum	210.00

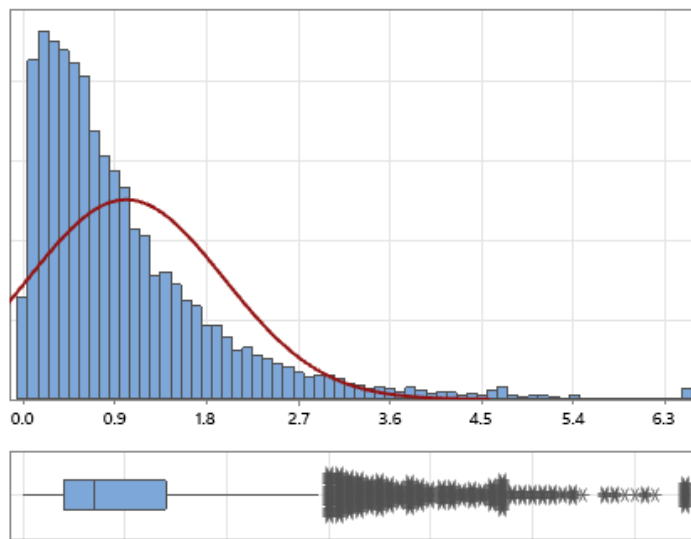
95% Confidence Interval for Mean
164.40 165.28

95% Confidence Interval for Median
165.00 166.00

95% Confidence Interval for StDev
20.98 21.61



Summary Report for st_depression



Anderson-Darling Normality Test

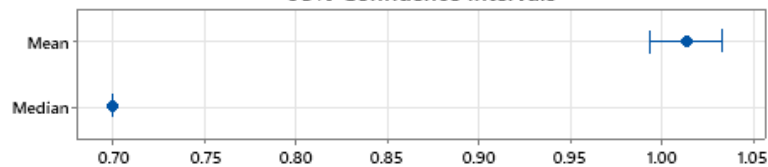
A-Squared	430.73
P-Value	<0.005
Mean	1.0129
StDev	0.9571
Variance	0.9160
Skewness	2.04091
Kurtosis	5.60642
N	9000
Minimum	0.0000
1st Quartile	0.4000
Median	0.7000
3rd Quartile	1.4000
Maximum	6.5000

95% Confidence Interval for Mean
0.9931 1.0327

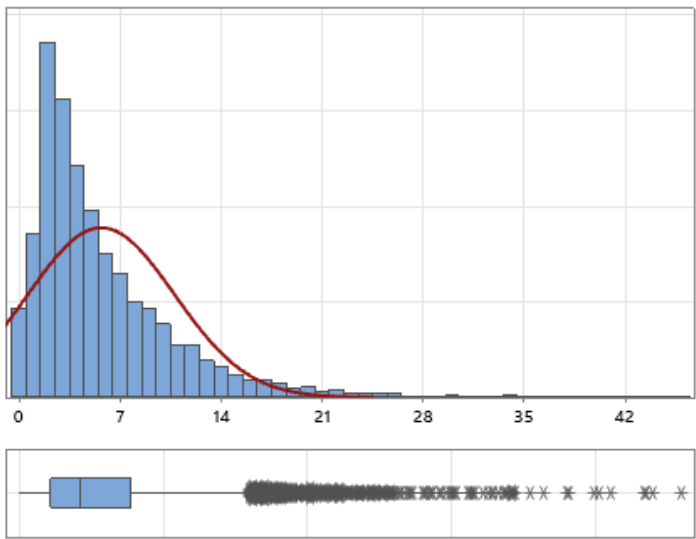
95% Confidence Interval for Median
0.7000 0.7000

95% Confidence Interval for StDev
0.9433 0.9713

95% Confidence Intervals



Summary Report for alcohol_units_per_week



Anderson-Darling Normality Test

A-Squared	389.67
P-Value	<0.005
Mean	5.7087
StDev	5.0706
Variance	25.7114
Skewness	2.02723
Kurtosis	6.28843
N	9000
Minimum	0.0000
1st Quartile	2.2000
Median	4.2000
3rd Quartile	7.7000
Maximum	45.9000

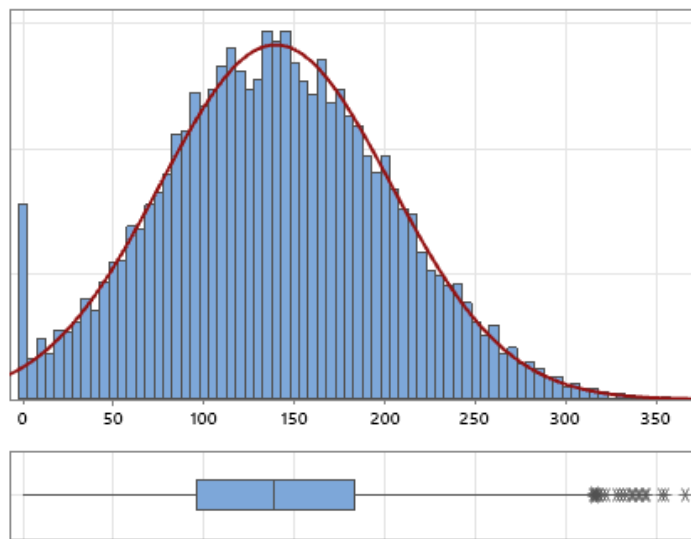
95% Confidence Interval for Mean
5.6040 5.8135

95% Confidence Interval for Median
4.0000 4.3000

95% Confidence Interval for StDev
4.9976 5.1458



Summary Report for exercise_minutes_per_week



Anderson-Darling Normality Test

A-Squared 1.94
P-Value <0.005

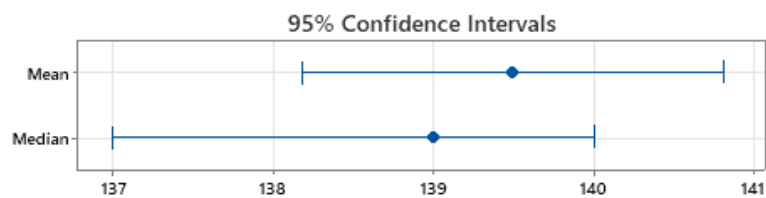
Mean 139.49
StDev 63.66
Variance 4052.65
Skewness 0.093647
Kurtosis -0.229501
N 9000

Minimum 0.00
1st Quartile 96.00
Median 139.00
3rd Quartile 183.00
Maximum 366.00

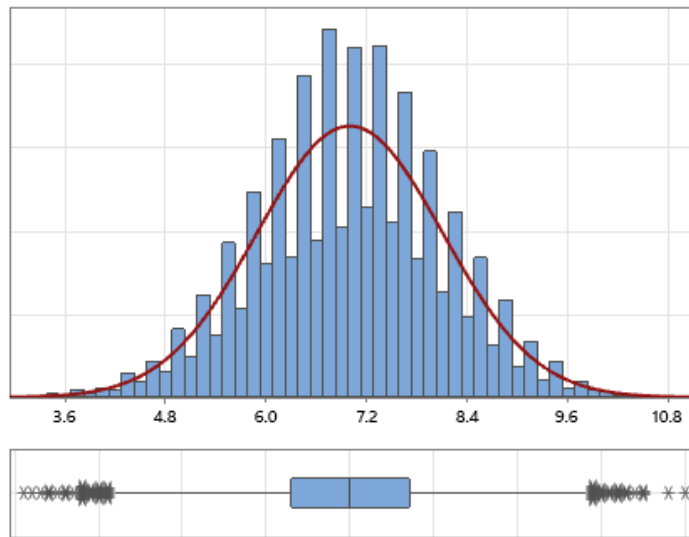
95% Confidence Interval for Mean
138.18 140.81

95% Confidence Interval for Median
137.00 140.00

95% Confidence Interval for StDev
62.74 64.60



Summary Report for sleep_hours



Anderson-Darling Normality Test

A-Squared 3.30
P-Value <0.005

Mean 6.9946
StDev 1.1042
Variance 1.2193
Skewness -0.0424554
Kurtosis -0.0308941
N 9000

Minimum 3.1000
1st Quartile 6.3000
Median 7.0000
3rd Quartile 7.7000
Maximum 11.0000

95% Confidence Interval for Mean

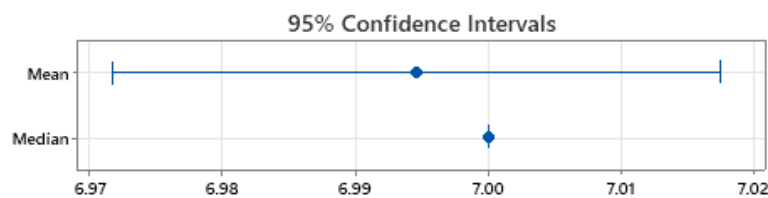
6.9718 7.0174

95% Confidence Interval for Median

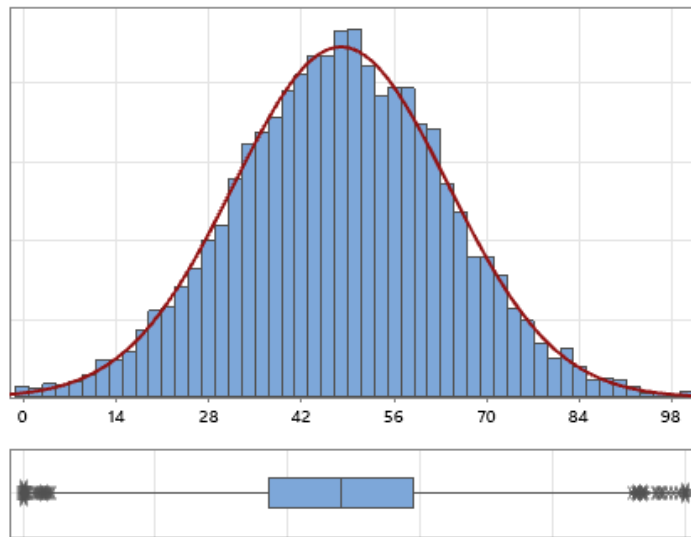
7.0000 7.0000

95% Confidence Interval for StDev

1.0883 1.1206



Summary Report for stress_score



Anderson-Darling Normality Test

A-Squared	0.42
P-Value	0.317

Mean	47.968
StDev	16.162
Variance	261.226
Skewness	-0.0129873
Kurtosis	-0.0138382
N	9000

Minimum	0.000
1st Quartile	37.100
Median	48.100
3rd Quartile	58.900
Maximum	100.000

95% Confidence Interval for Mean

47.634	48.302
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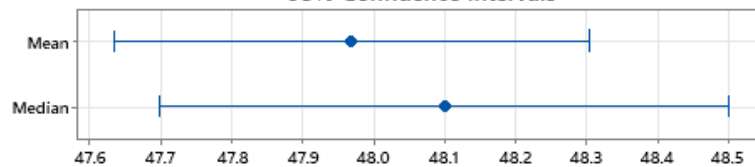
95% Confidence Interval for Median

47.700	48.500
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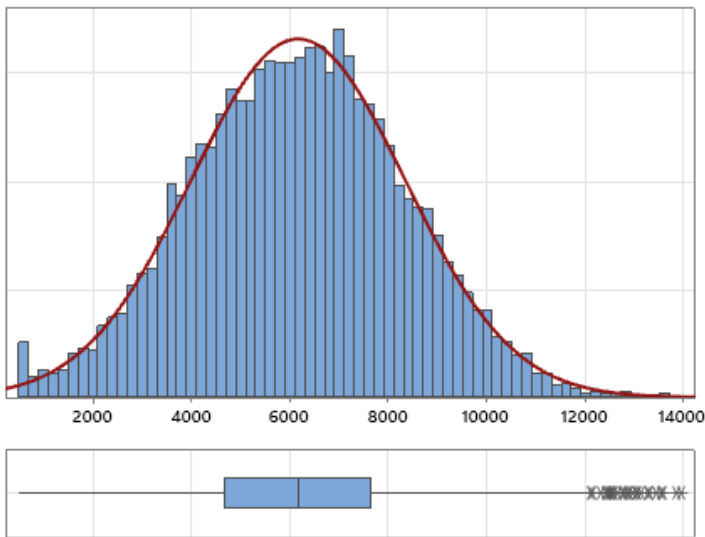
95% Confidence Interval for StDev

15.930	16.402
--------	--------

95% Confidence Intervals



Summary Report for daily_steps



Anderson-Darling Normality Test

A-Squared	0.82
P-Value	0.035
Mean	6163.7
StDev	2170.7
Variance	4711877.7
Skewness	0.024890
Kurtosis	-0.134160
N	9000
Minimum	500.0
1st Quartile	4658.0
Median	6178.0
3rd Quartile	7634.3
Maximum	13950.0

95% Confidence Interval for Mean

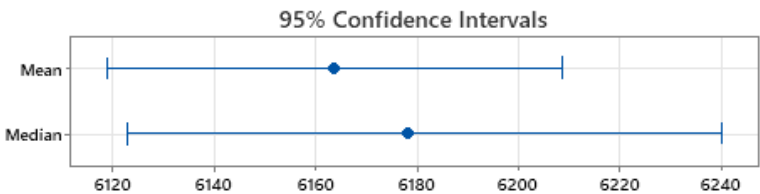
6118.8	6208.5
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95% Confidence Interval for Median

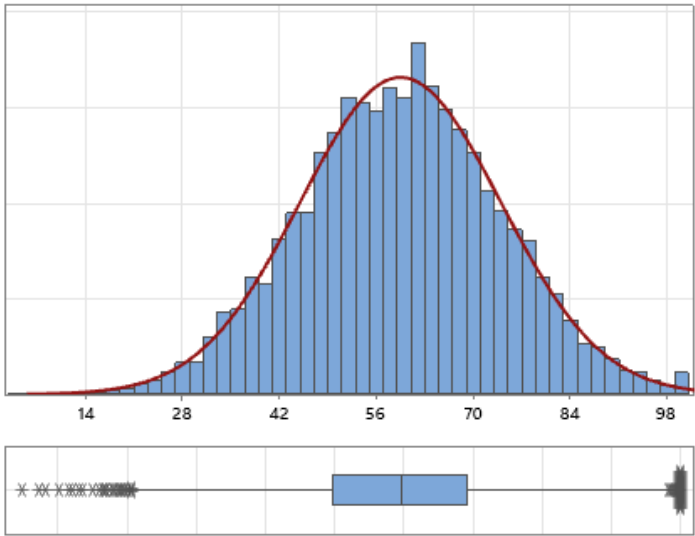
6123.0	6240.0
--------	--------

95% Confidence Interval for StDev

2139.4	2202.9
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Summary Report for diet_quality_score



Anderson-Darling Normality Test

A-Squared	0.36
P-Value	0.456
Mean	59.467
StDev	14.476
Variance	209.556
Skewness	0.0012270
Kurtosis	-0.0769348
N	9000
Minimum	4.800
1st Quartile	49.700
Median	59.600
3rd Quartile	69.100
Maximum	100.000

95% Confidence Interval for Mean

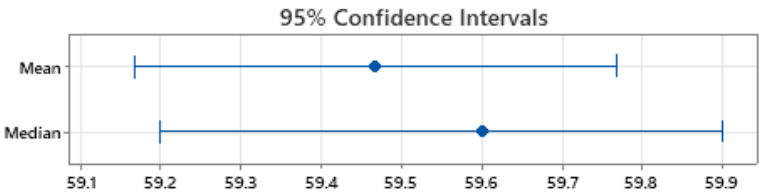
59.168	59.767
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95% Confidence Interval for Median

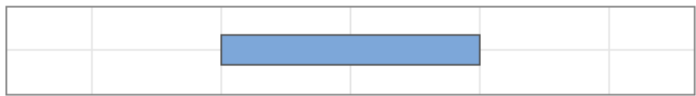
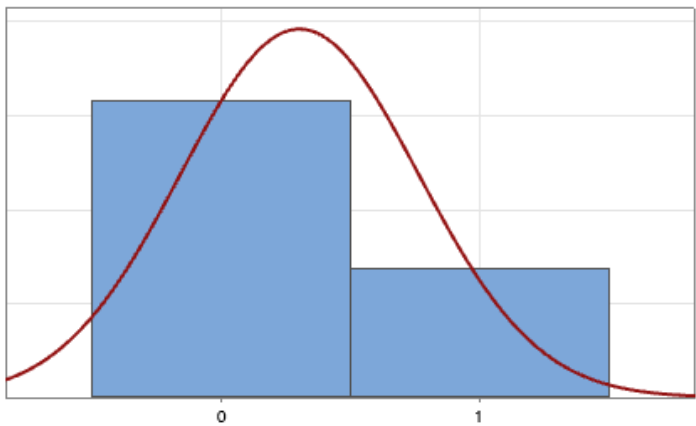
59.200	59.900
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95% Confidence Interval for StDev

14.268	14.691
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Summary Report for has_heart_disease



Anderson-Darling Normality Test

A-Squared	1953.72
P-Value	<0.005
Mean	0.30300
StDev	0.45958
Variance	0.21121
Skewness	0.85749
Kurtosis	-1.26499
N	9000
Minimum	0.00000
1st Quartile	0.00000
Median	0.00000
3rd Quartile	1.00000
Maximum	1.00000

95% Confidence Interval for Mean
0.29350 0.31250

95% Confidence Interval for Median
0.00000 0.00000

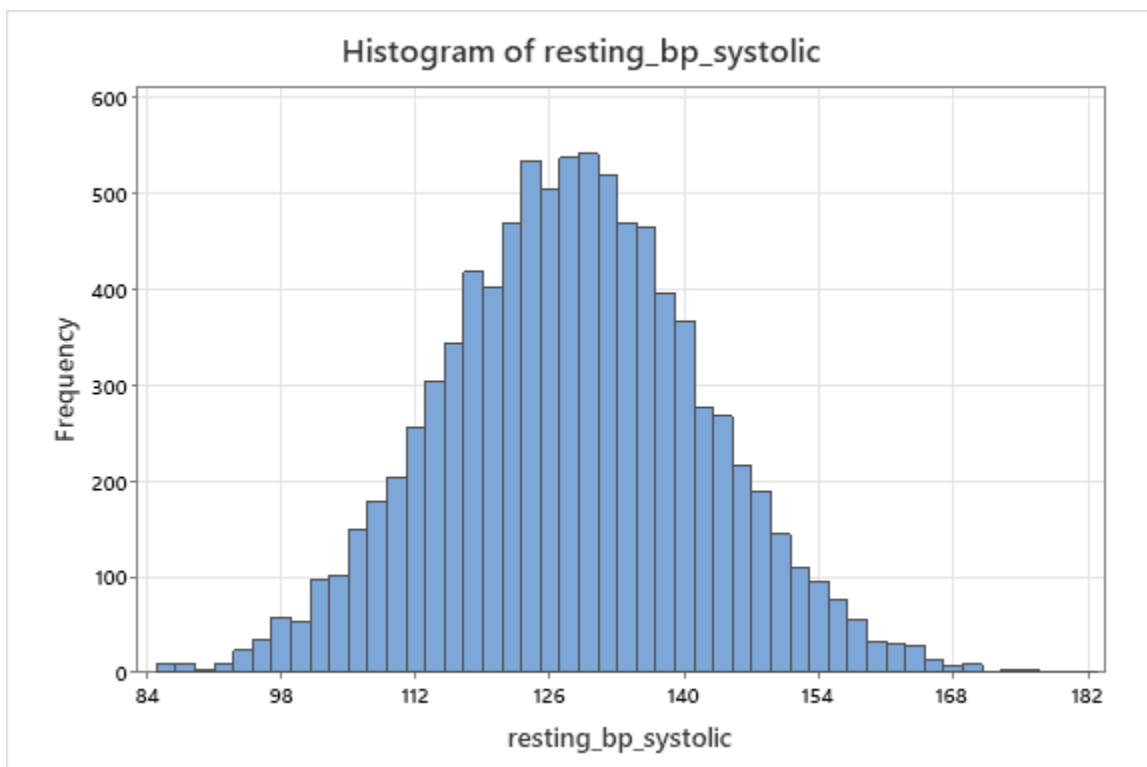
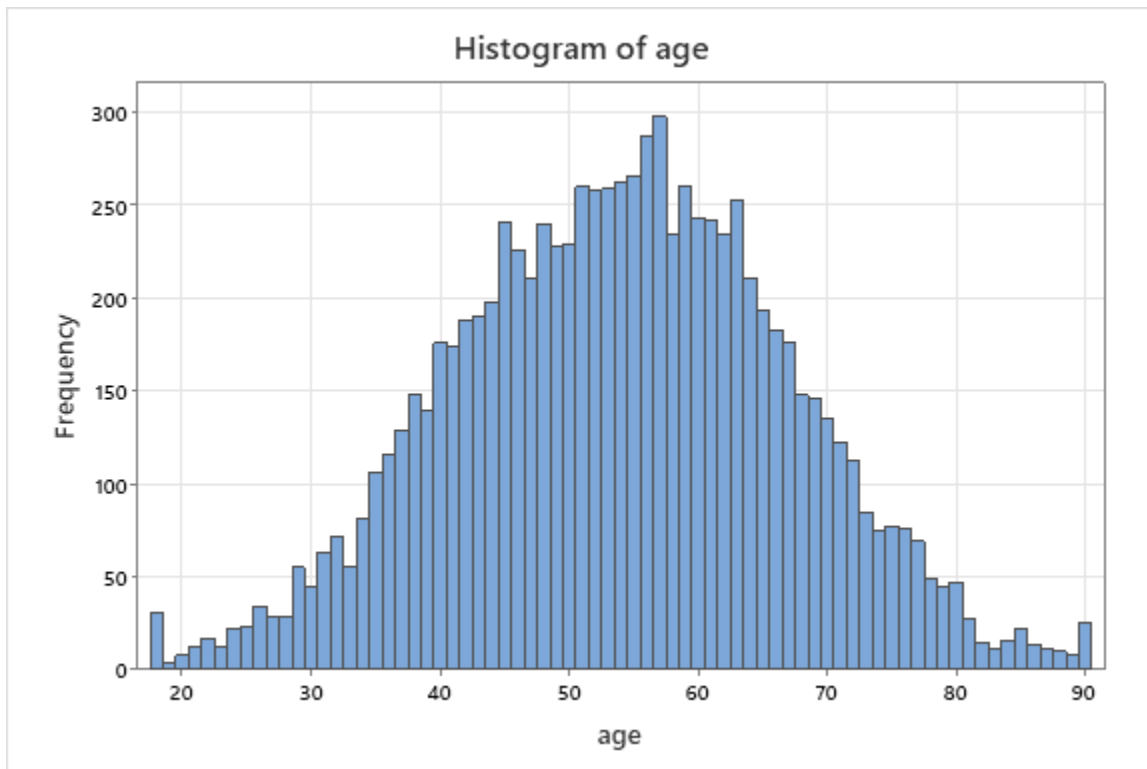
95% Confidence Interval for StDev
0.45296 0.46640

Descriptive Statistics:

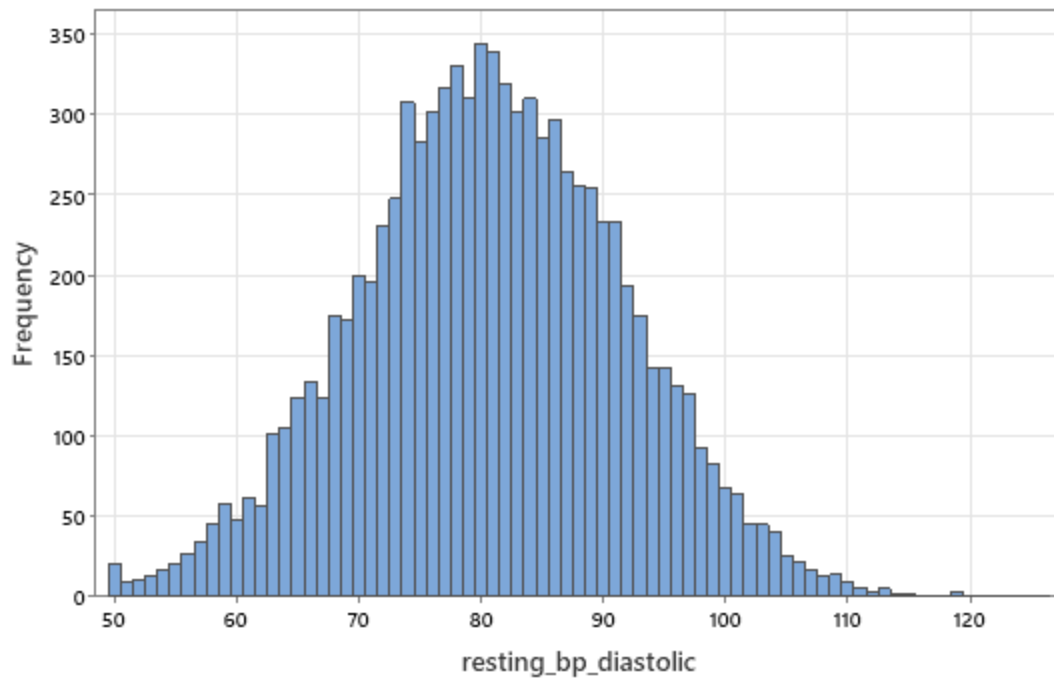
Statistics

Variable	N	N*	Mean	SE Mean	StDev	Variance	Sum	
age	9000	0	53.9773	0.137117	13.0081	169.209	485796	
resting_bp_systolic	9000	0	127.850	0.145119	13.7672	189.537	1150648	
resting_bp_diastolic	9000	0	80.7888	0.115317	10.9399	119.682	727099	
cholesterol_total	9000	0	188.754	0.340173	32.2716	1041.46	1698782	
hdl	9000	0	55.2153	0.130648	12.3944	153.621	496938	
ldl	9000	0	103.276	0.288212	27.3422	747.596	929480	
triglycerides	9000	0	152.045	0.542314	51.4484	2646.94	1368405	
fasting_blood_sugar	9000	0	119.470	0.240102	22.7781	518.841	1075234	
hba1c	9000	0	5.79163	0.0073011	0.692645	0.479758	52124.7	
bmi	9000	0	25.2649	0.0460799	4.37152	19.1102	227384	
resting_heart_rate	9000	0	80.9127	0.0907013	8.60468	74.0406	728214	
max_heart_rate_achieved	9000	0	164.838	0.224424	21.2908	453.297	1483545	
st_depression	9000	0	1.01291	0.0100886	0.957090	0.916022	9116.2	
alcohol_units_per_week	9000	0	5.70873	0.0534493	5.07064	25.7114	51378.6	
exercise_minutes_per_week	9000	0	139.493	0.671040	63.6604	4052.65	1255441	
sleep_hours	9000	0	6.99458	0.0116394	1.10421	1.21929	62951.2	
stress_score	9000	0	47.9680	0.170368	16.1625	261.226	431712	
daily_steps	9000	0	6163.66	22.8810	2170.69	4711878	55472934	
diet_quality_score	9000	0	59.4674	0.152591	14.4761	209.556	535207	
has_heart_disease	9000	0	0.303	0.0048444	0.459581	0.211214	2727	
Variable	Minimum	Q1	Median	Q3	Maximum	Range	IQR	Mode
age	18	45	54	63	90	72	18	57
resting_bp_systolic	85	119	128	137	181	96	18	127
resting_bp_diastolic	50	74	81	88	126	76	14	80
cholesterol_total	90	167	189	211	314	224	44	187
hdl	18	47	55	64	110	92	17	58
ldl	35	85	103	122	207	172	37	99
triglycerides	35	117	151	186	390	355	69	35
fasting_blood_sugar	60	104	119	135	204	144	31	119
hba1c	4	5.3	5.8	6.3	8.6	4.6	1	5.8, 5.9
bmi	15	22.3	25.3	28.2	43.3	28.3	5.9	25.8
resting_heart_rate	48	75	81	87	111	63	12	81
max_heart_rate_achieved	93	150	166	180	210	117	30	168
st_depression	0	0.4	0.7	1.4	6.5	6.5	1	0.2
alcohol_units_per_week	0	2.2	4.2	7.7	45.9	45.9	5.5	1.6
exercise_minutes_per_week	0	96	139	183	366	366	87	0
sleep_hours	3.1	6.3	7	7.7	11	7.9	1.4	7.2
stress_score	0	37.1	48.1	58.9	100	100	21.8	48.7
daily_steps	500	4658	6178	7634.25	13950	13450	2976.25	500
diet_quality_score	4.8	49.7	59.6	69.1	100	95.2	19.4	52.8
has_heart_disease	0	0	0	1	1	1	1	0
Variable	N for Mode	Skewness	Kurtosis					
age	298	-0.02	-0.21					
resting_bp_systolic	276	0.07	0.03					
resting_bp_diastolic	344	0.01	-0.08					
cholesterol_total	138	0.00	-0.03					
hdl	316	0.02	-0.03					
ldl	159	0.04	-0.10					
triglycerides	114	0.13	-0.02					
fasting_blood_sugar	182	0.04	-0.14					
hba1c	549	0.03	-0.08					
bmi	98	0.09	-0.12					
resting_heart_rate	431	-0.02	-0.02					
max_heart_rate_achieved	173	-0.18	-0.37					

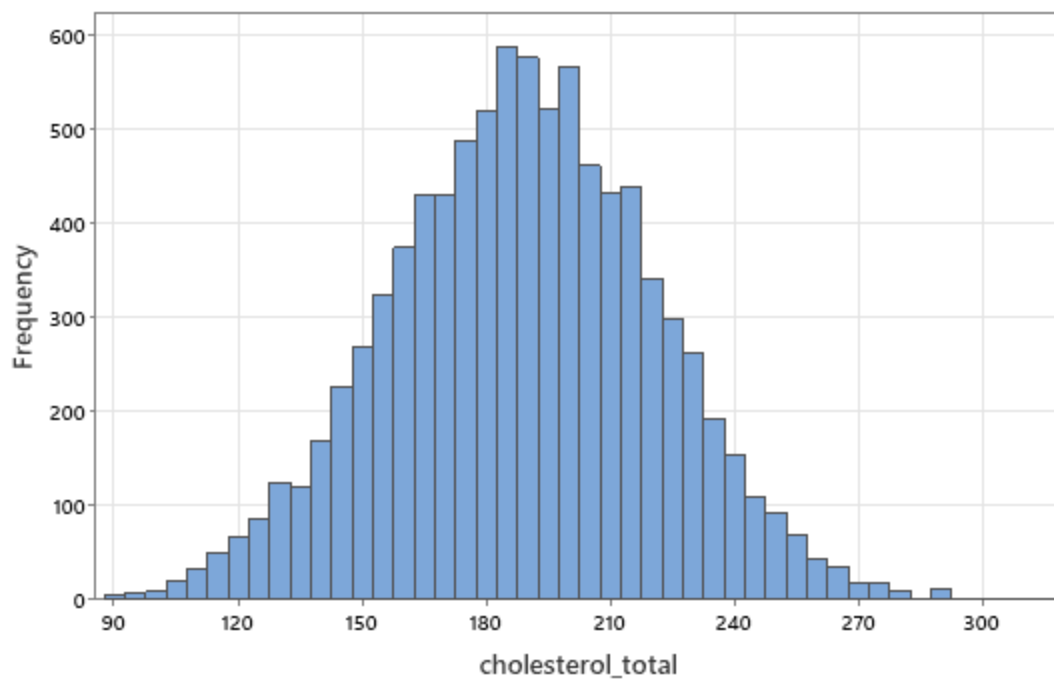
st_depression	692	2.04	5.61
alcohol_units_per_week	192	2.03	6.29
exercise_minutes_per_week	146	0.09	-0.23
sleep_hours	342	-0.04	-0.03
stress_score	40	-0.01	-0.01
daily_steps	39	0.02	-0.13
diet_quality_score	38	0.00	-0.08
has_heart_disease	6273	0.86	-1.26

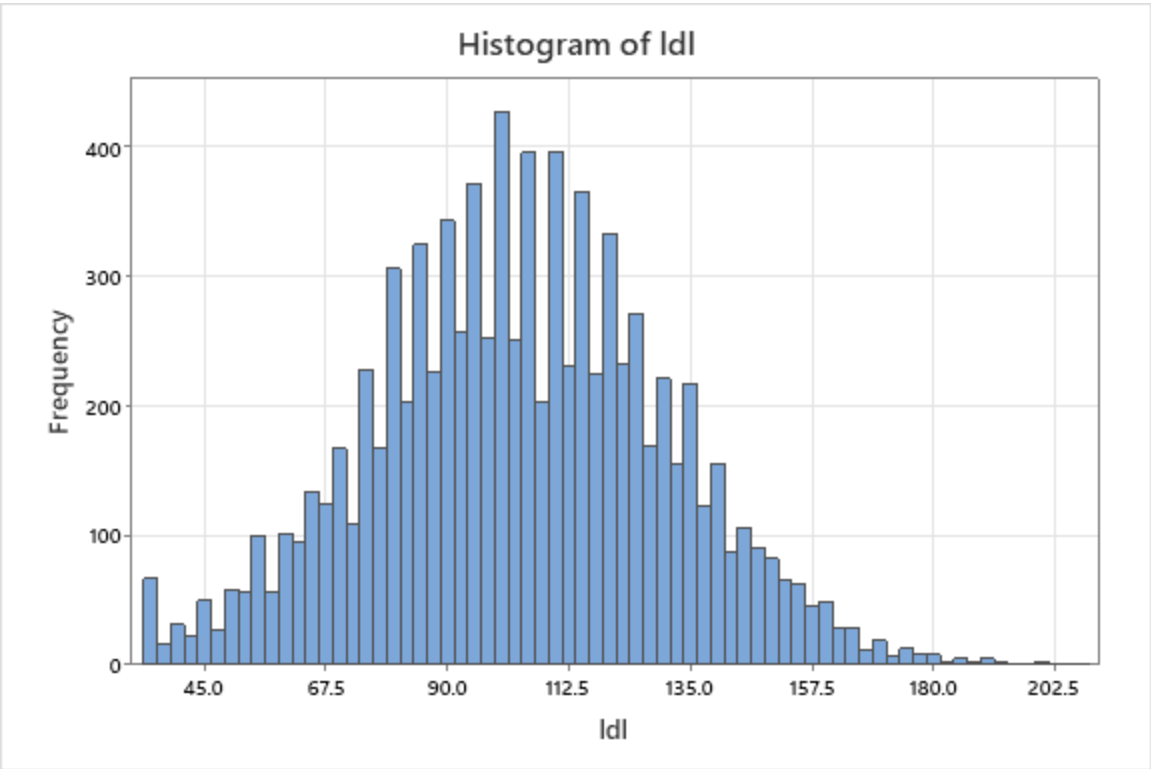
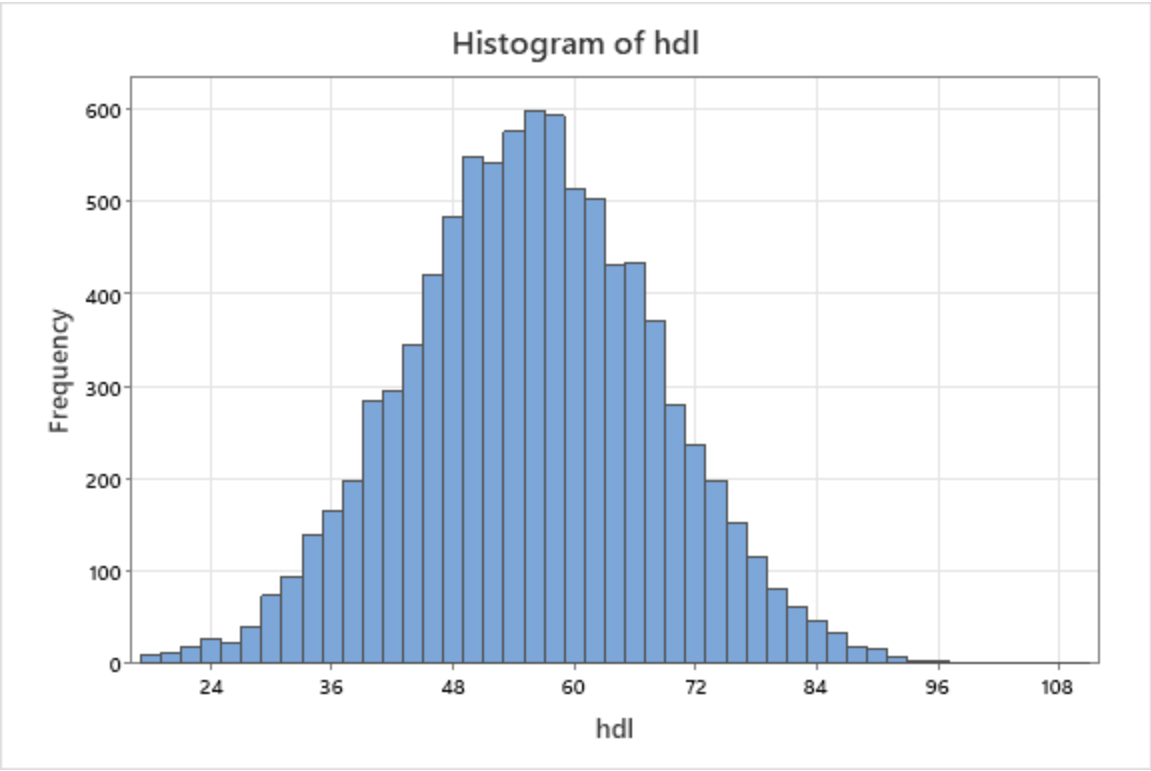


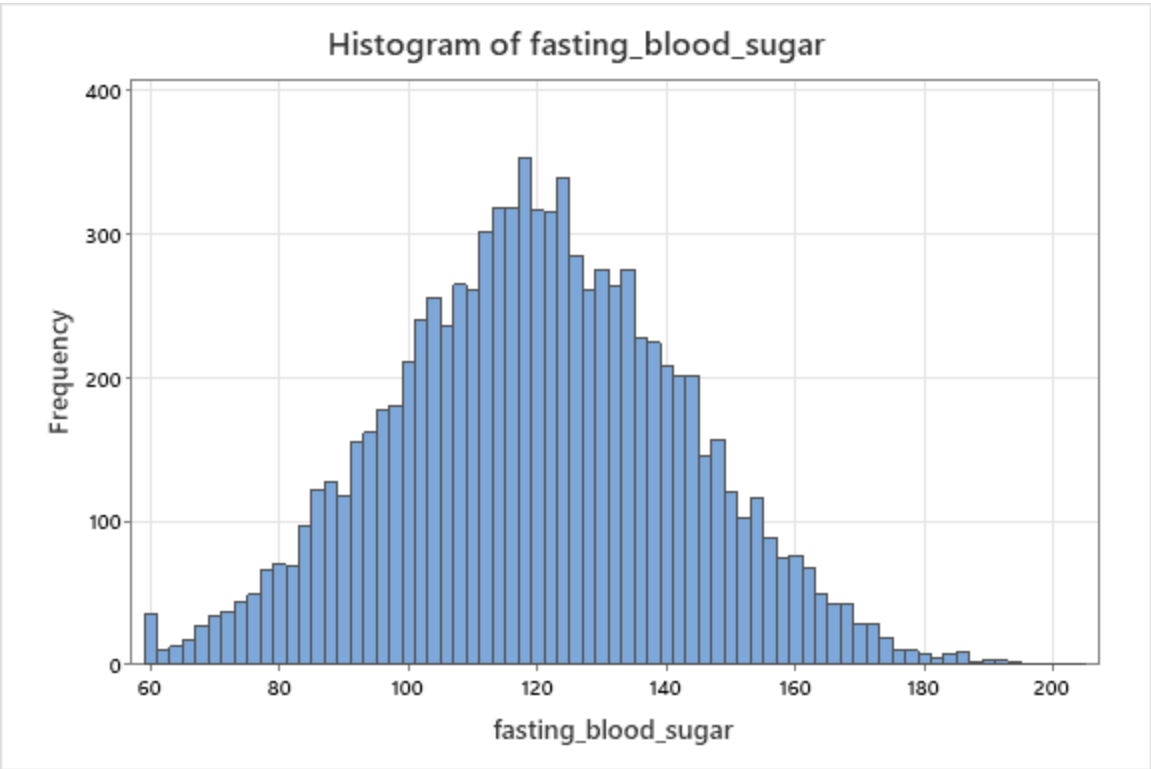
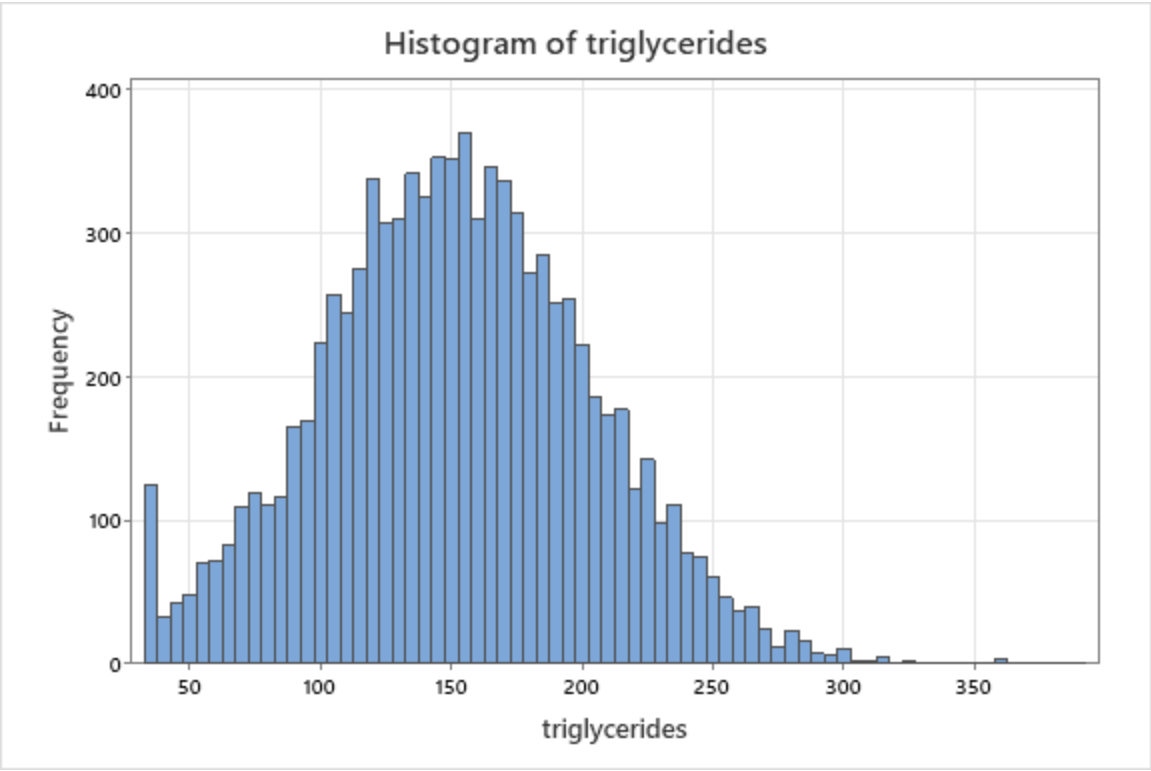
Histogram of resting_bp_diastolic

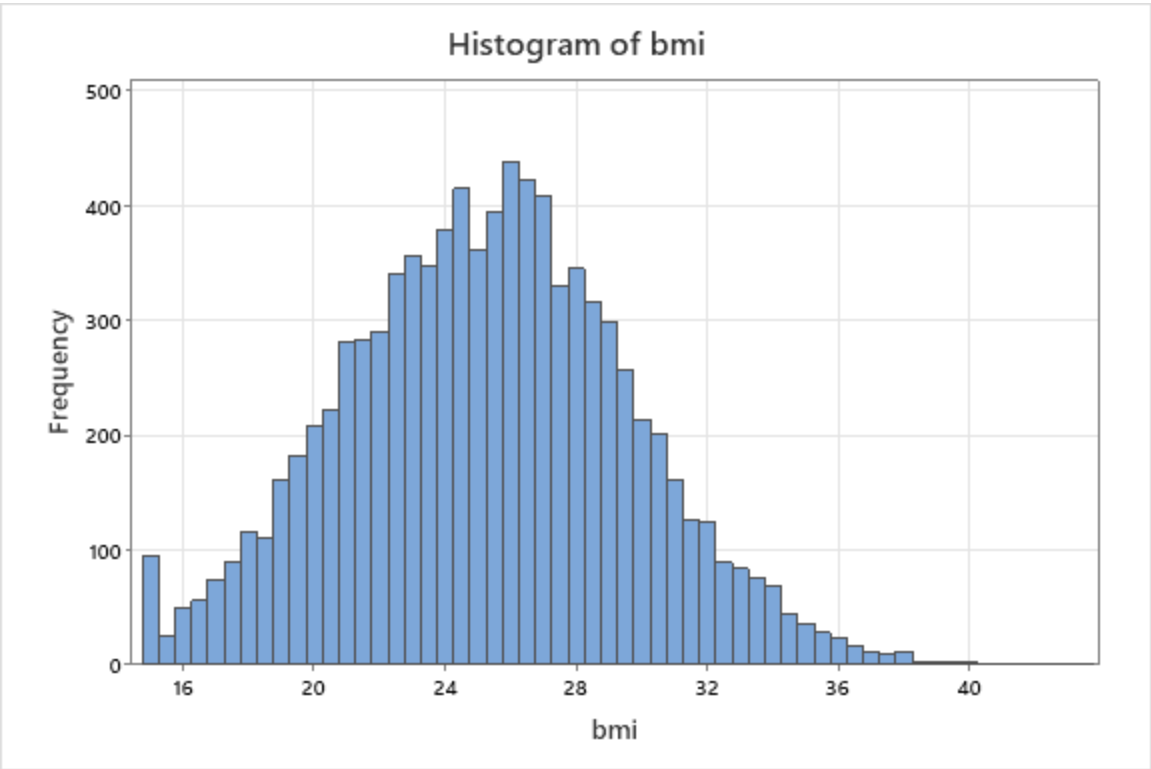
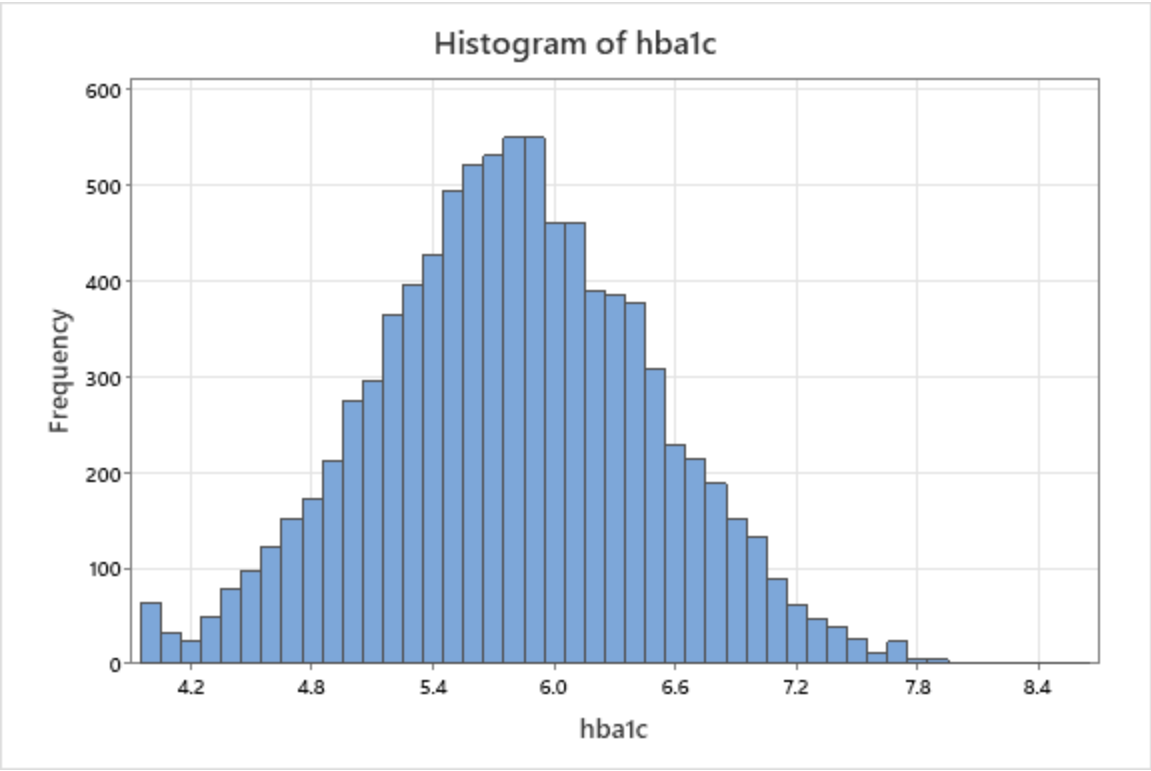


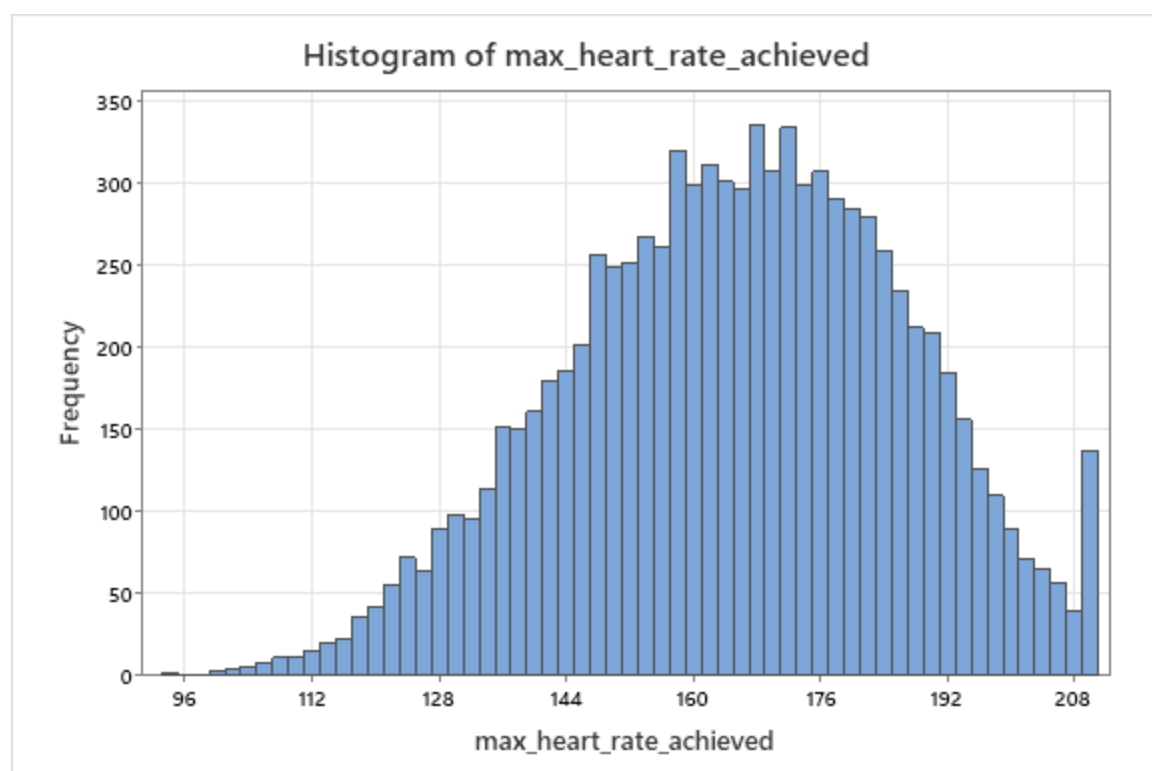
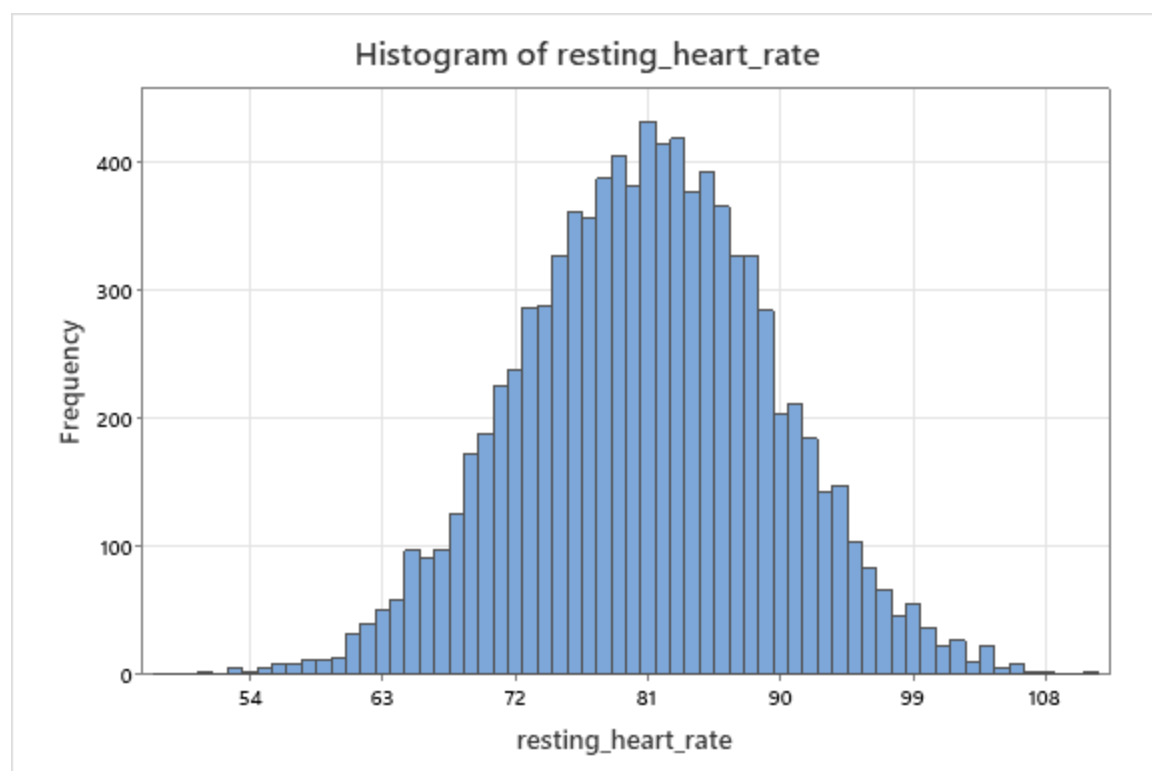
Histogram of cholesterol_total

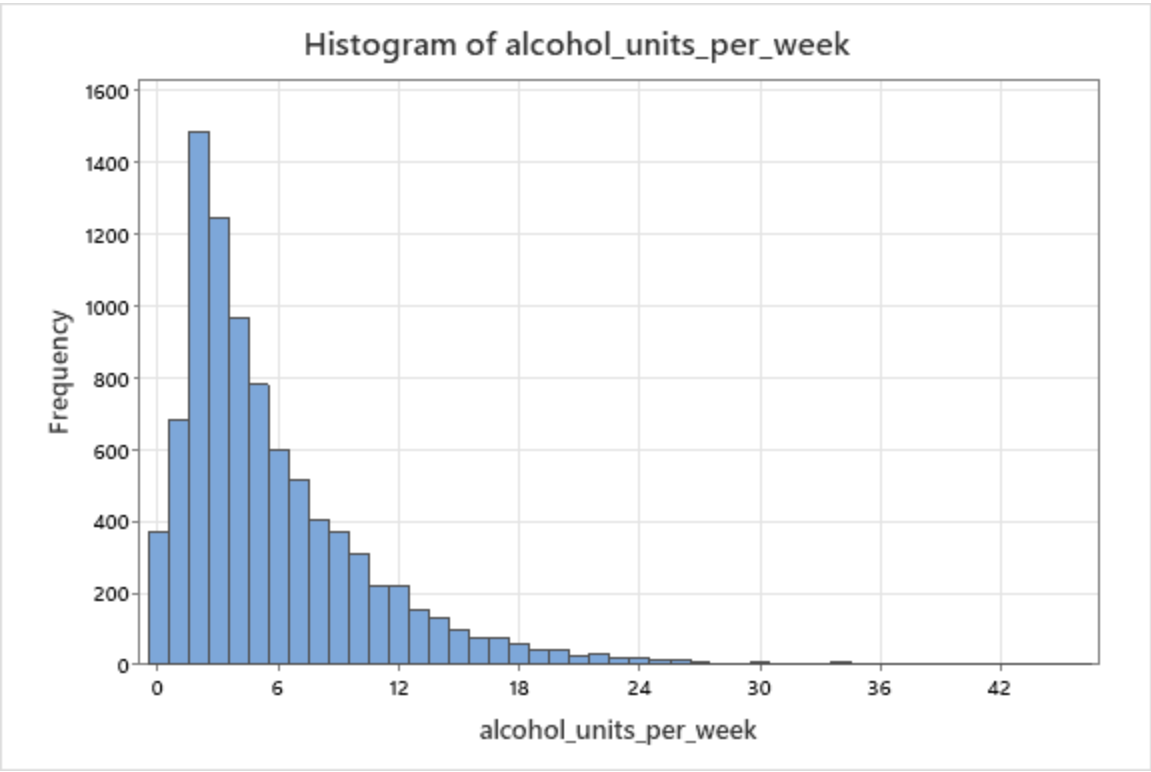
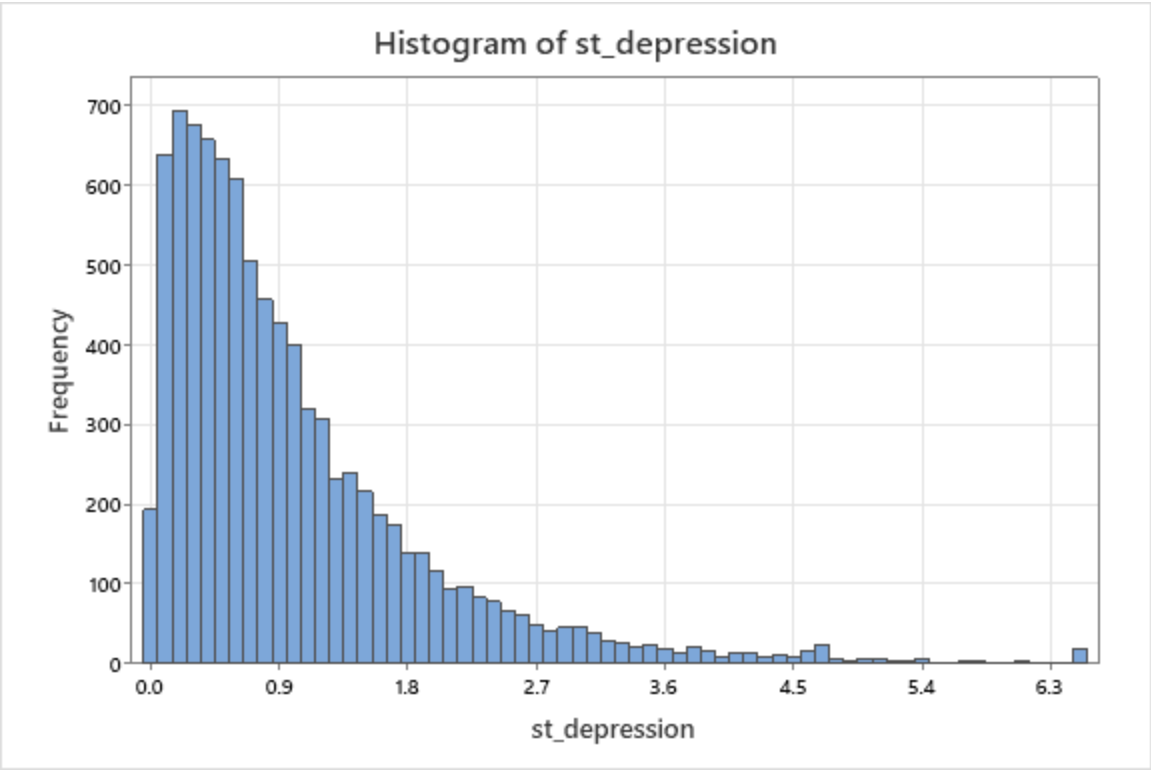


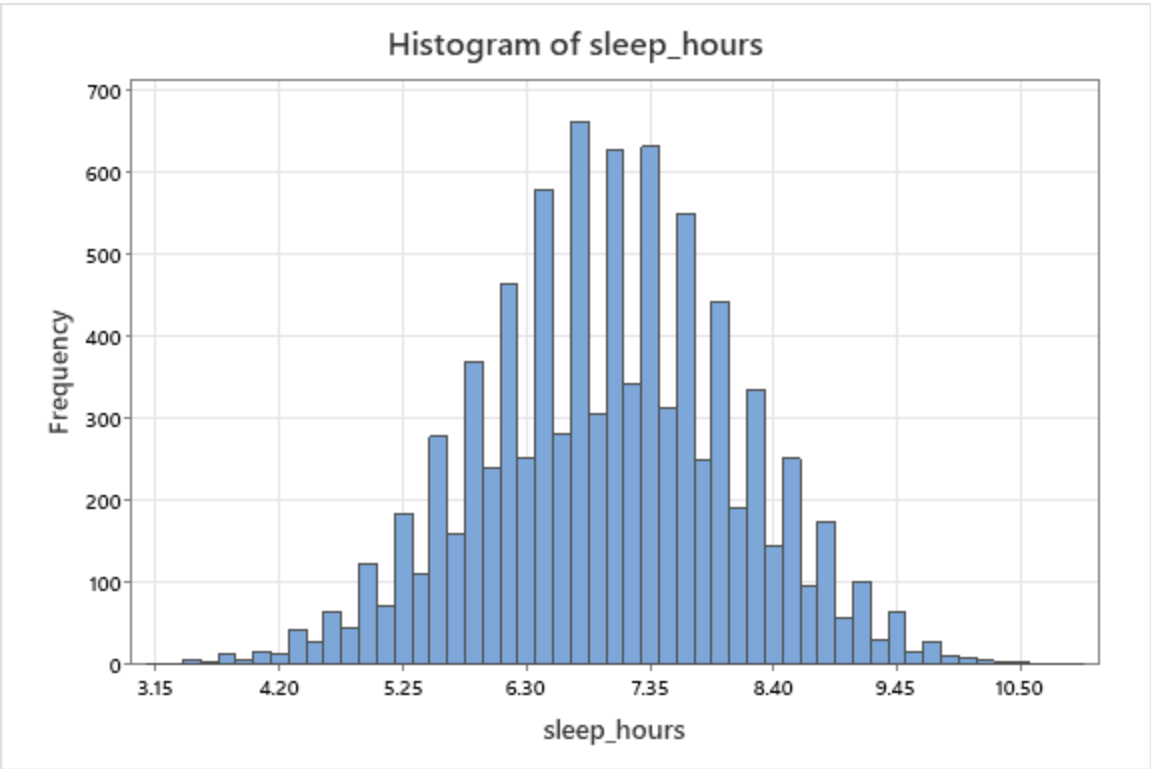
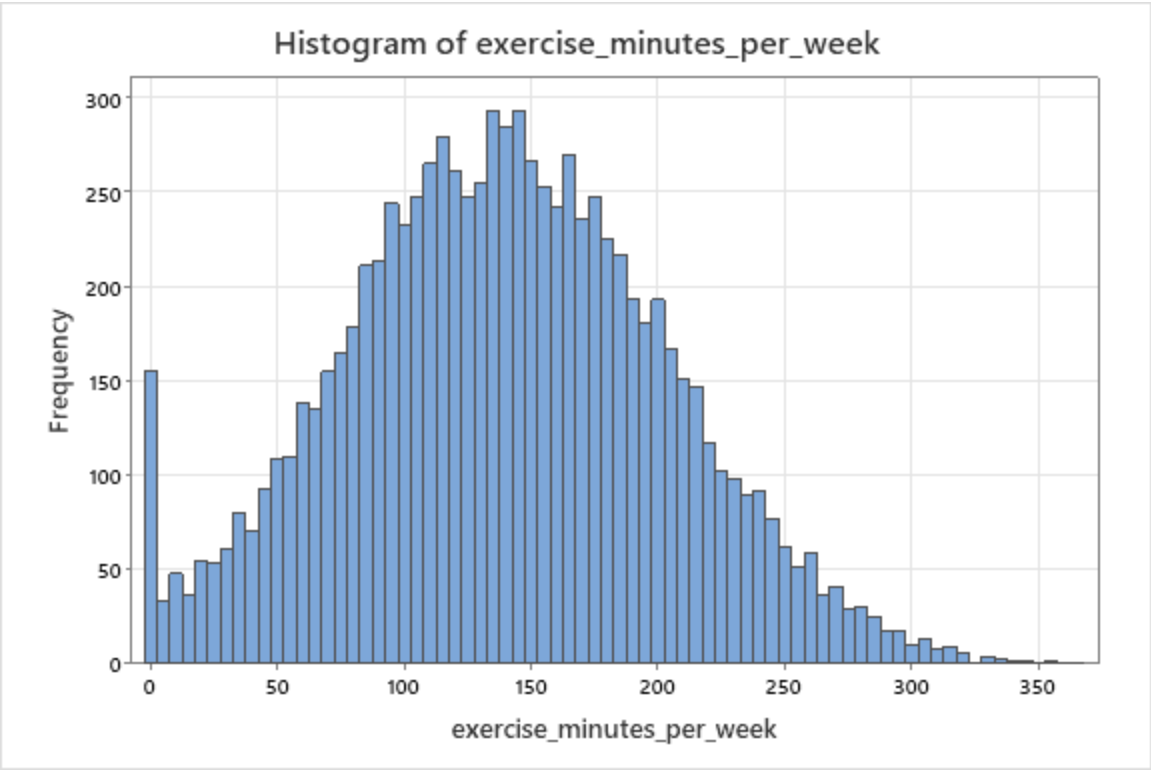


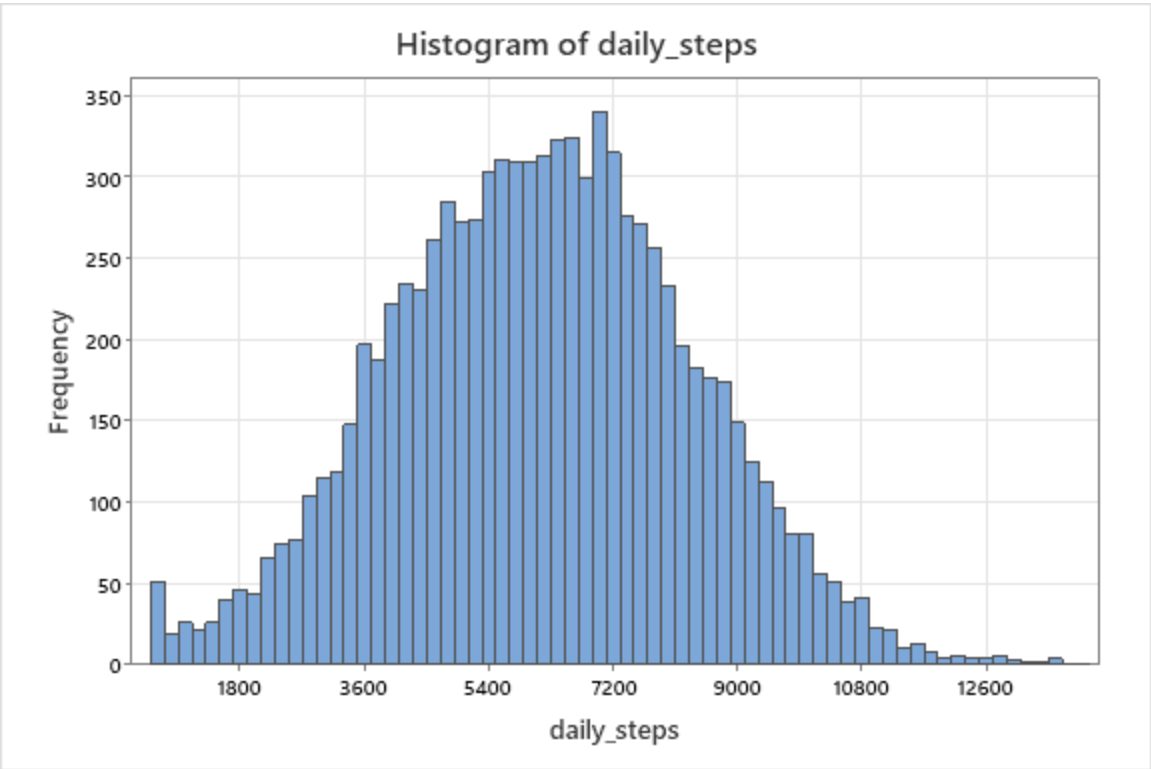
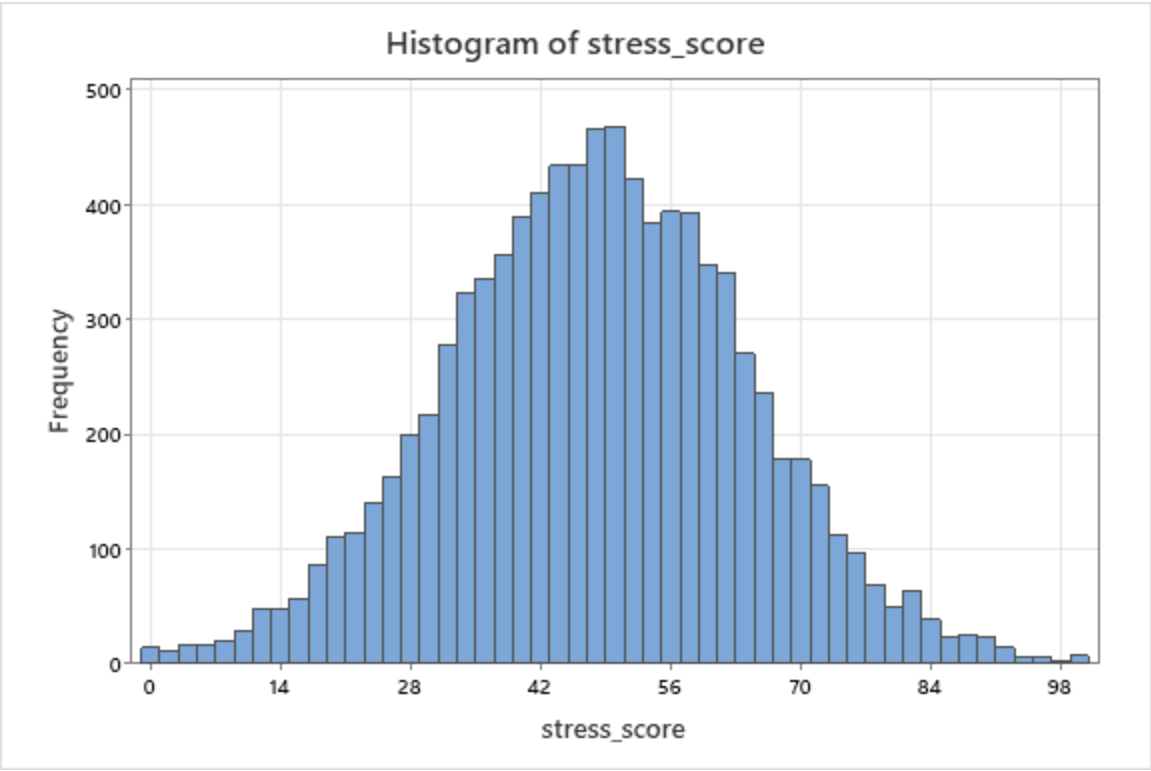


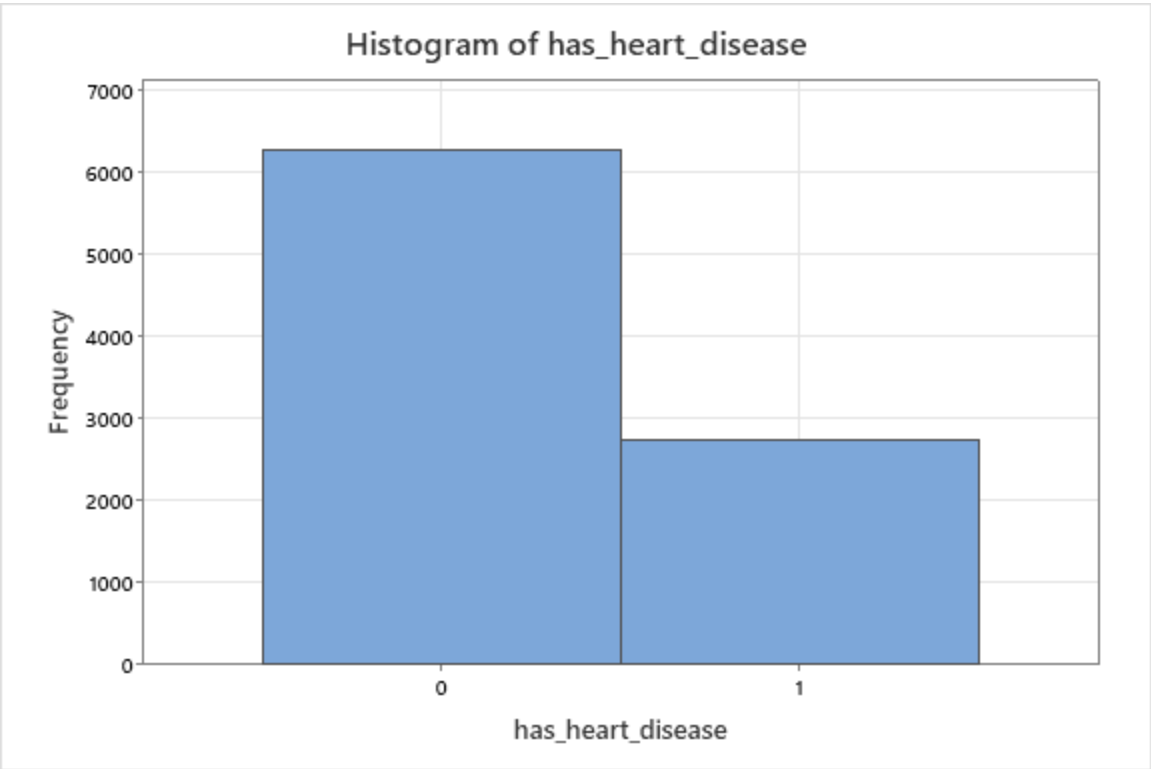


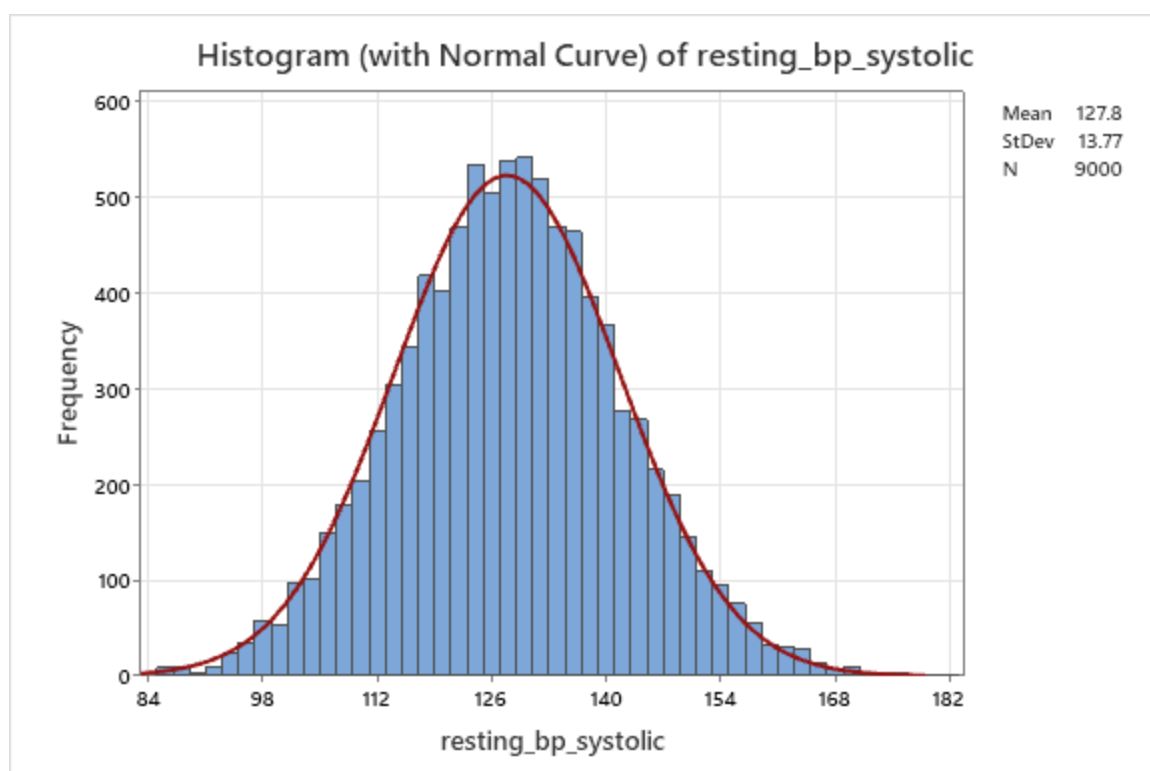
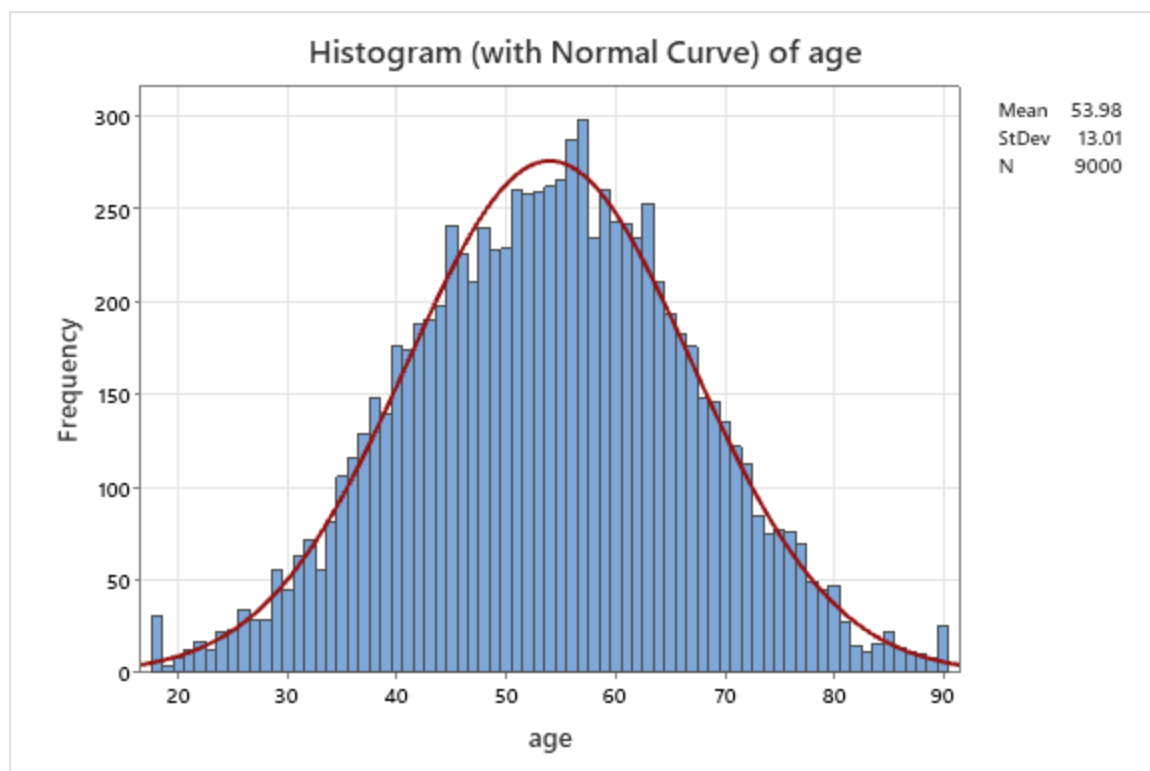




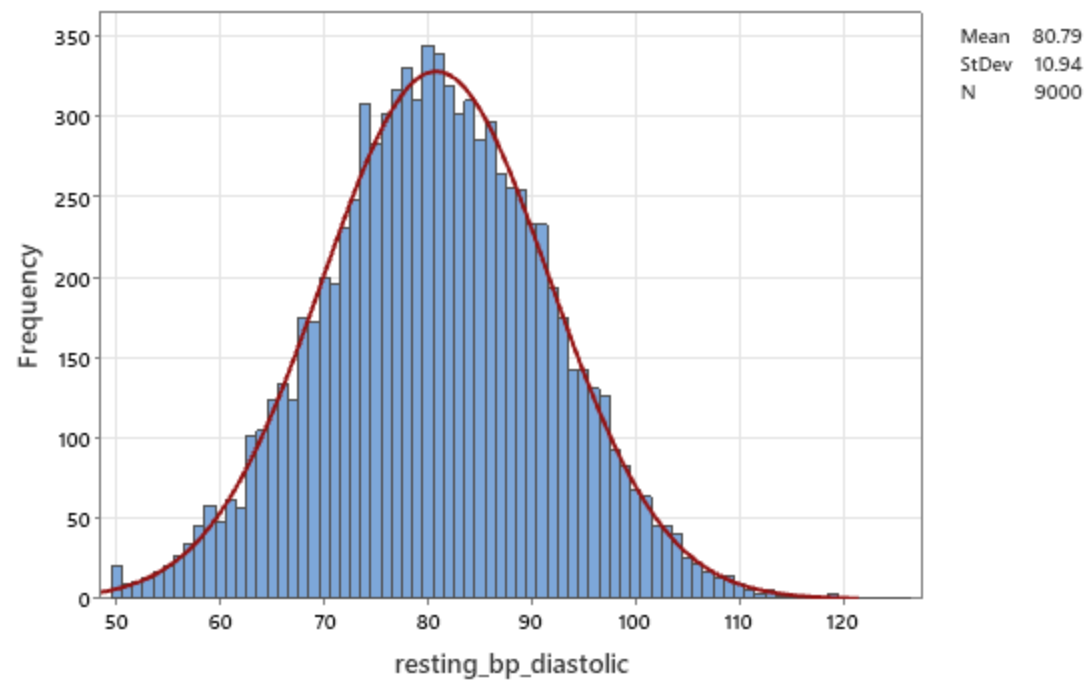




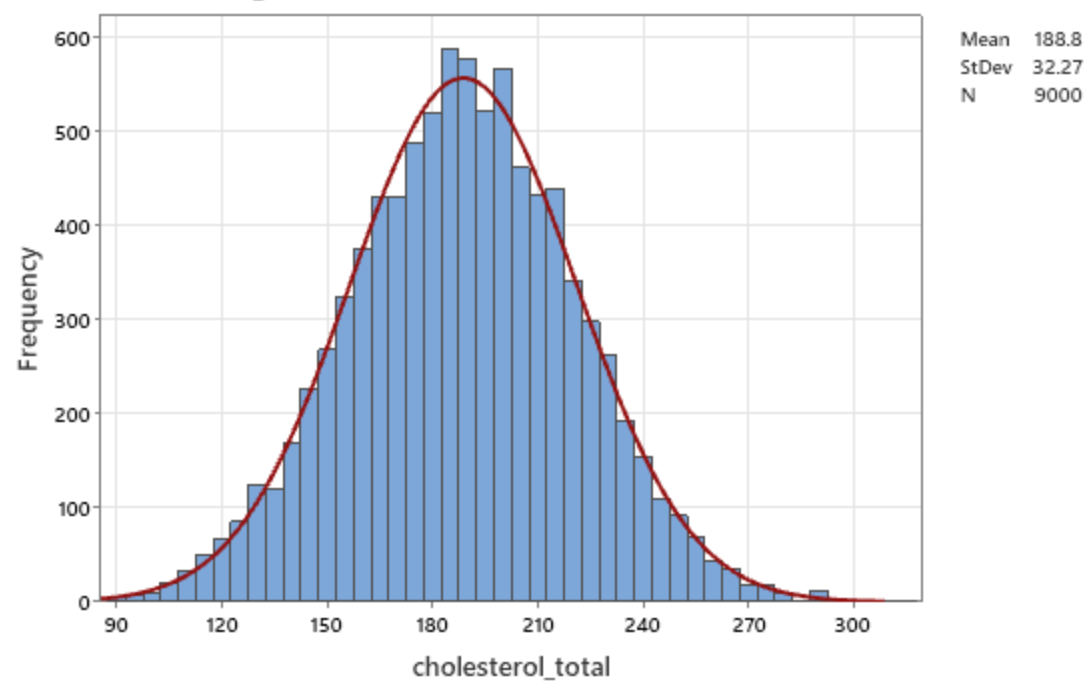


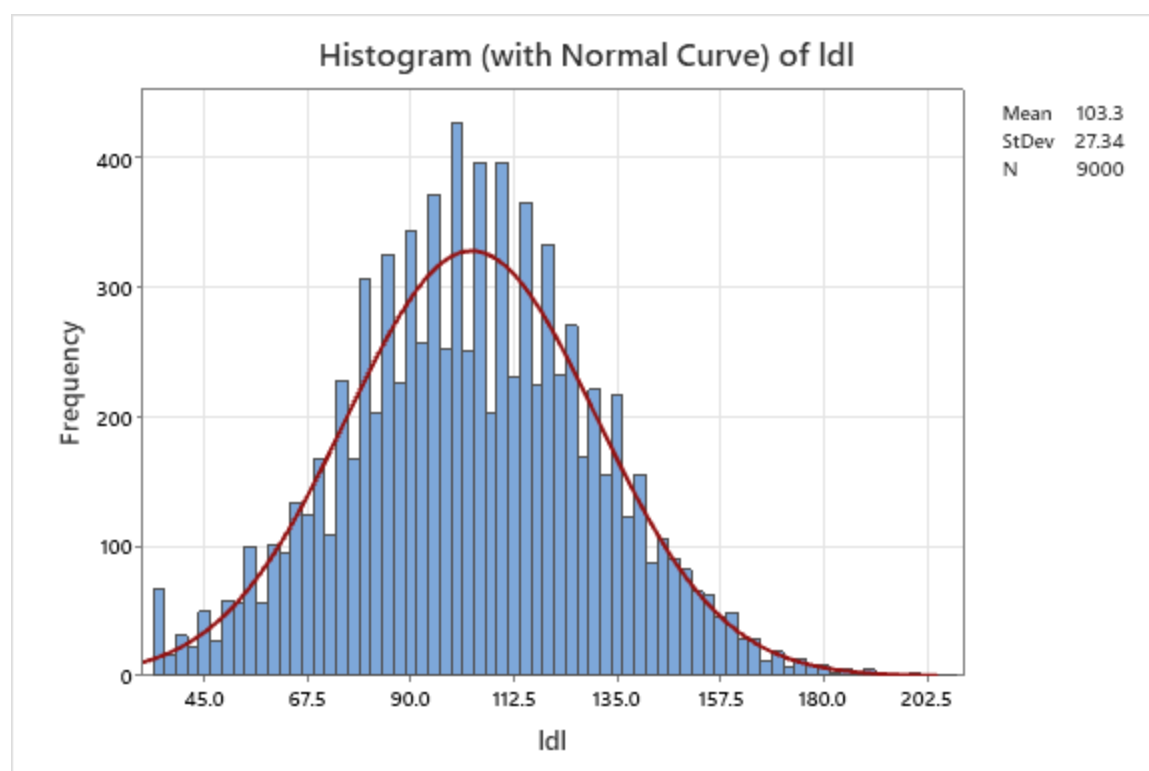
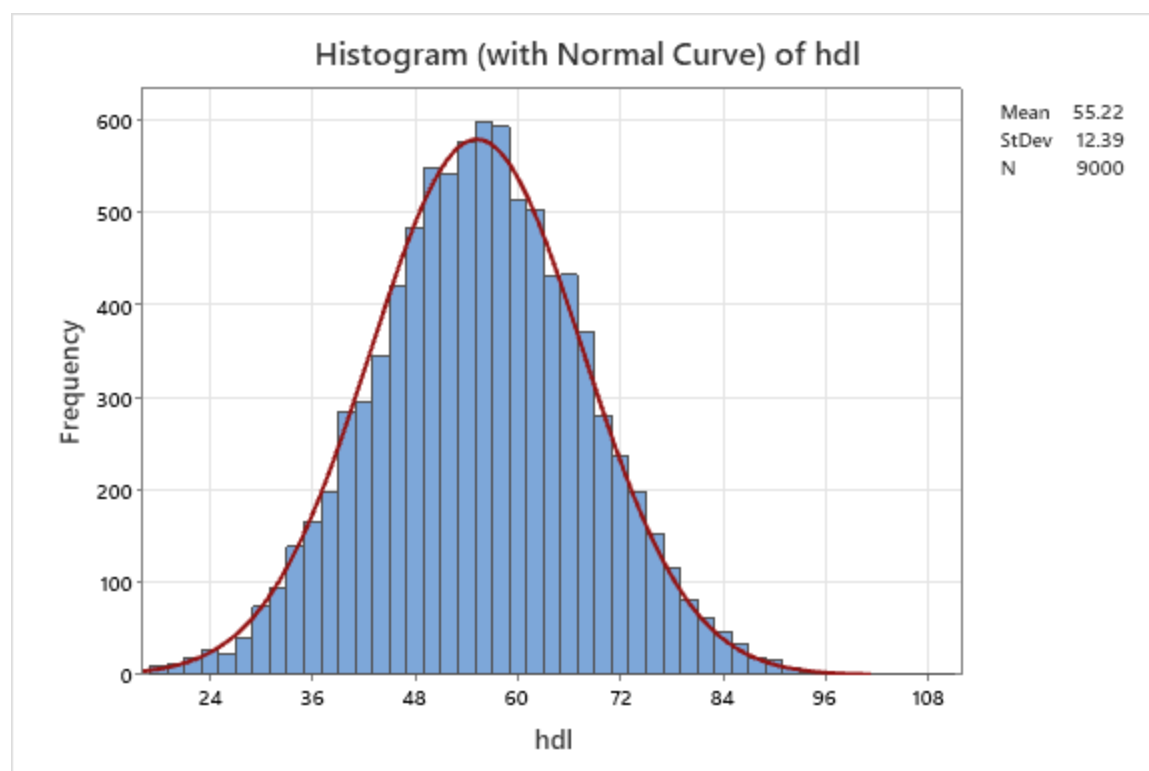


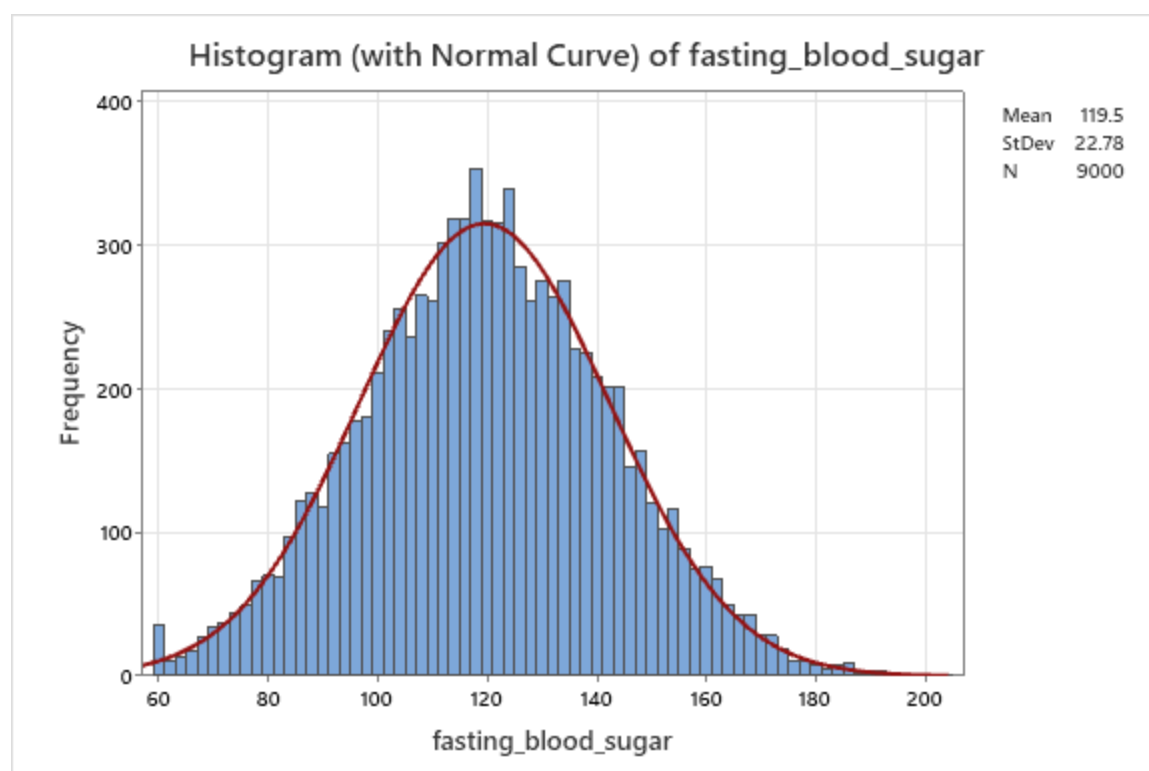
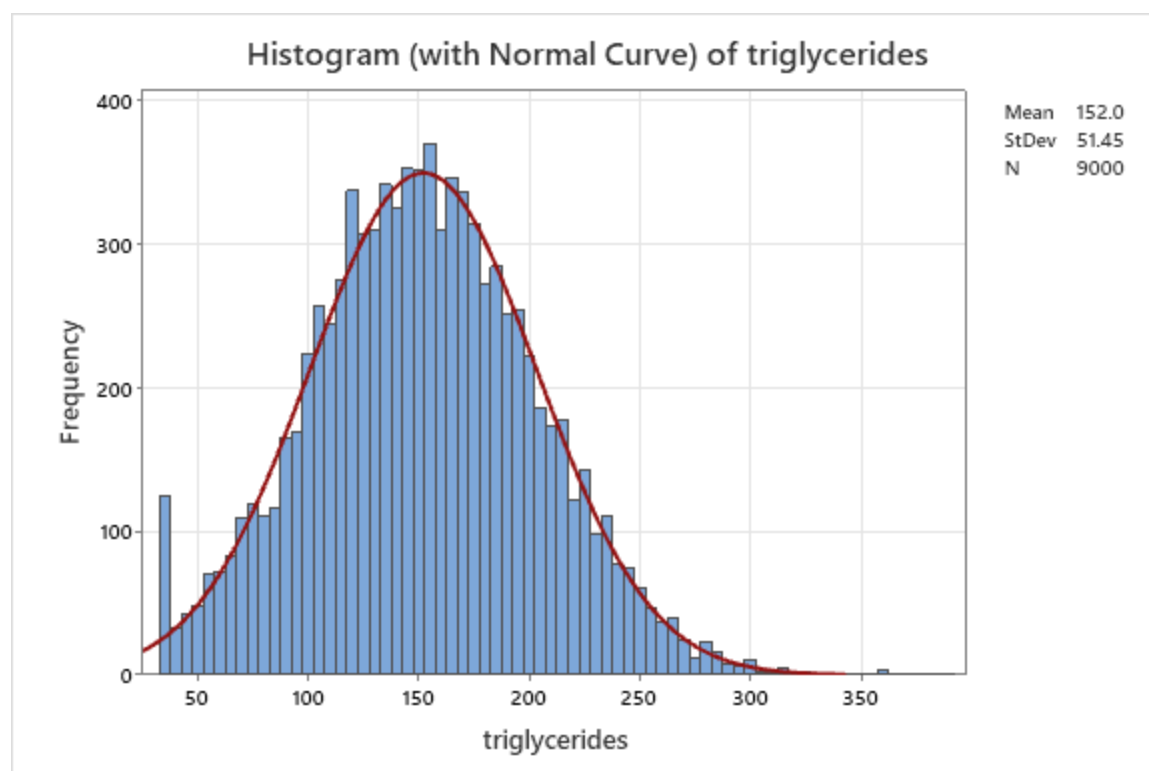
Histogram (with Normal Curve) of resting_bp_diastolic

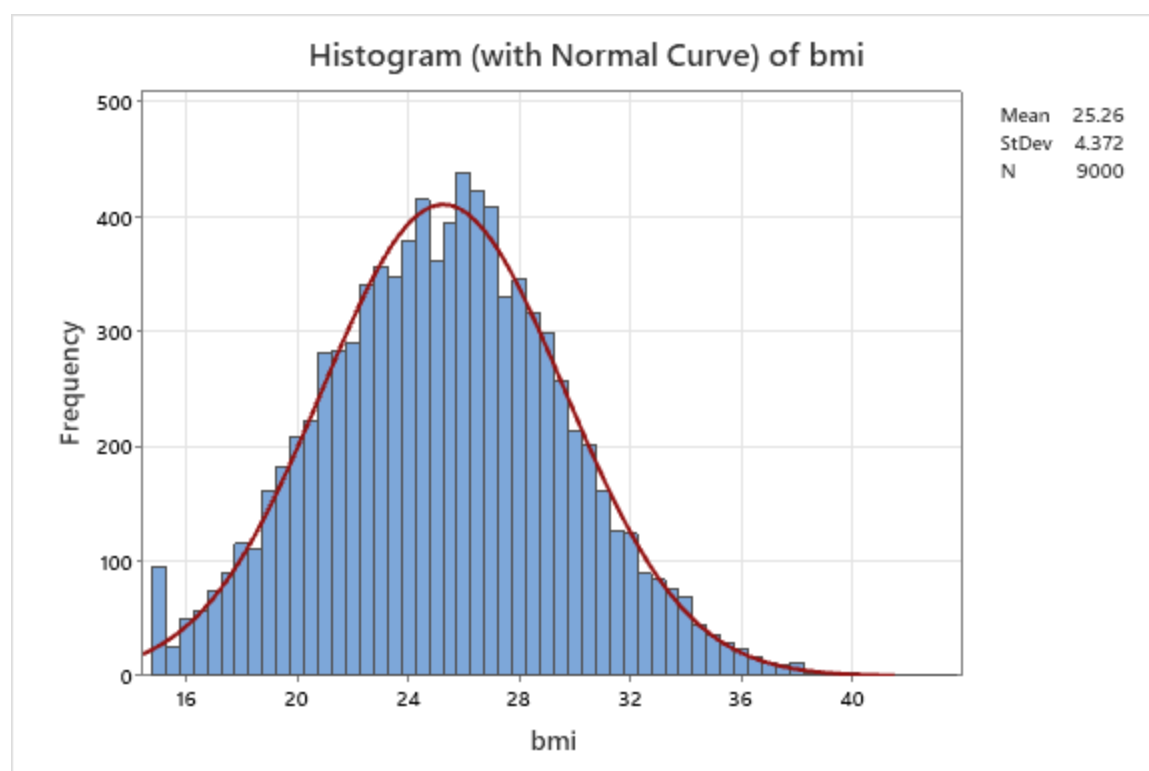
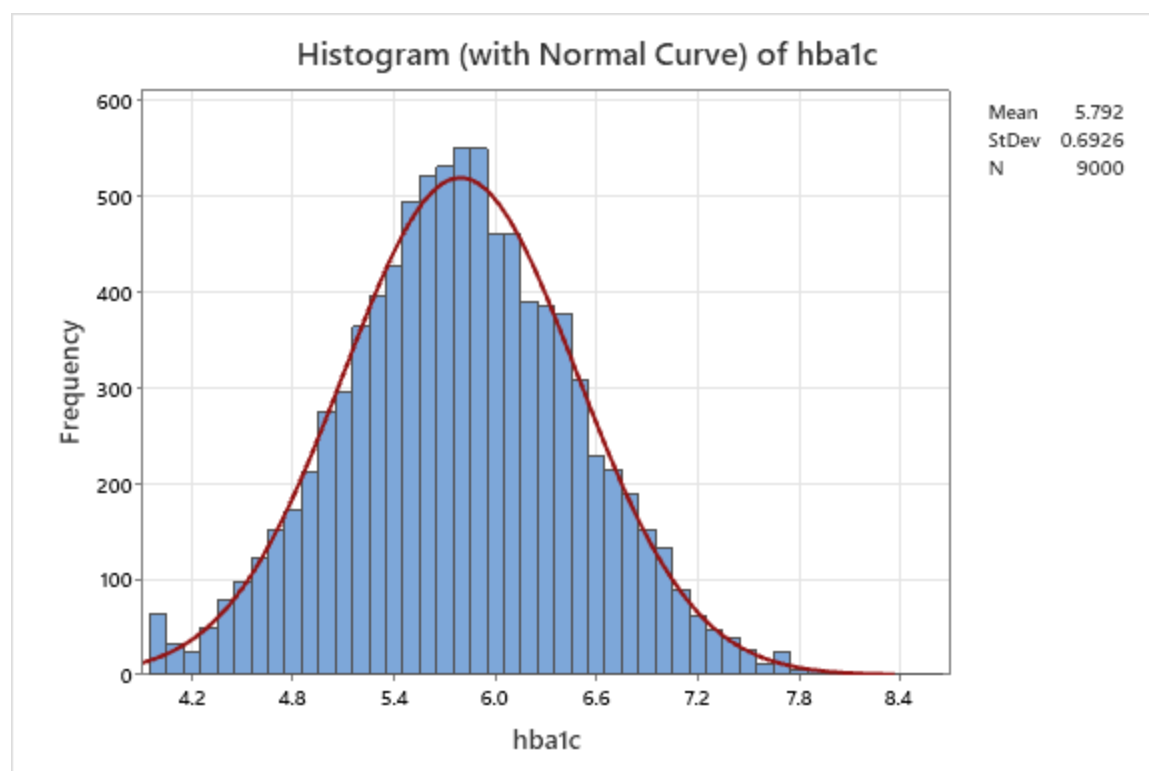


Histogram (with Normal Curve) of cholesterol_total

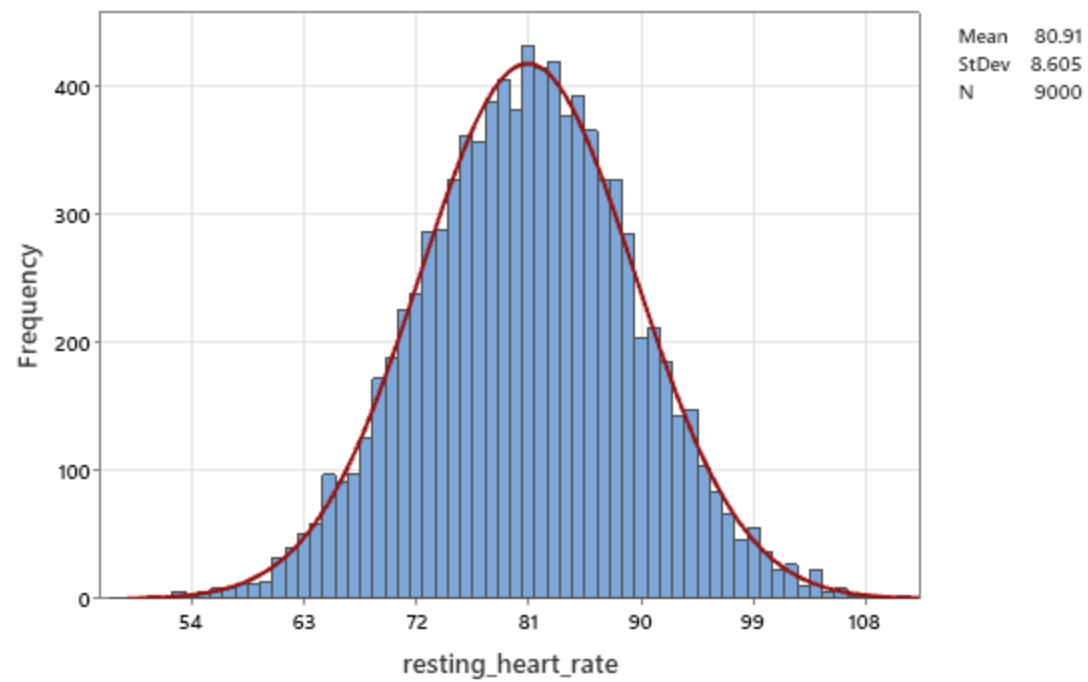




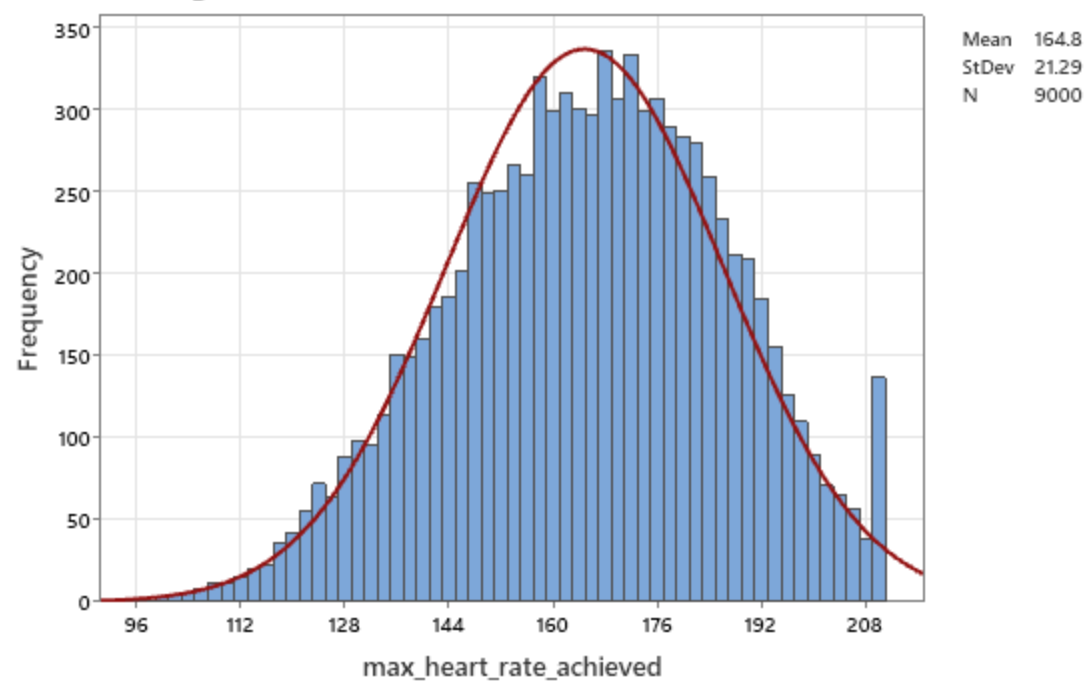


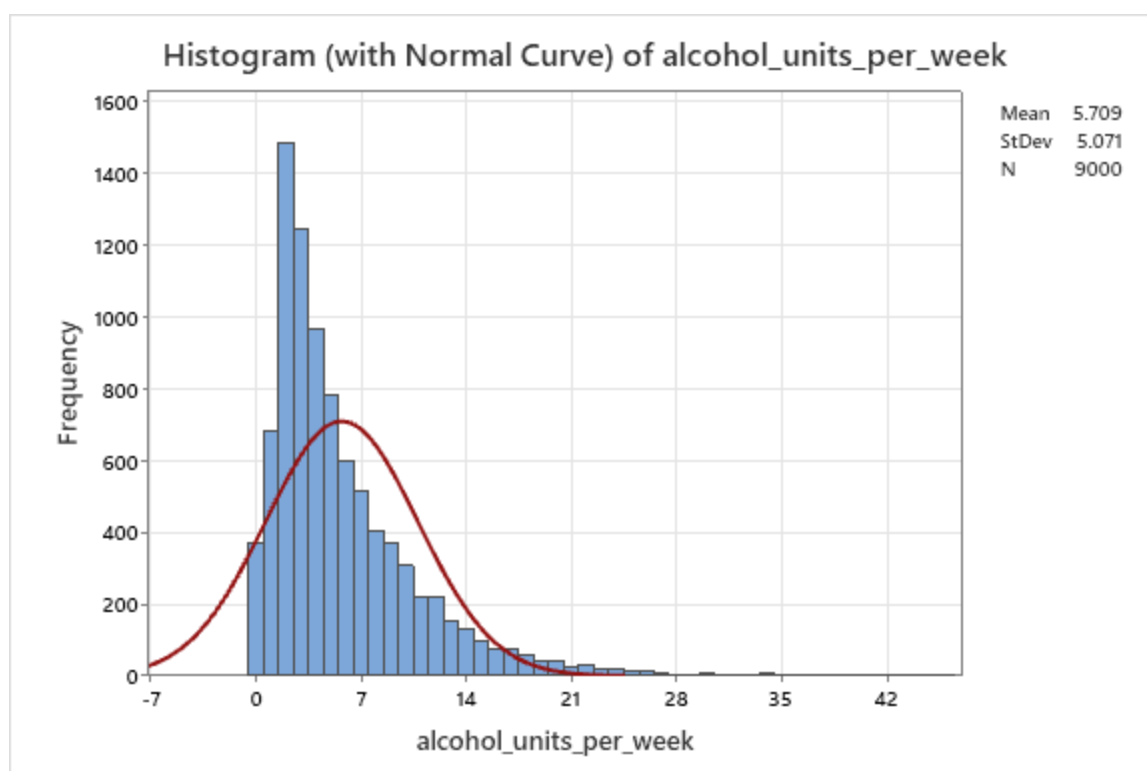
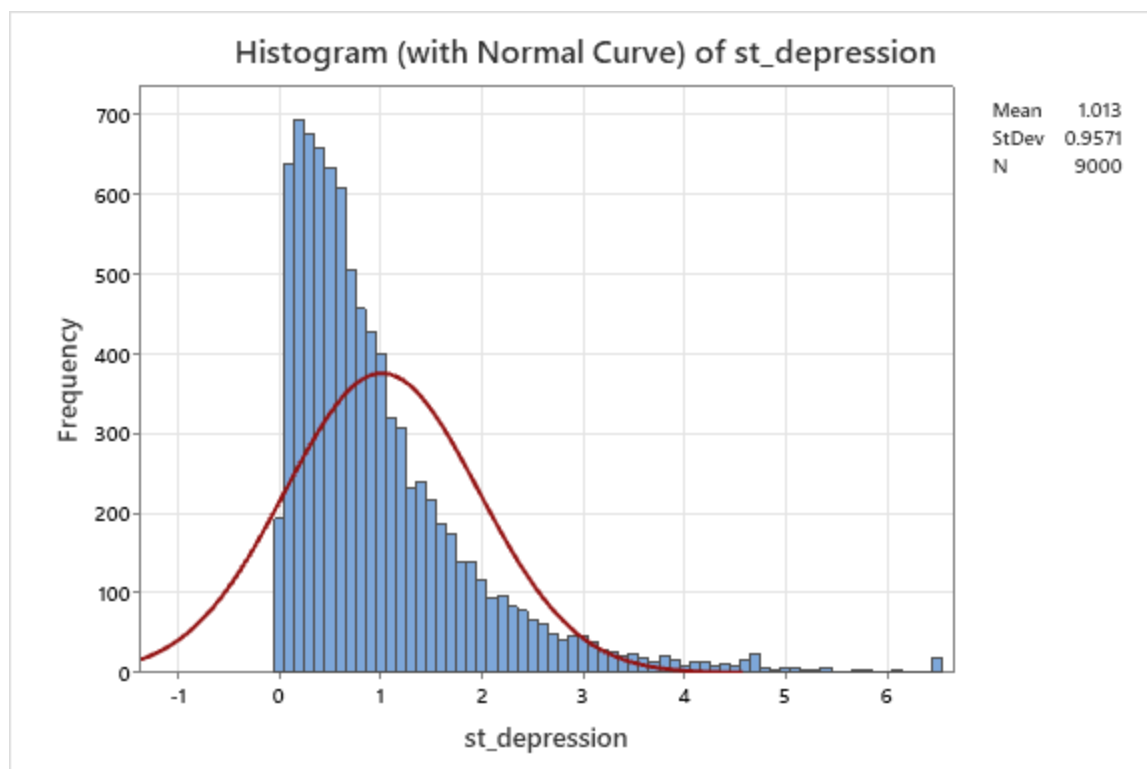


Histogram (with Normal Curve) of resting_heart_rate

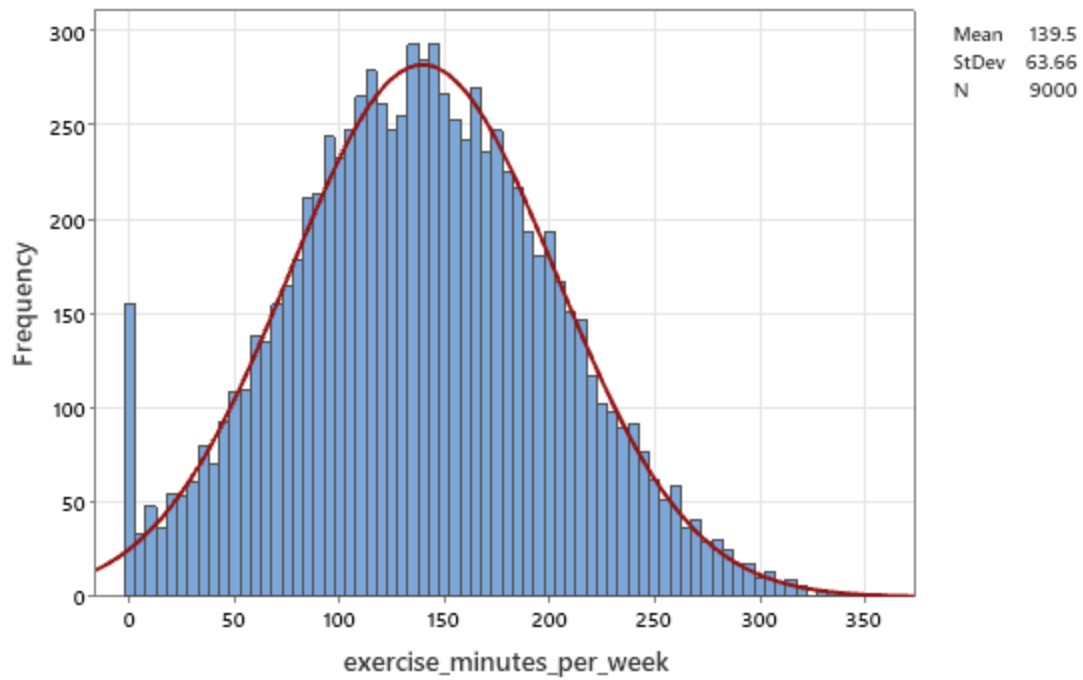


Histogram (with Normal Curve) of max_heart_rate_achieved

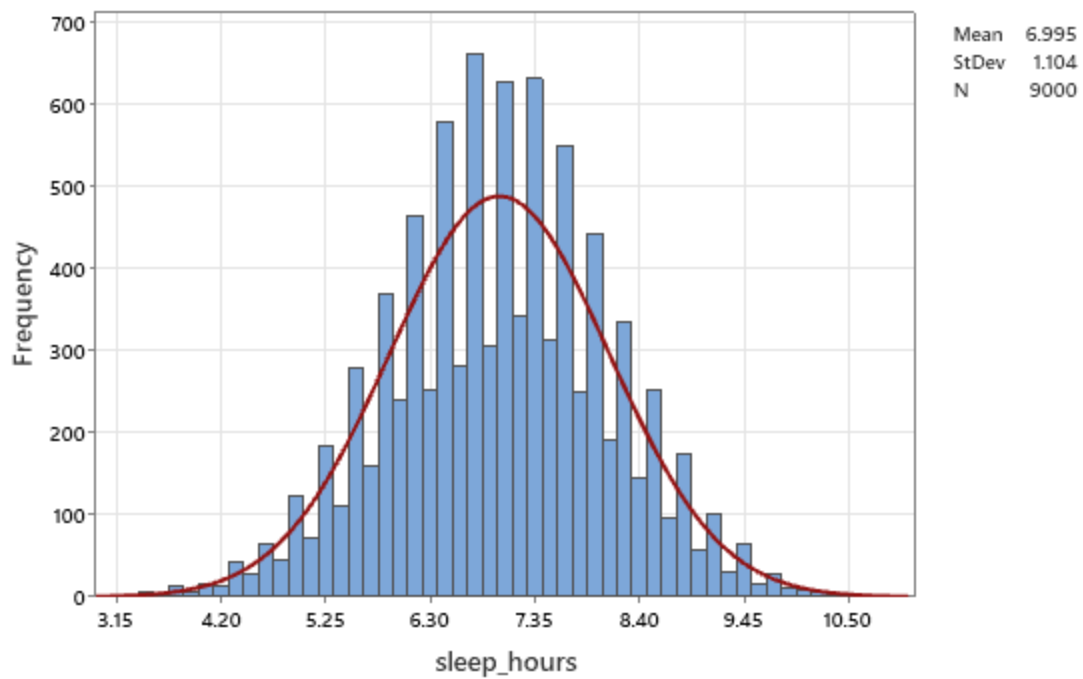




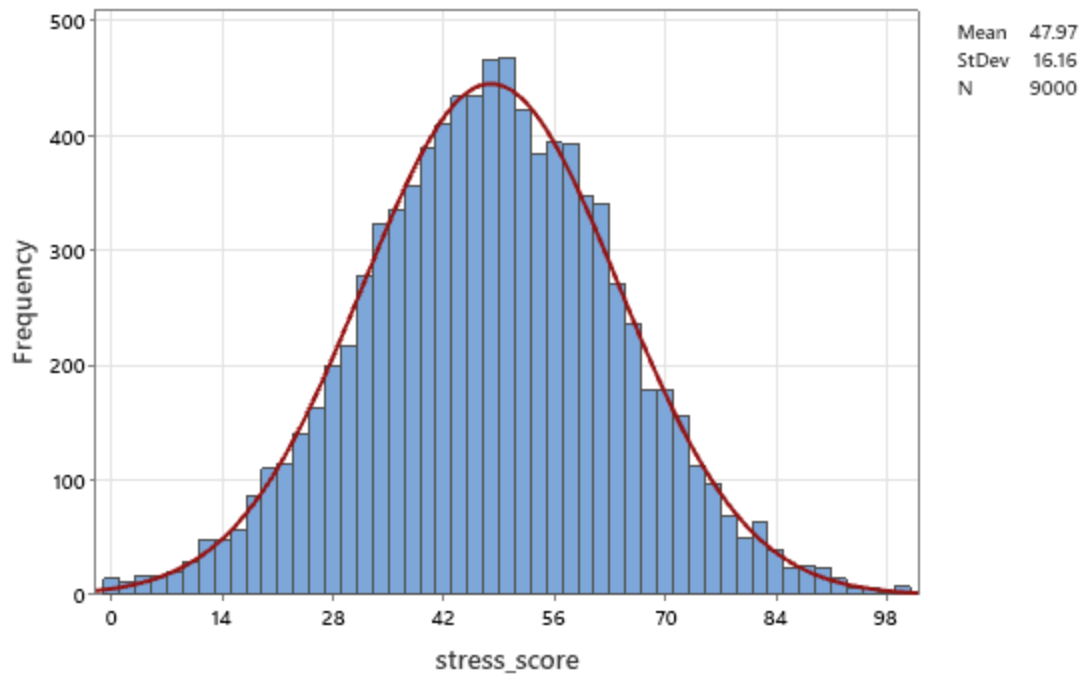
Histogram (with Normal Curve) of exercise_minutes_per_week



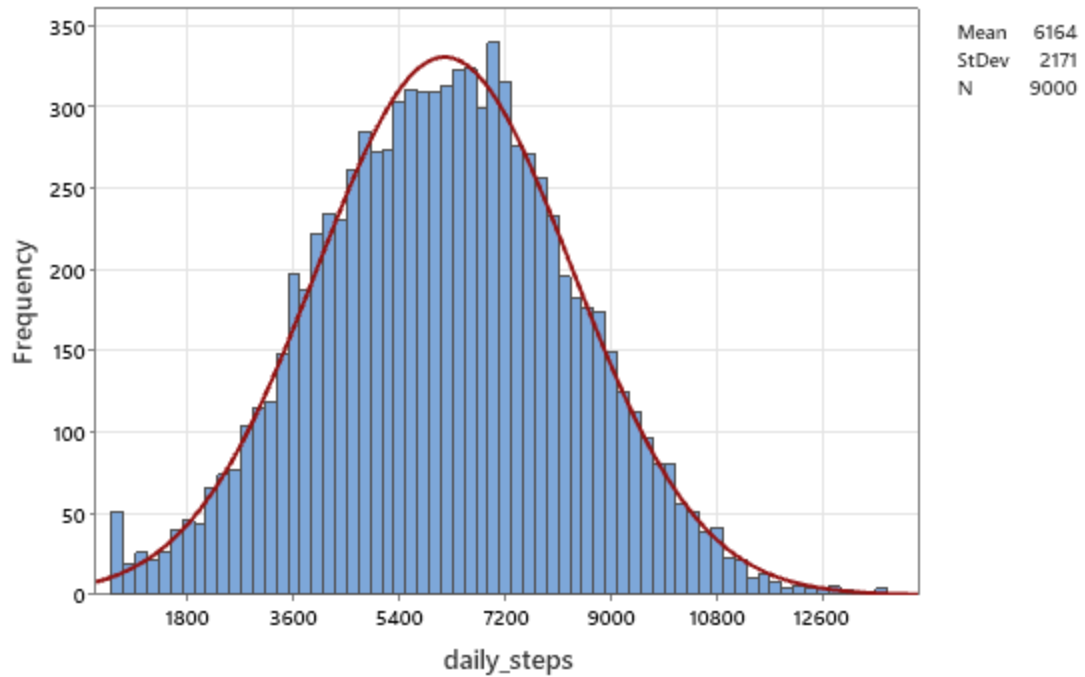
Histogram (with Normal Curve) of sleep_hours

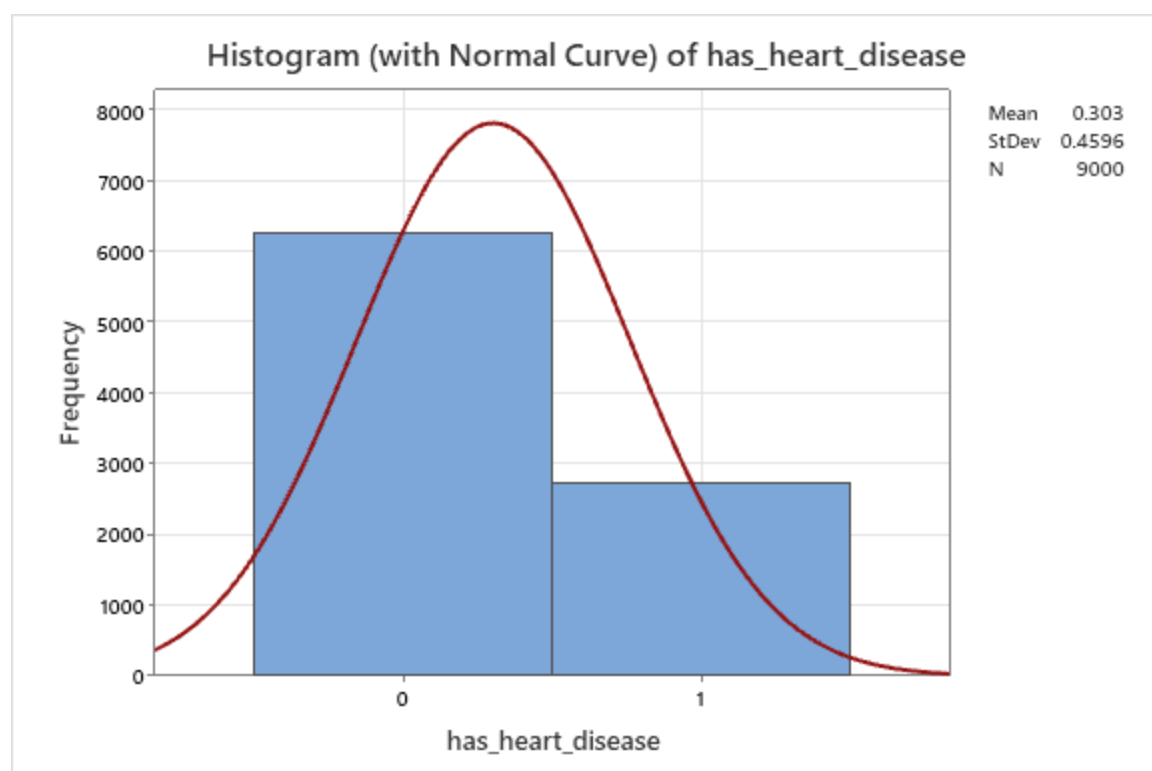
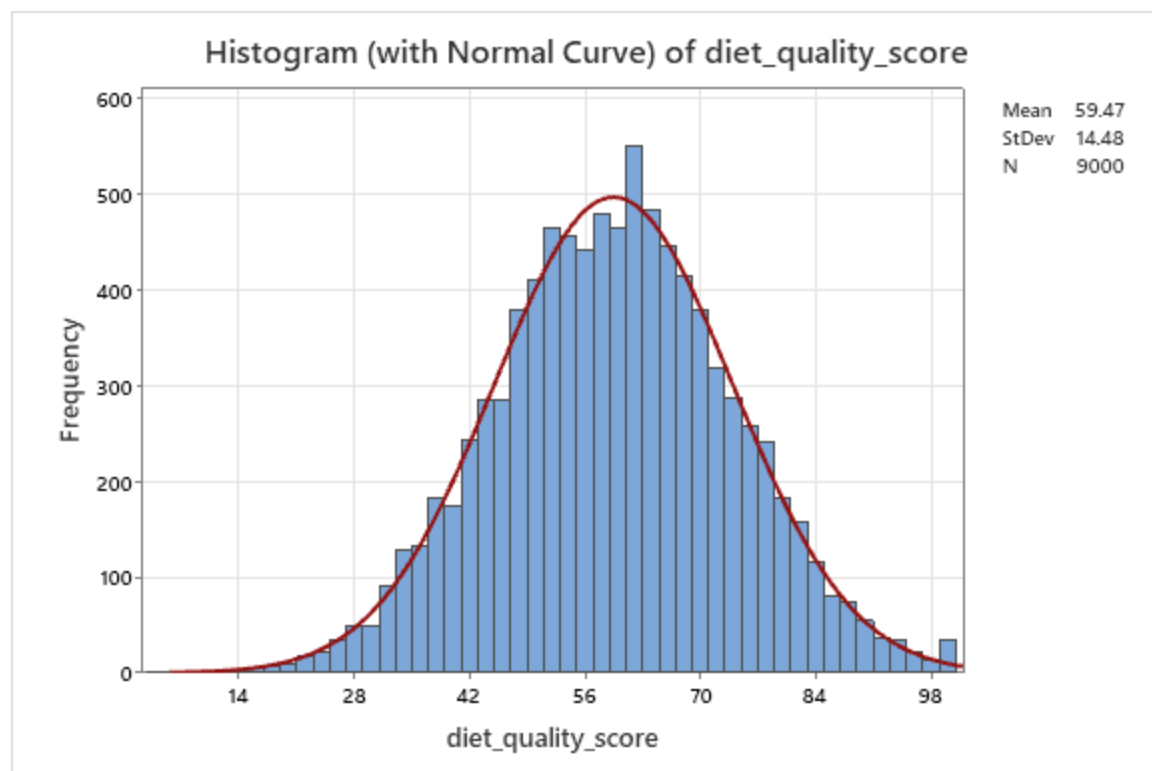


Histogram (with Normal Curve) of stress_score

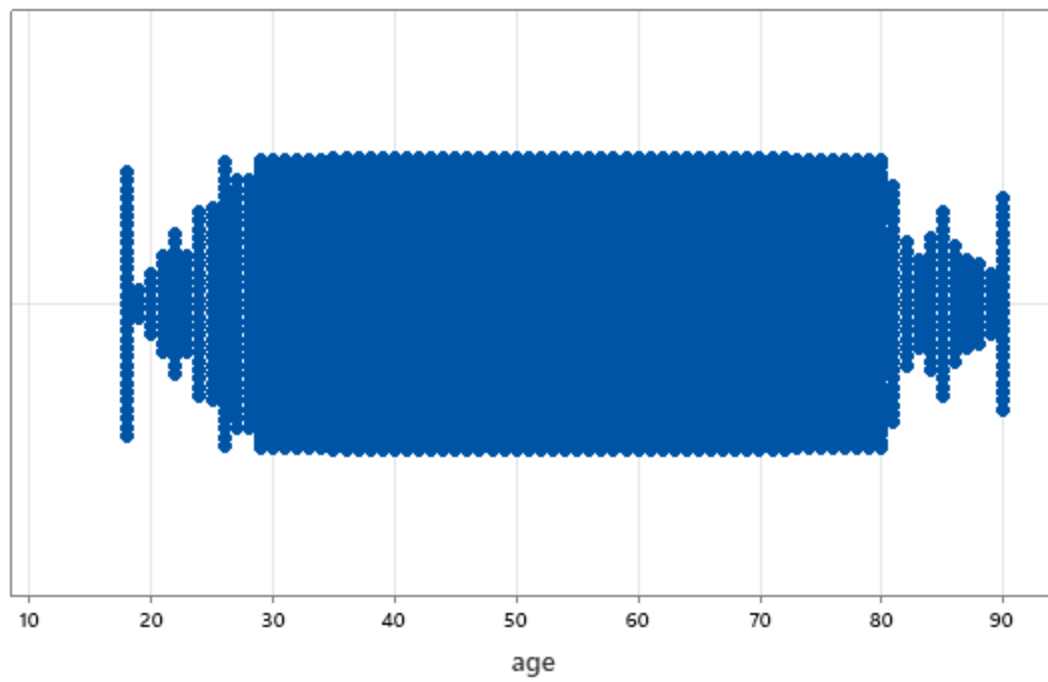


Histogram (with Normal Curve) of daily_steps

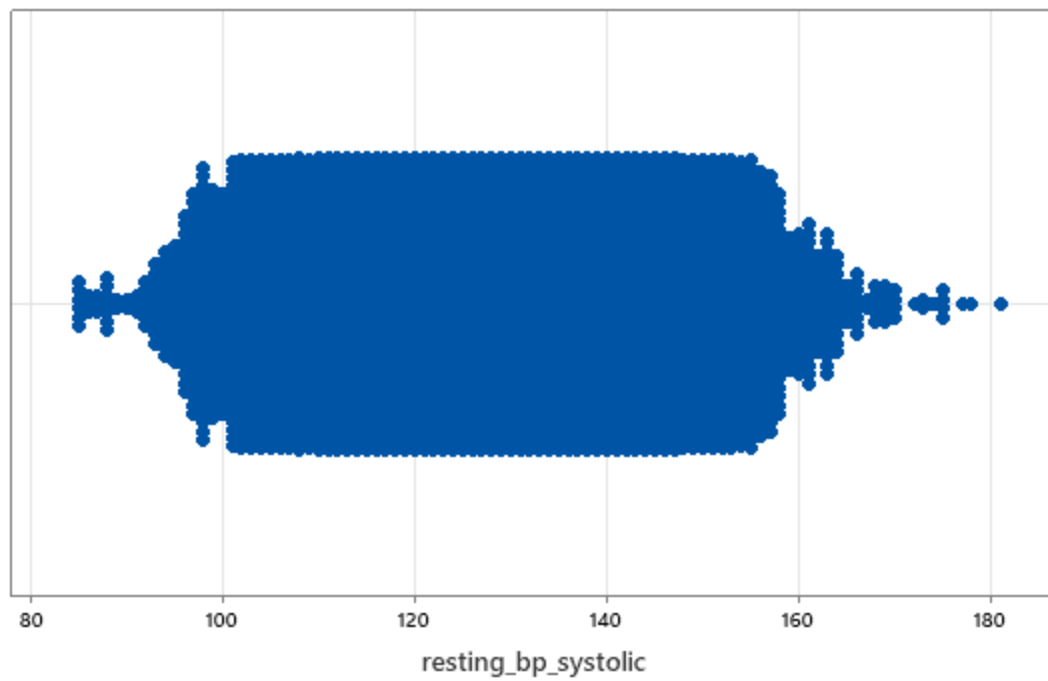




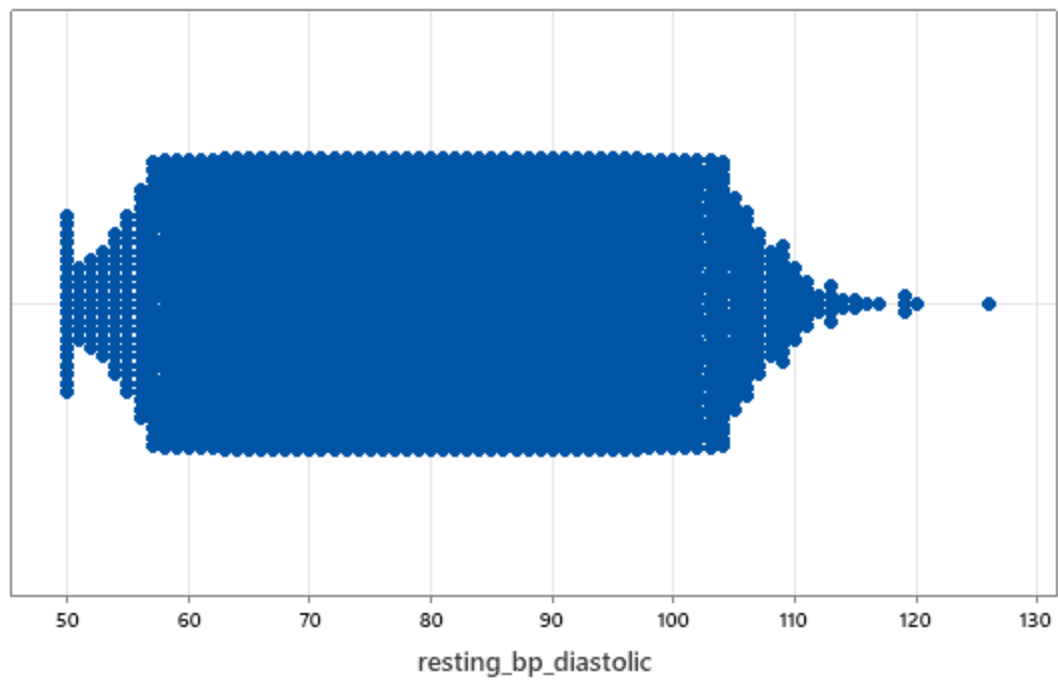
Individual Value Plot of age



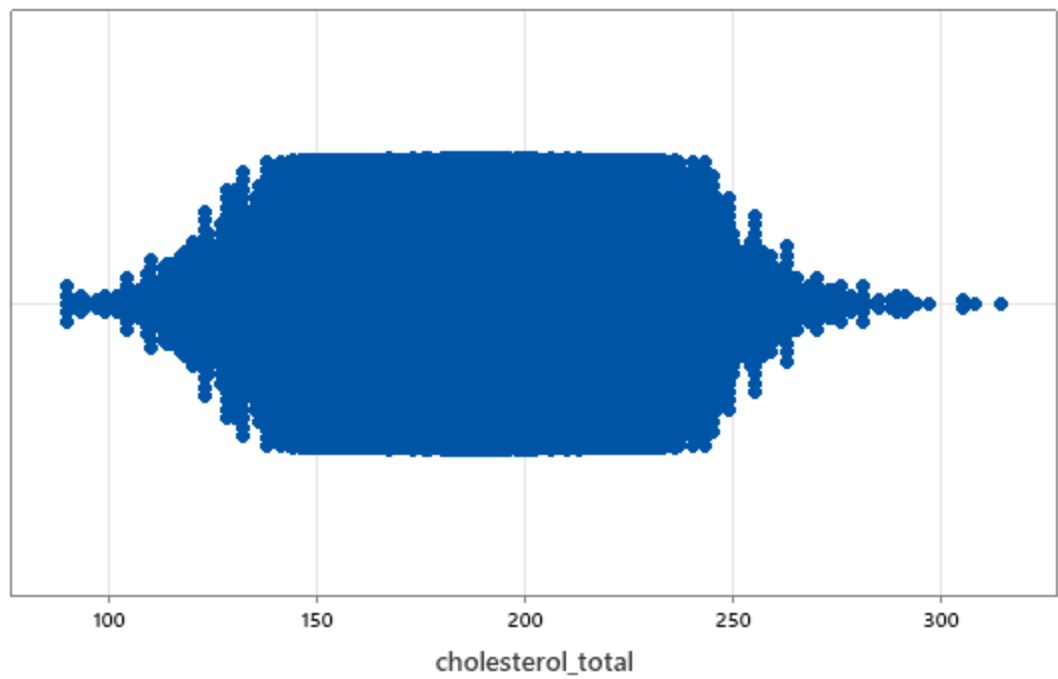
Individual Value Plot of resting_bp_systolic



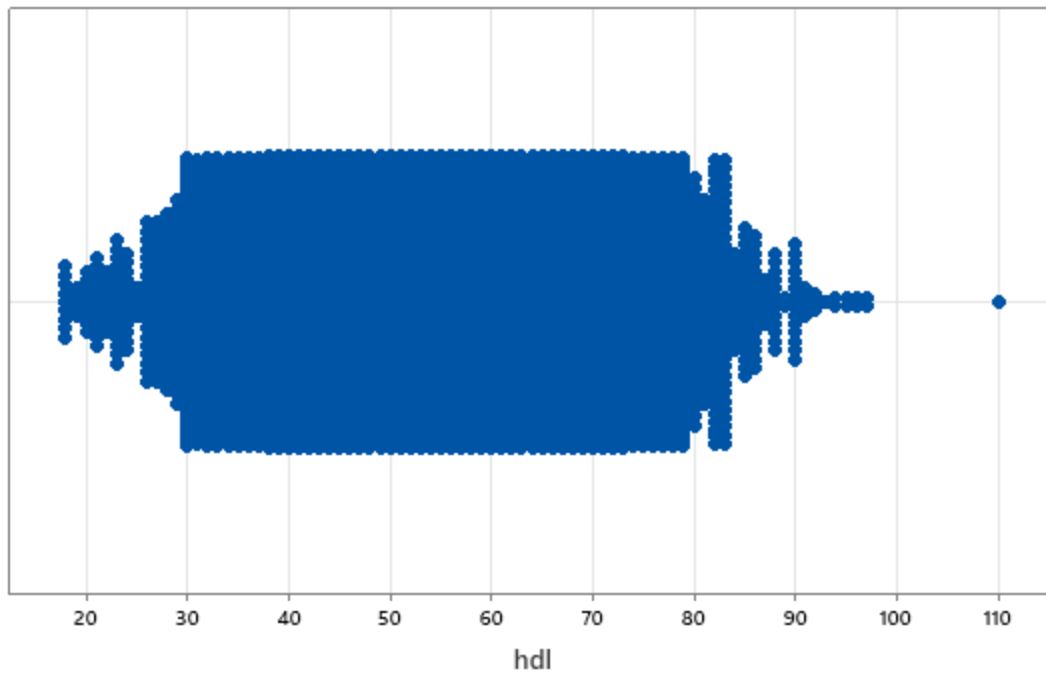
Individual Value Plot of resting_bp_diastolic



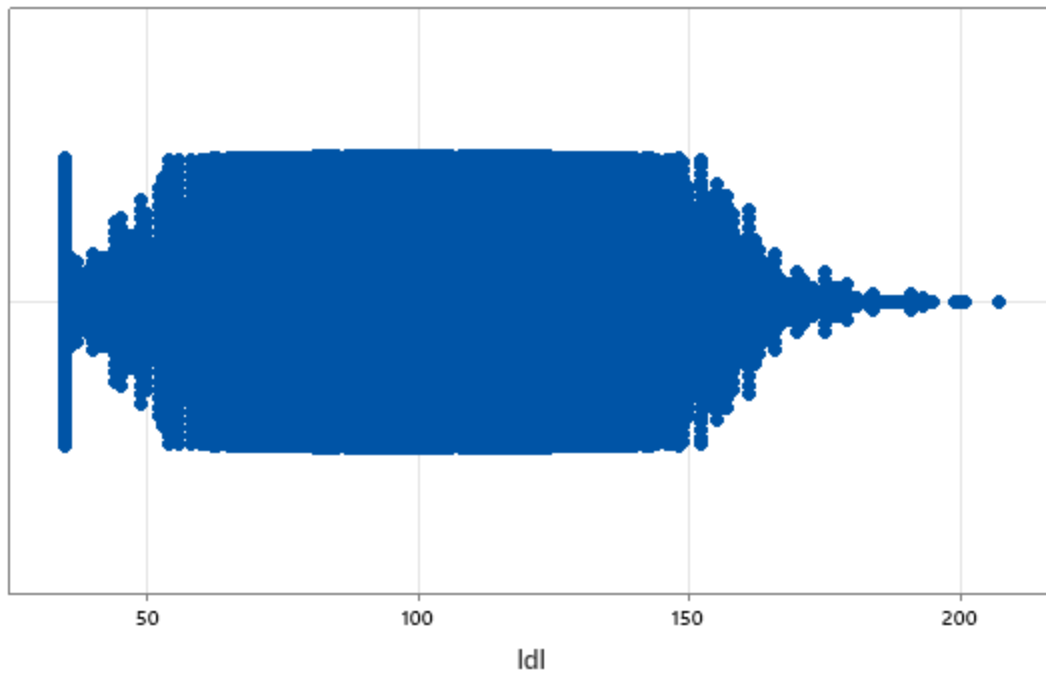
Individual Value Plot of cholesterol_total



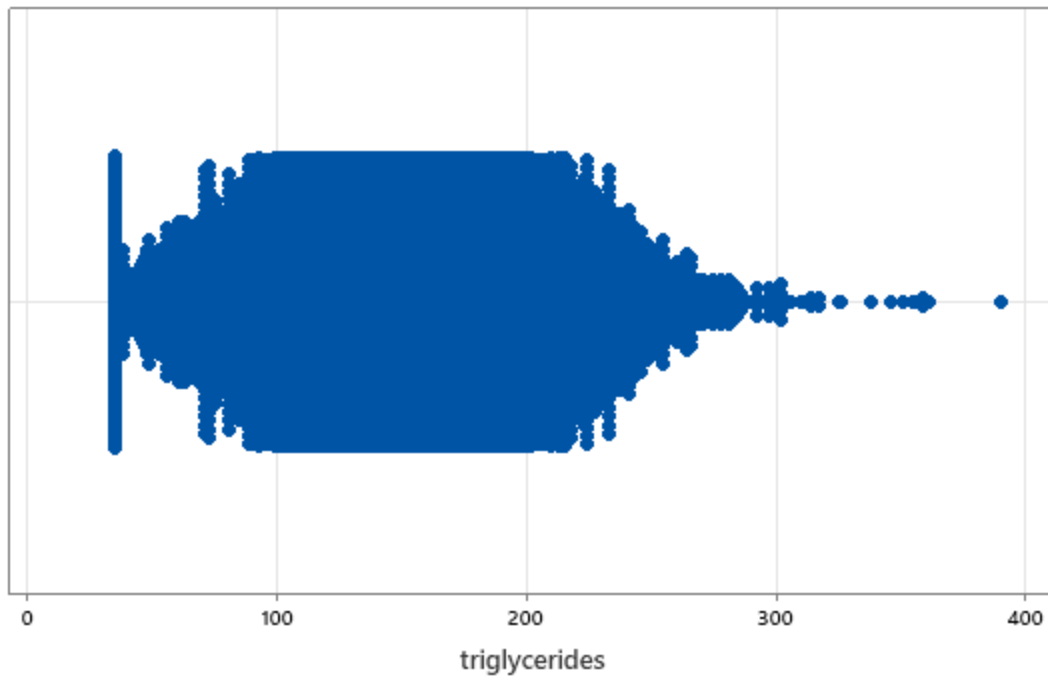
Individual Value Plot of hdl



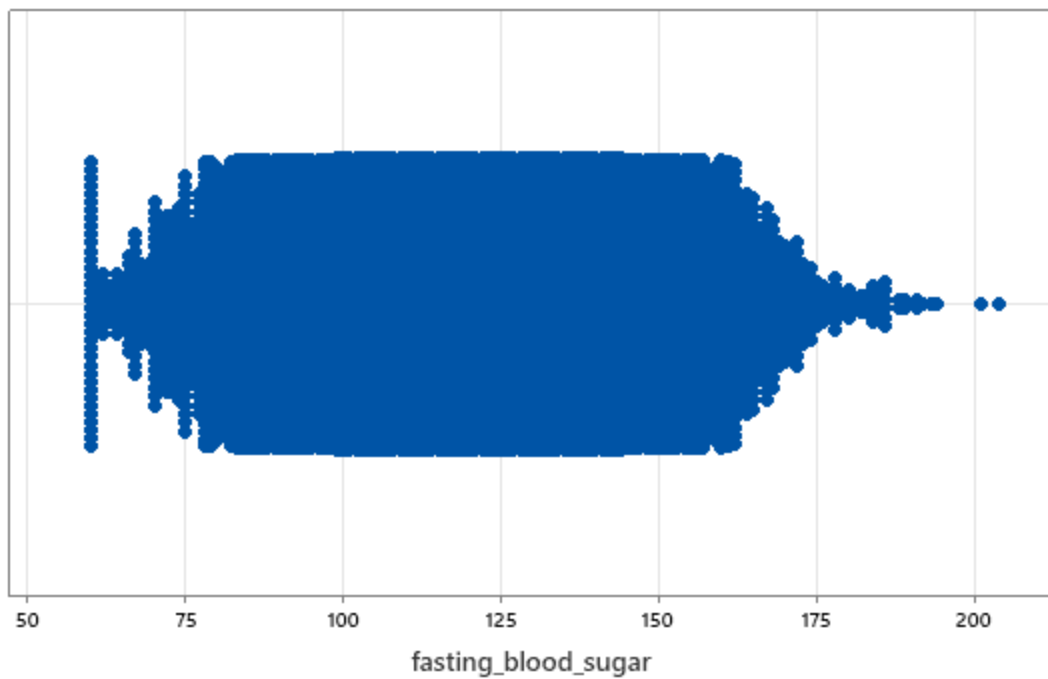
Individual Value Plot of ldl



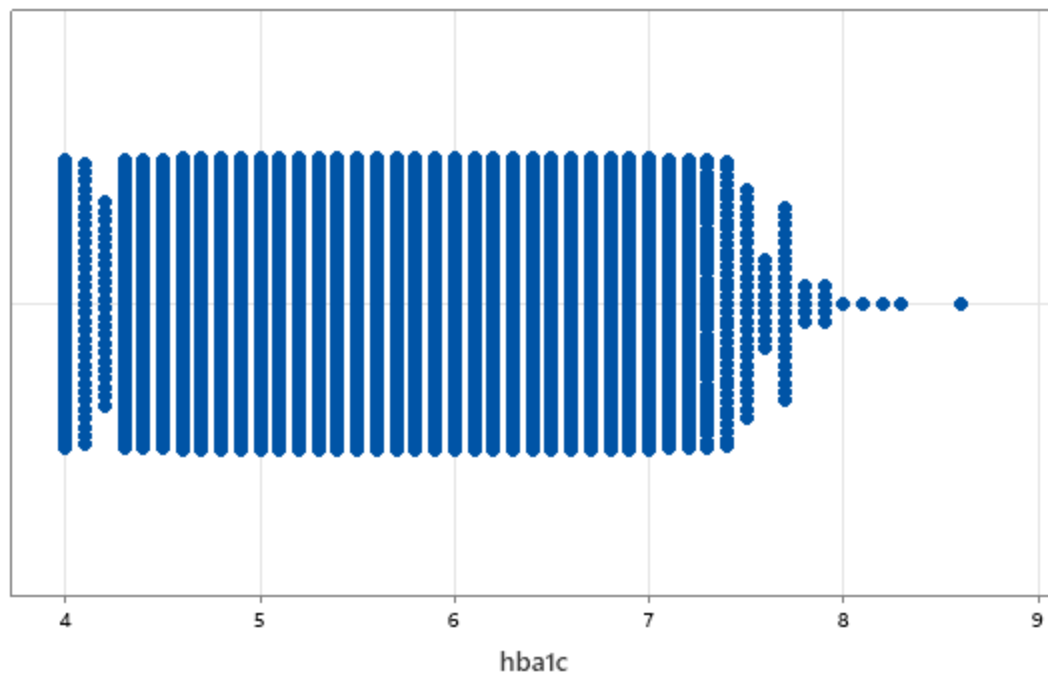
Individual Value Plot of triglycerides



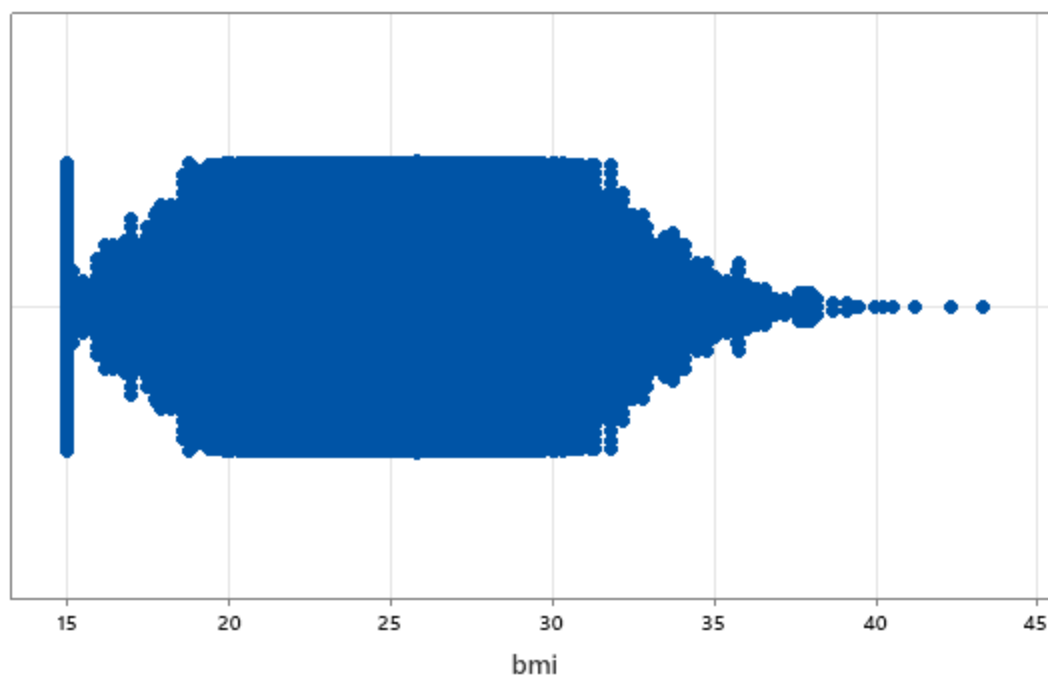
Individual Value Plot of fasting_blood_sugar



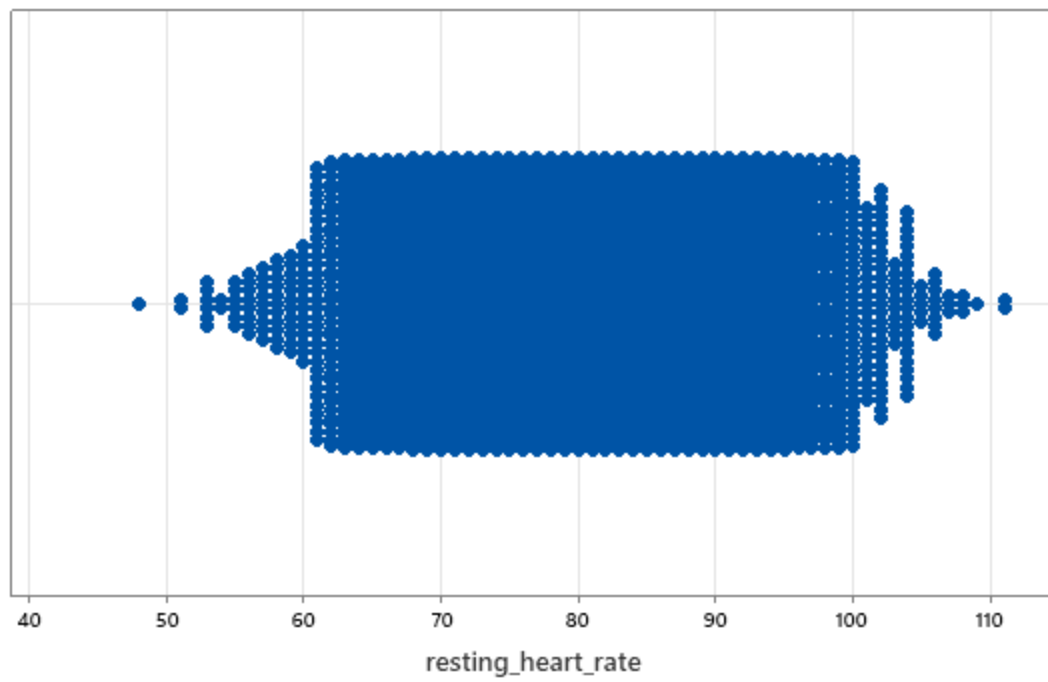
Individual Value Plot of hba1c



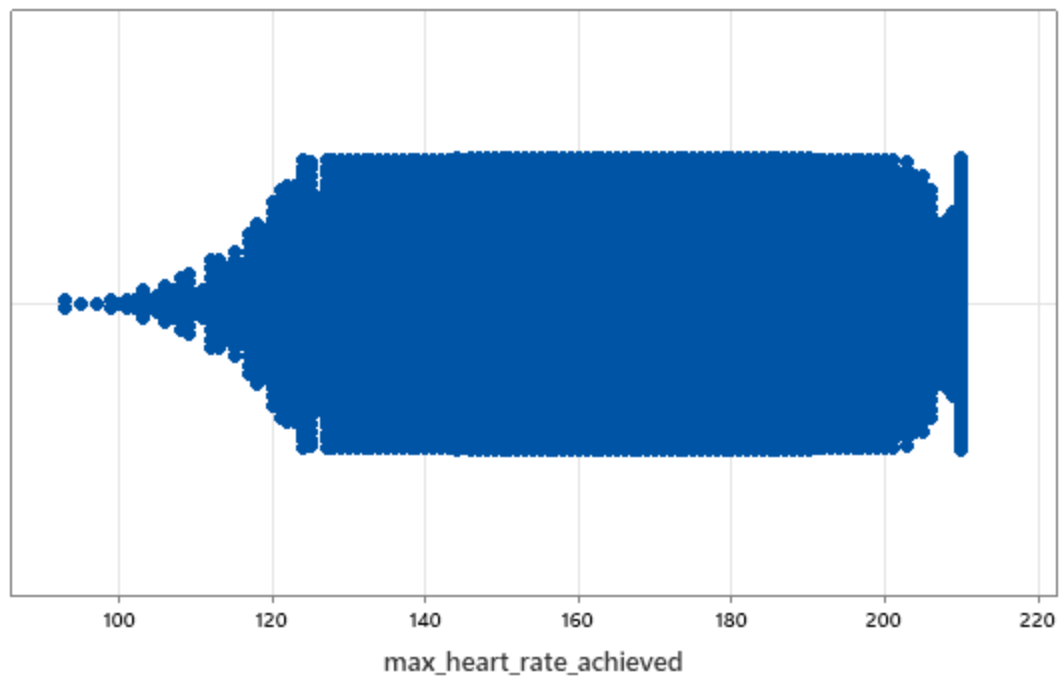
Individual Value Plot of bmi



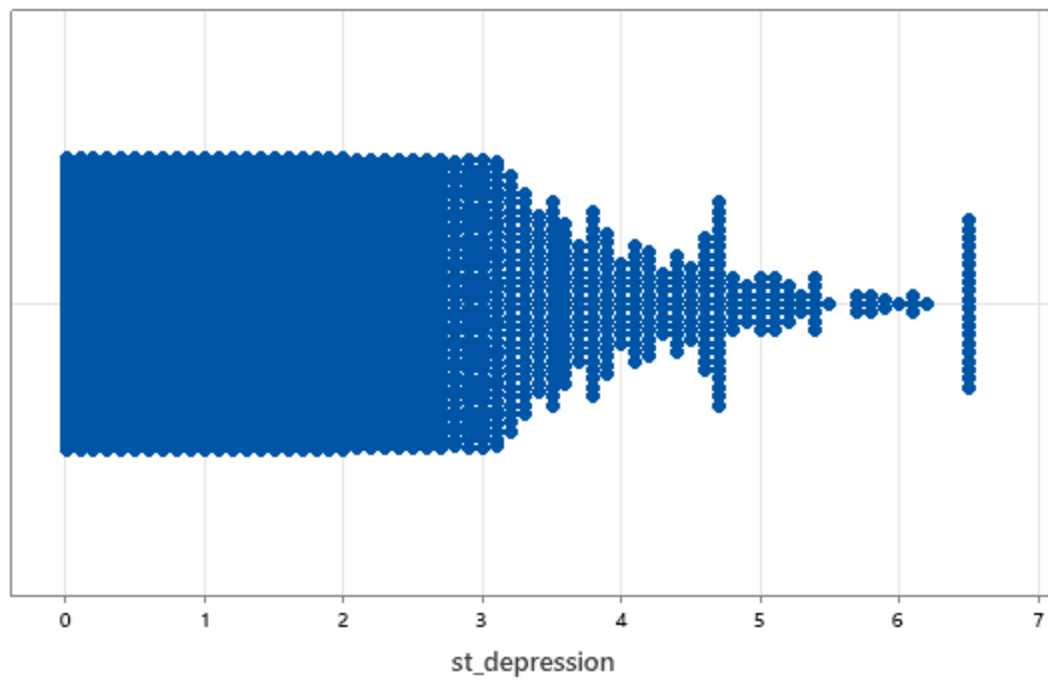
Individual Value Plot of resting_heart_rate



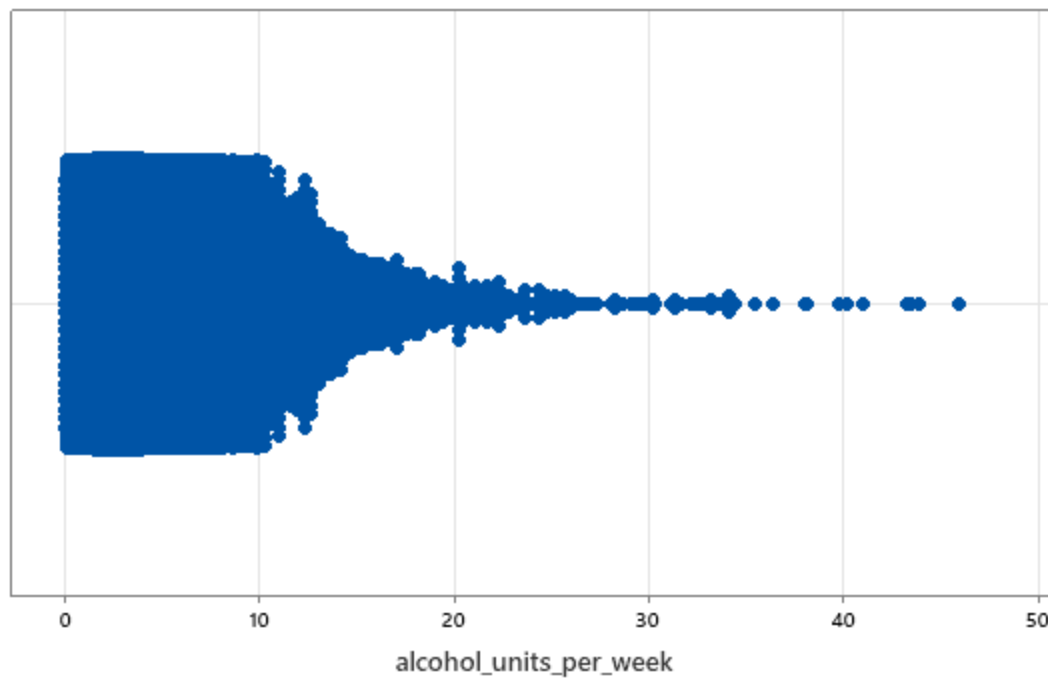
Individual Value Plot of max_heart_rate_achieved



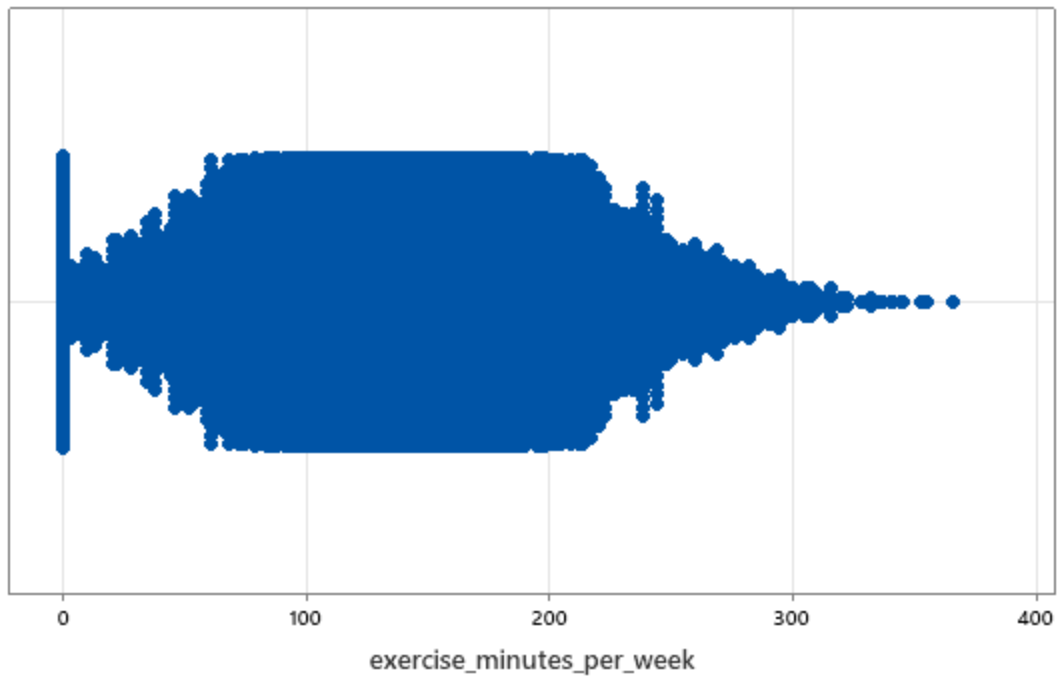
Individual Value Plot of st_depression



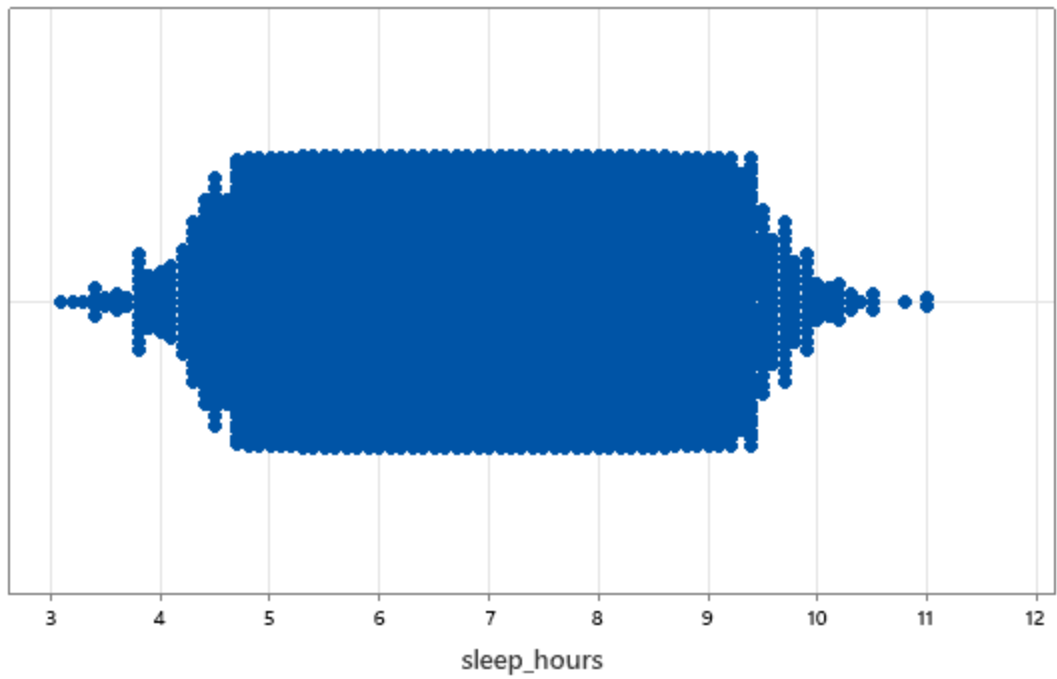
Individual Value Plot of alcohol_units_per_week



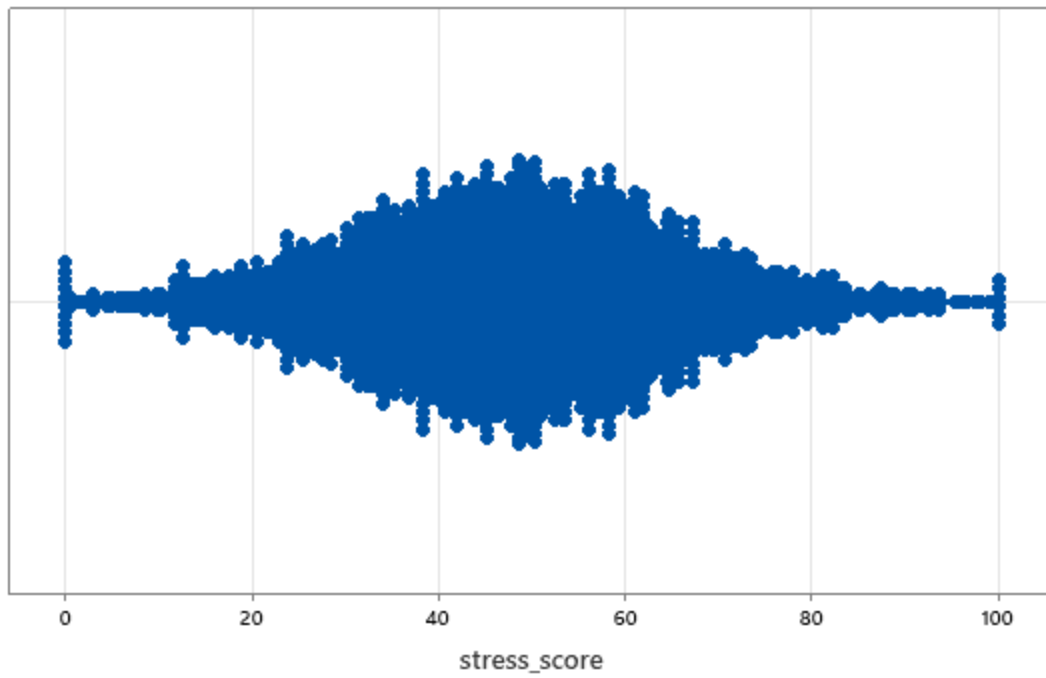
Individual Value Plot of exercise_minutes_per_week



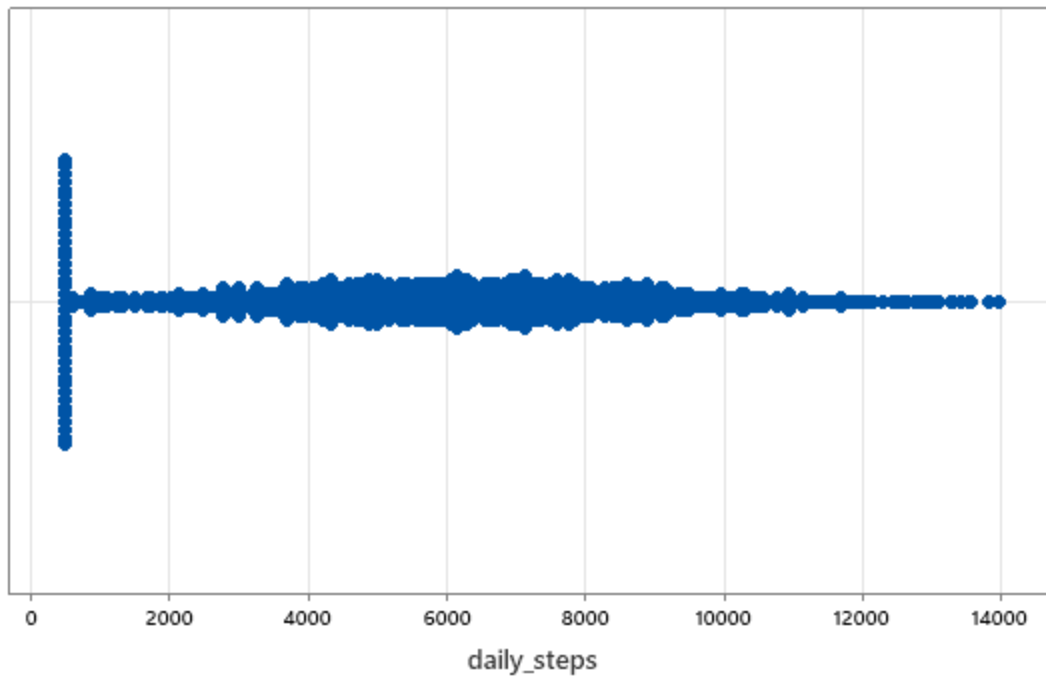
Individual Value Plot of sleep_hours



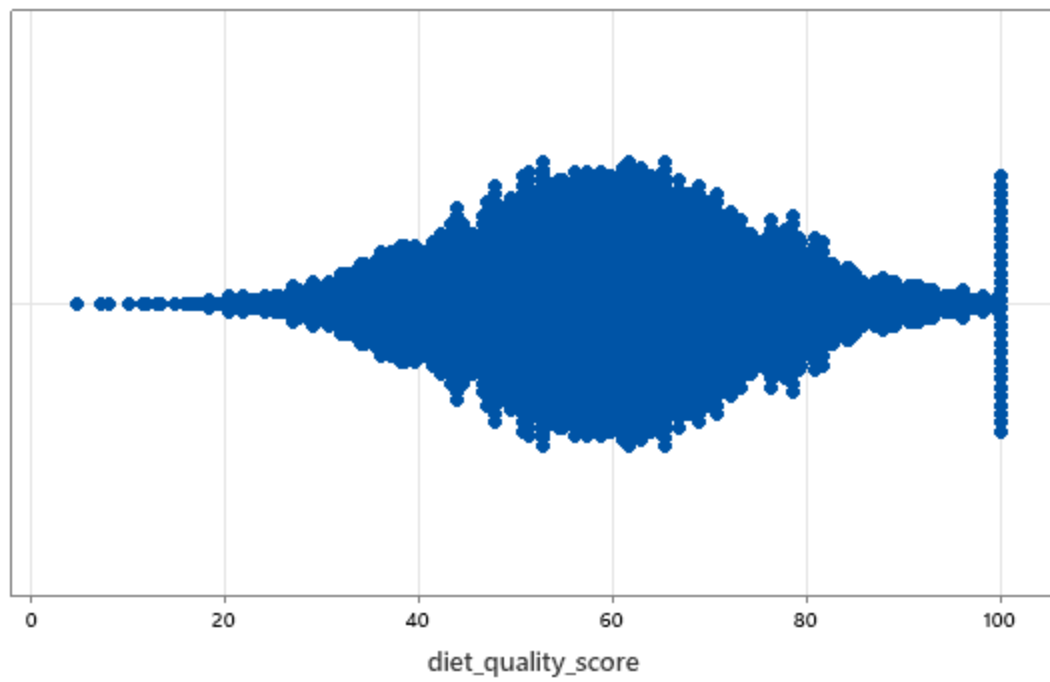
Individual Value Plot of stress_score



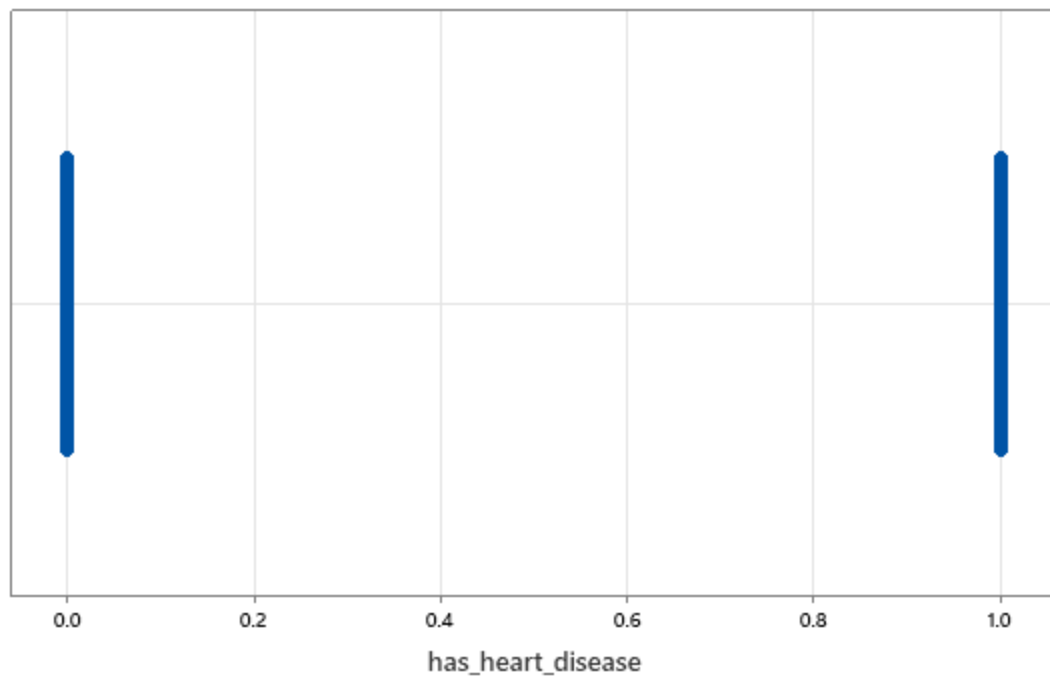
Individual Value Plot of daily_steps

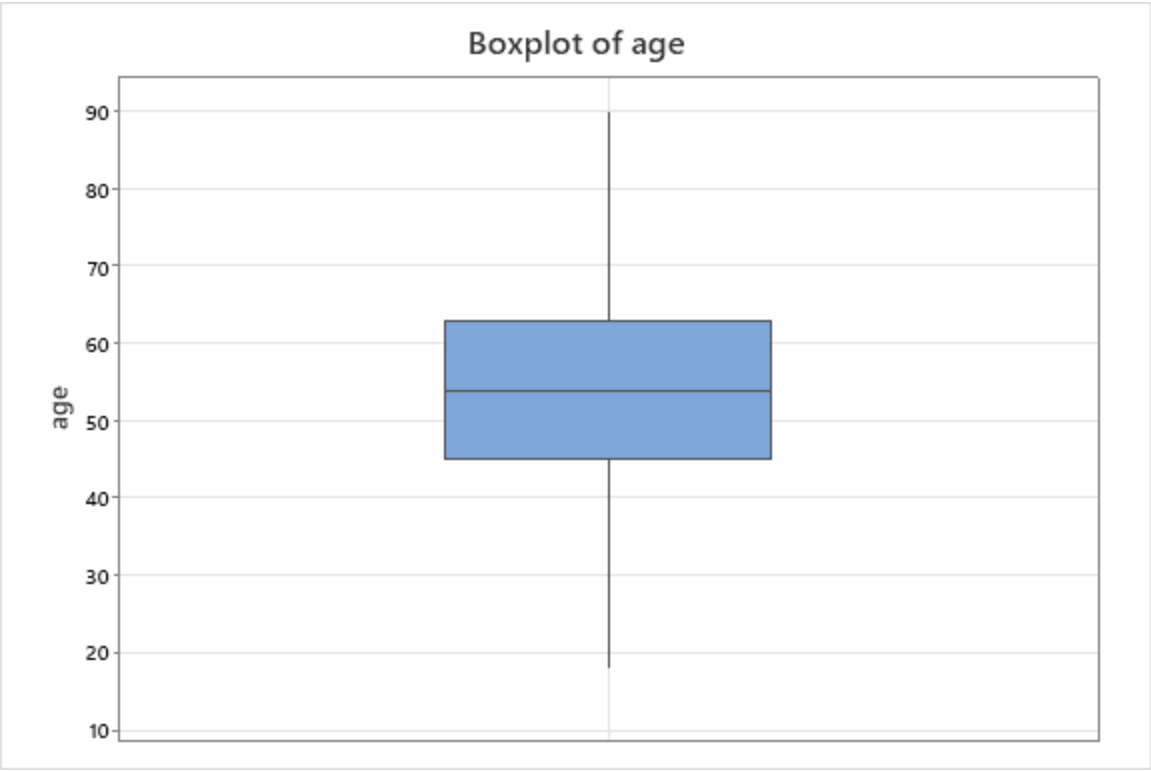


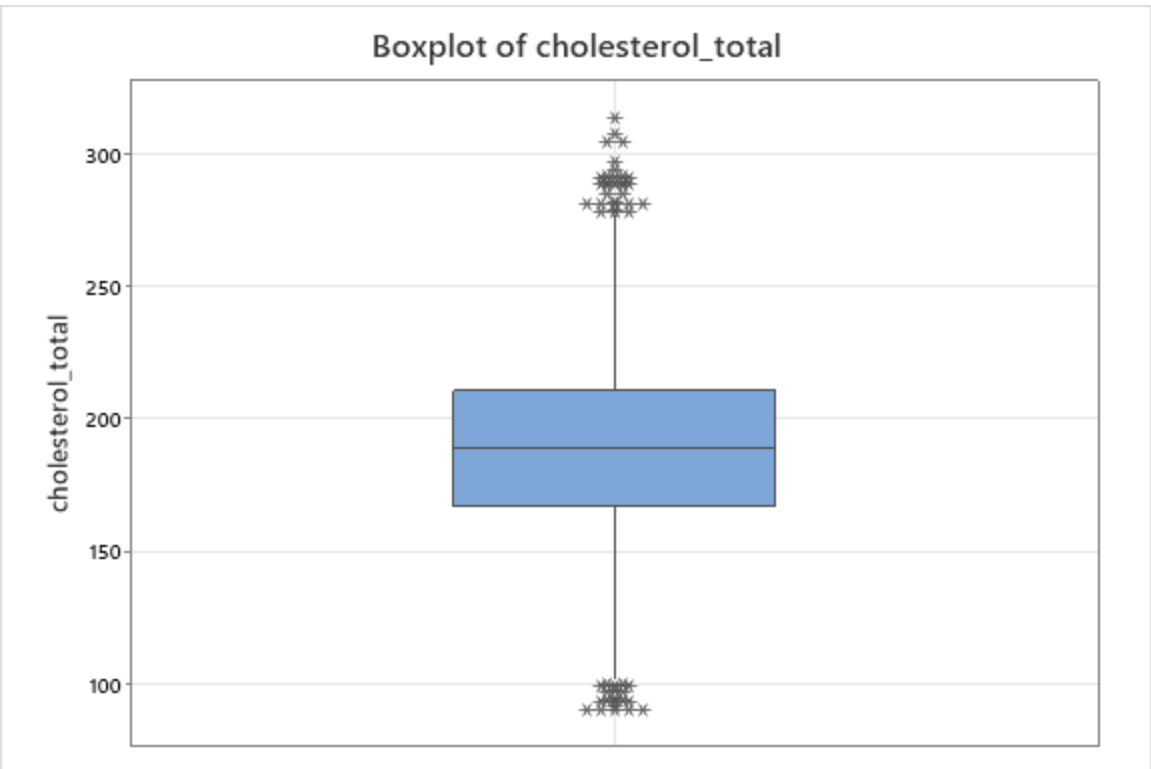
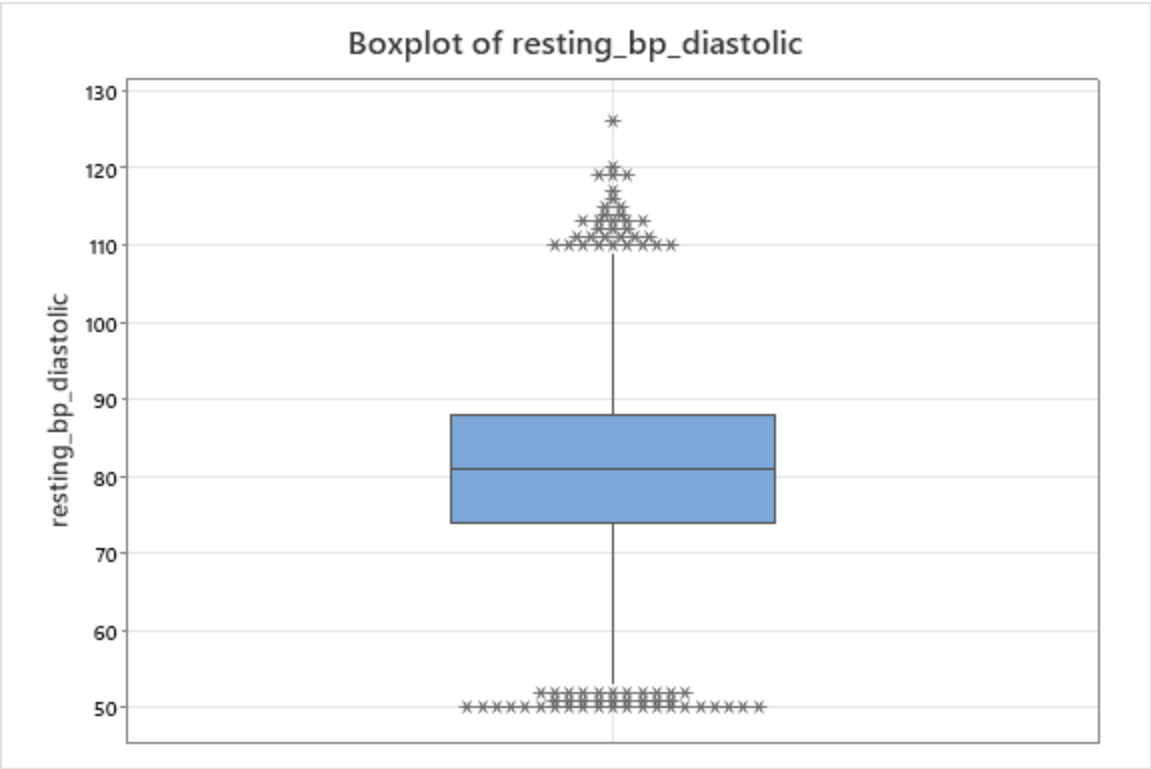
Individual Value Plot of diet_quality_score

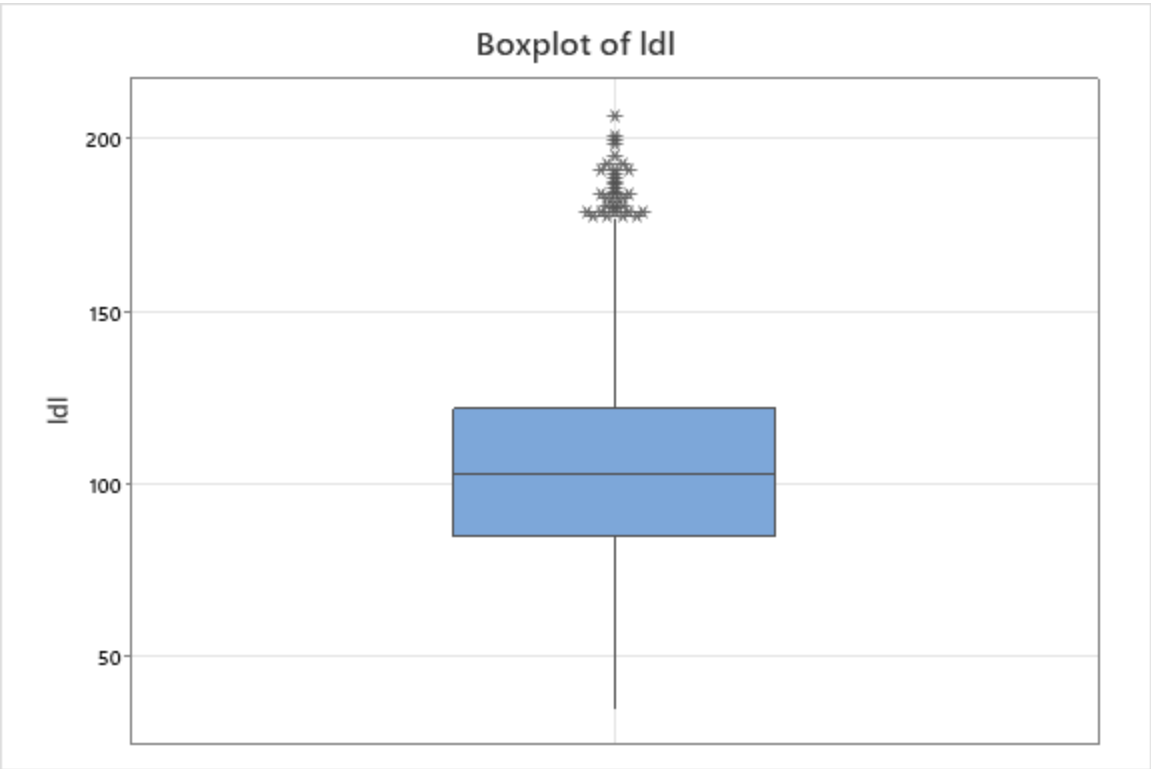
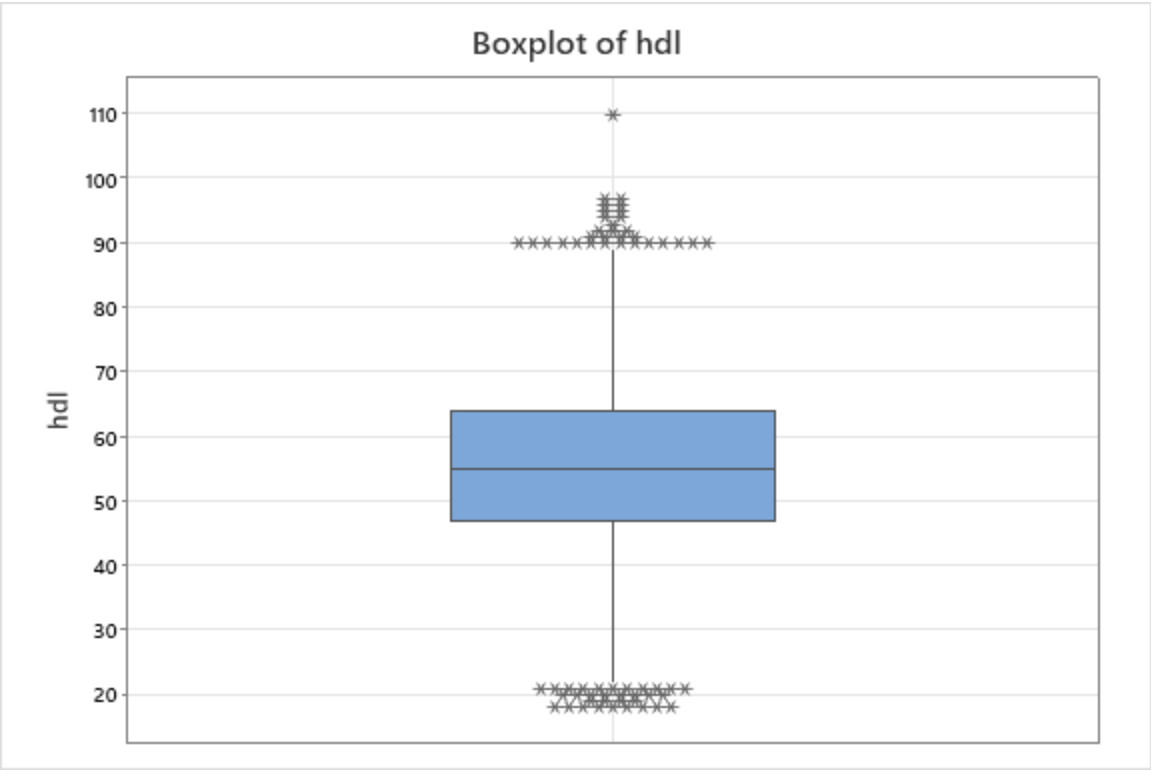


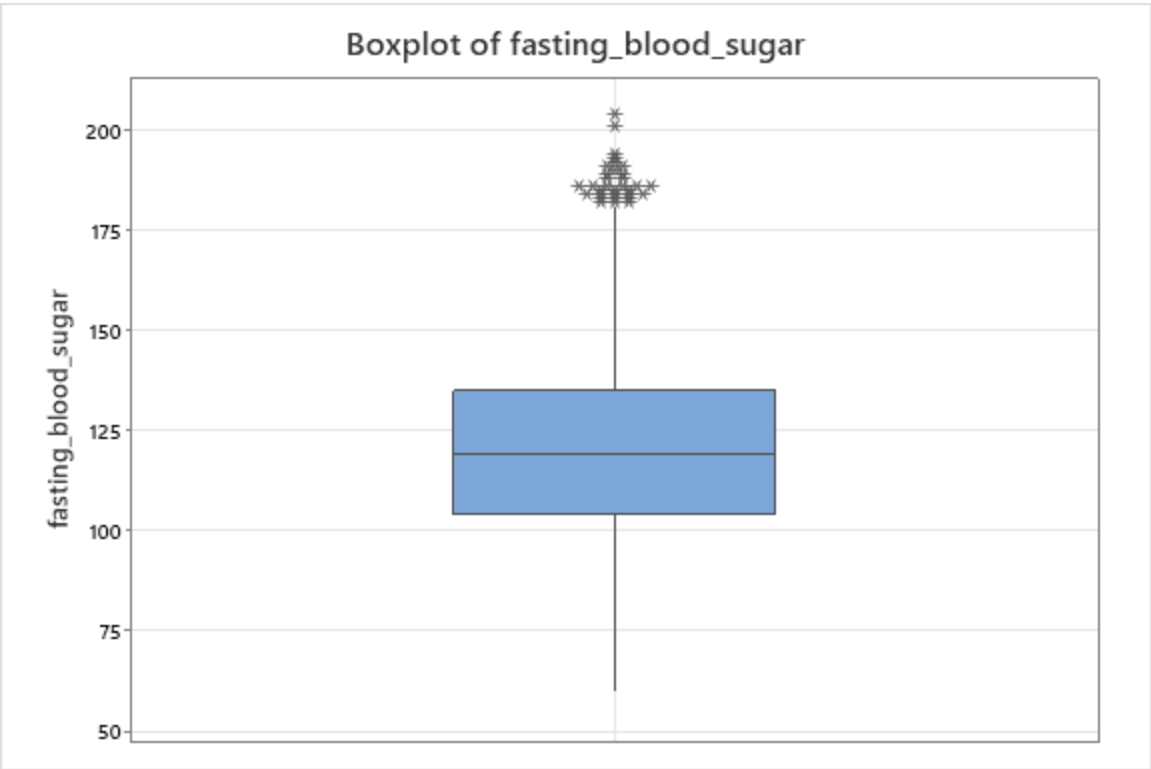
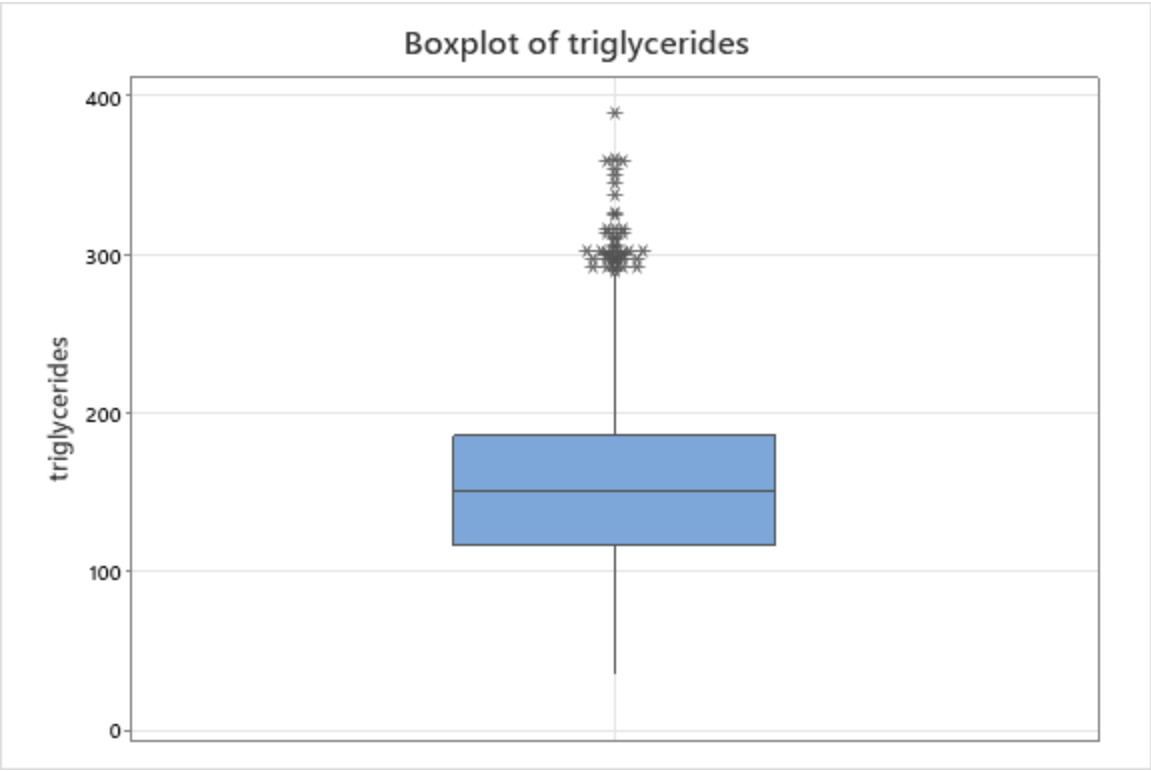
Individual Value Plot of has_heart_disease

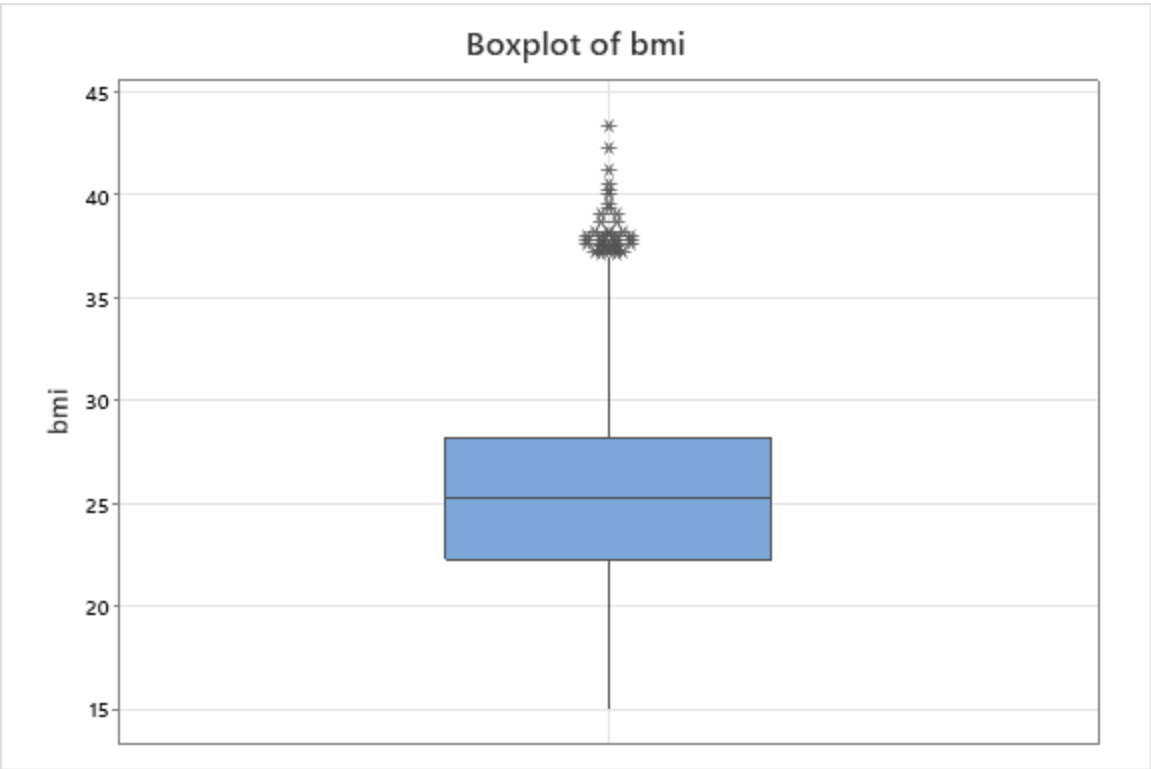
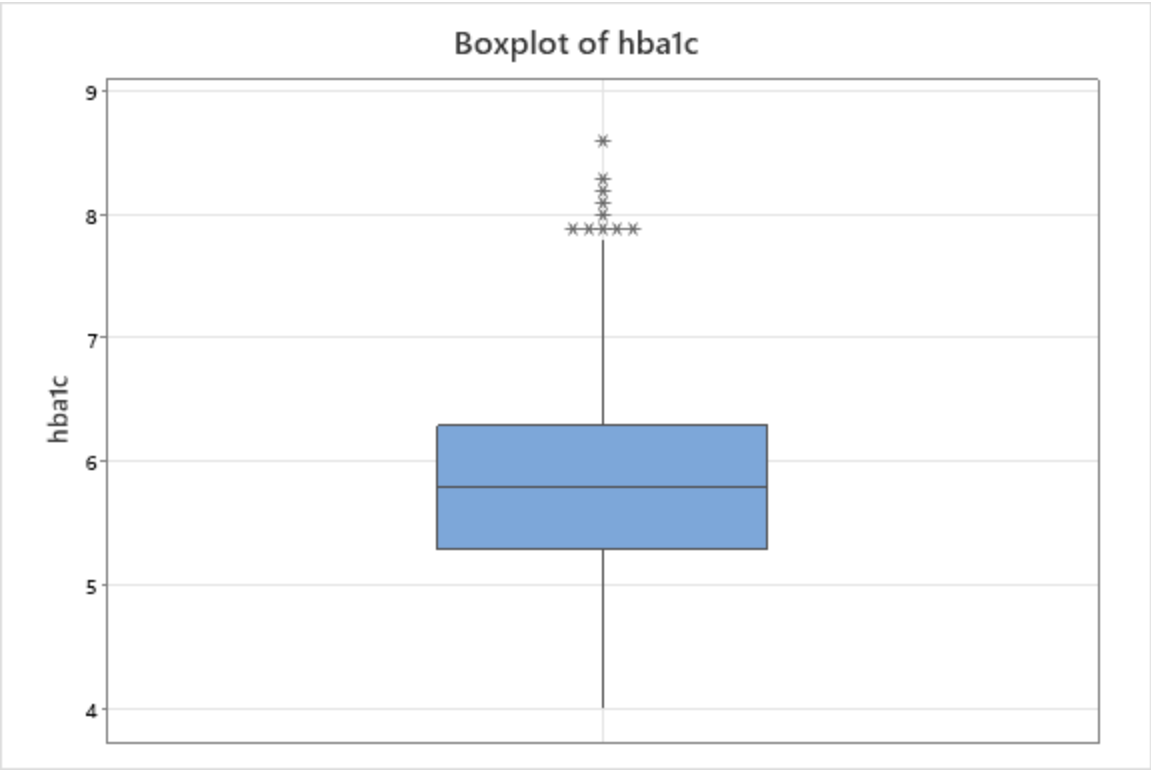


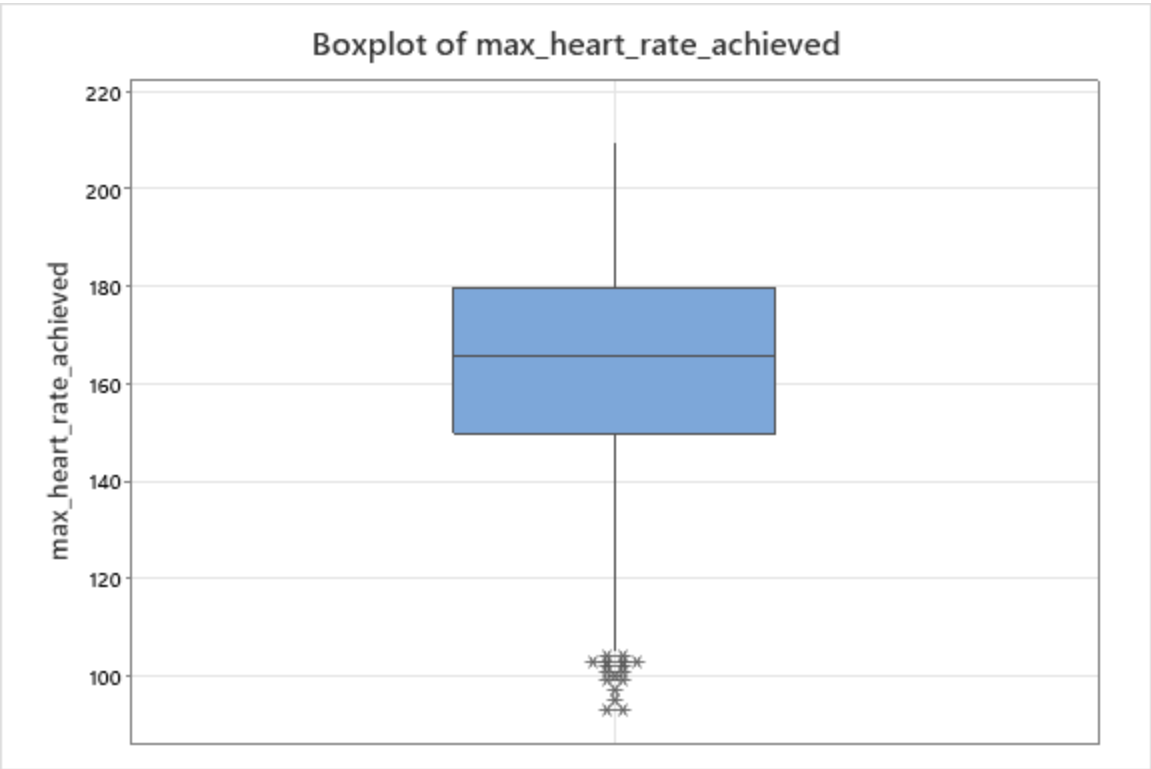
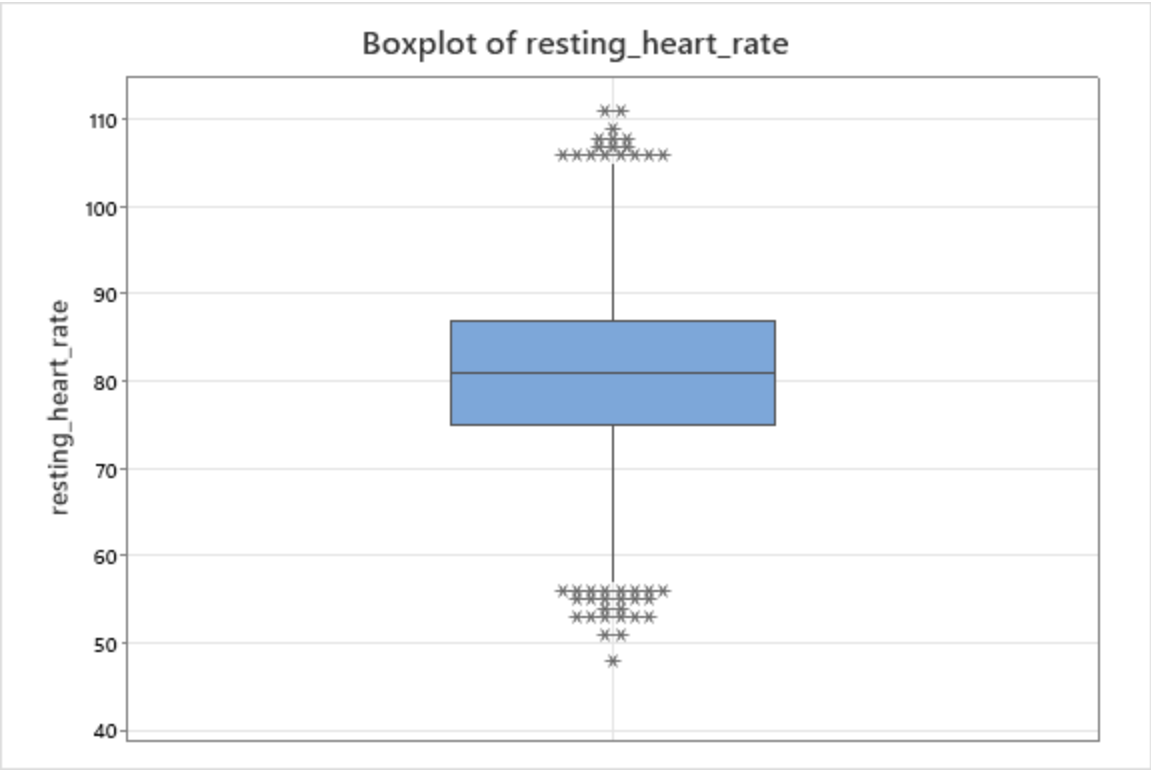


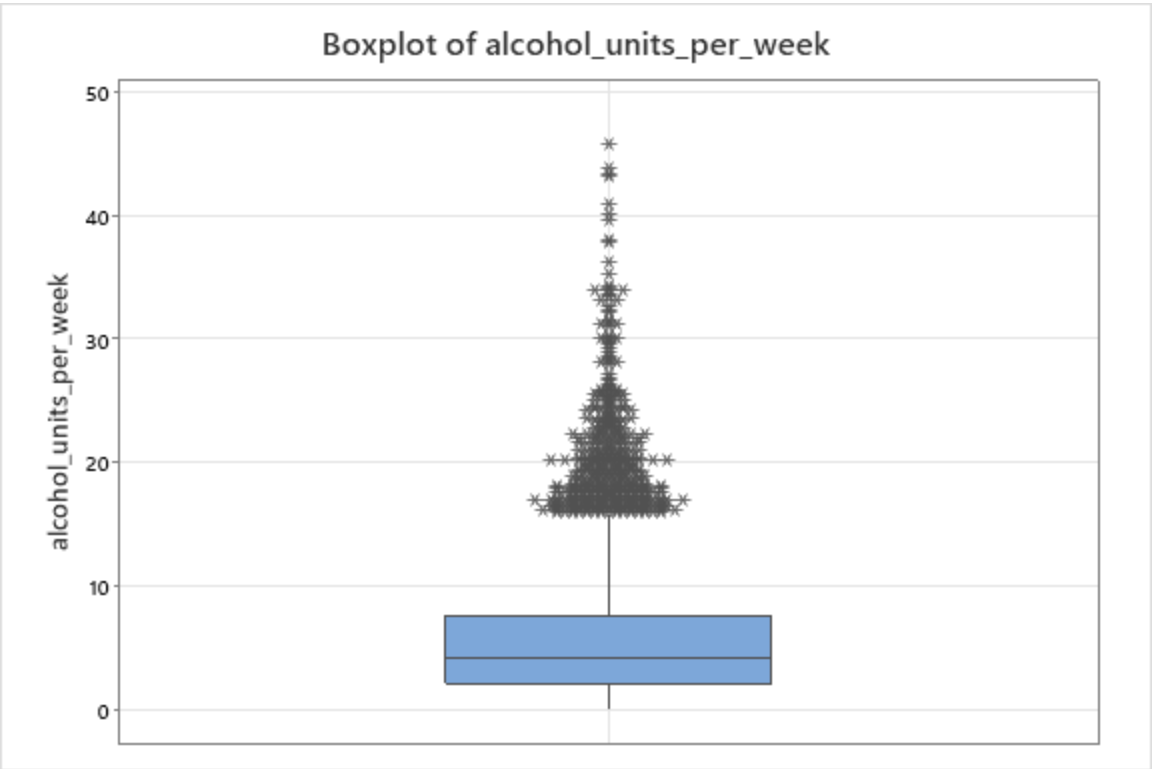
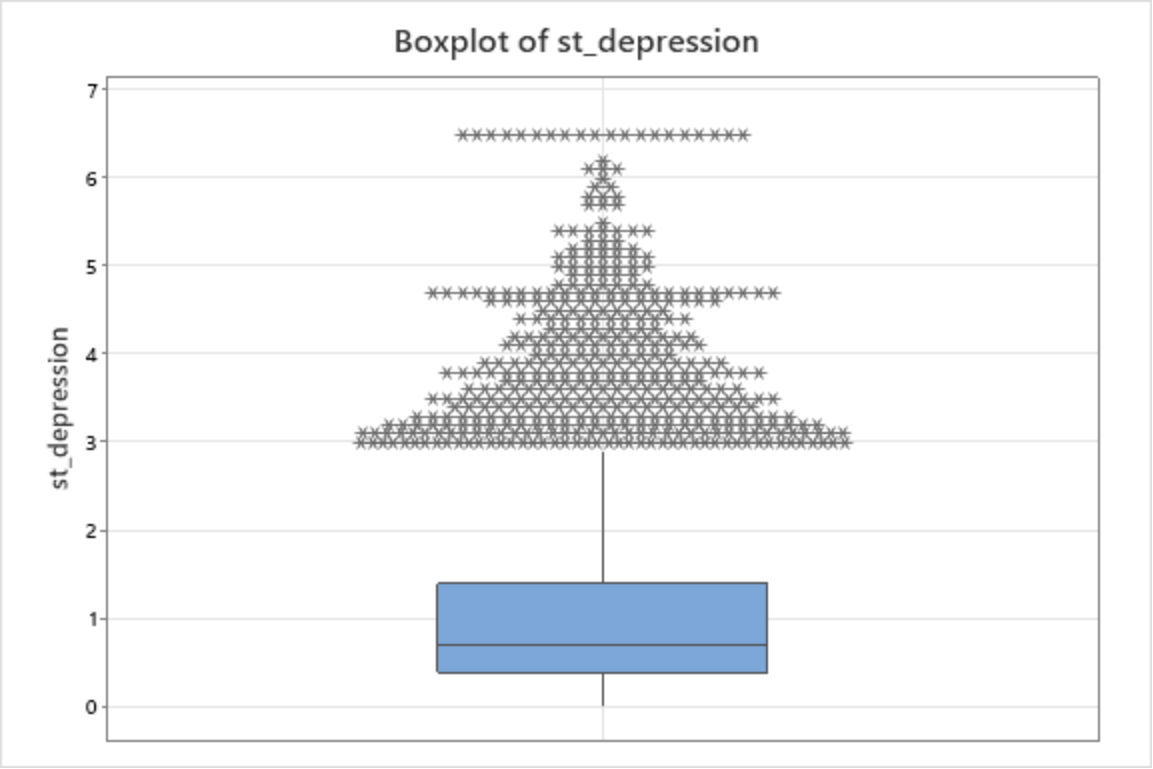


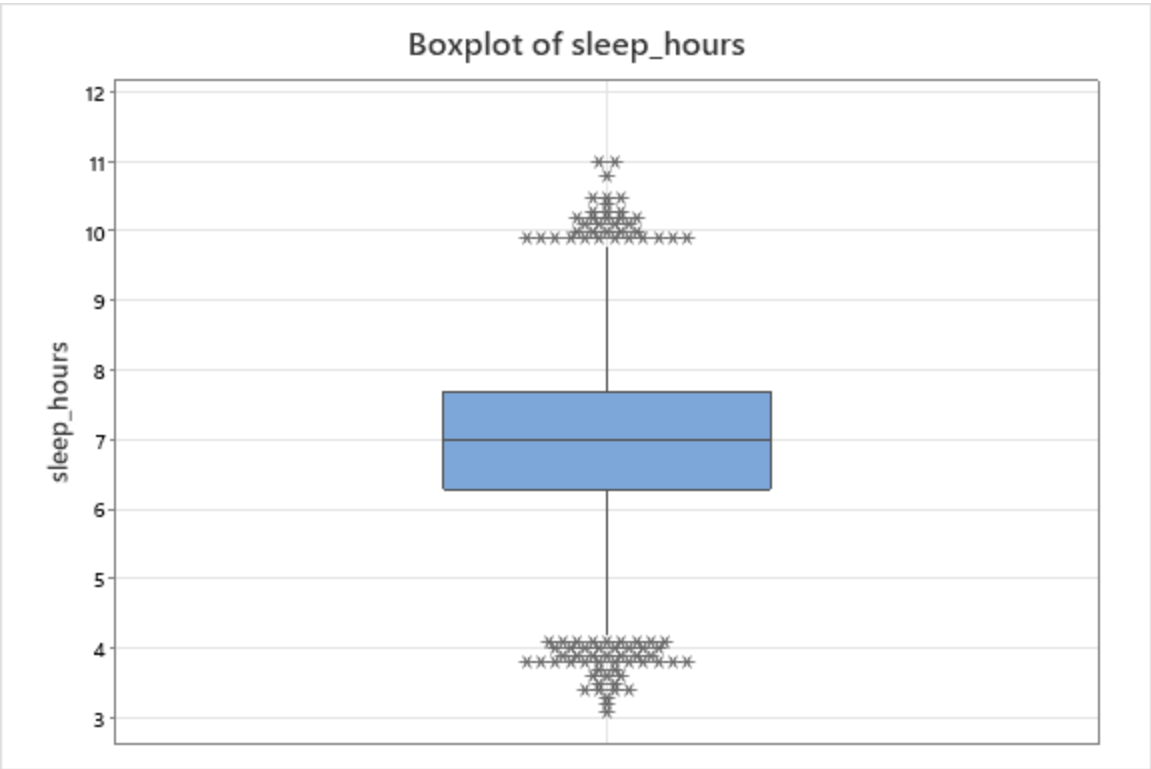
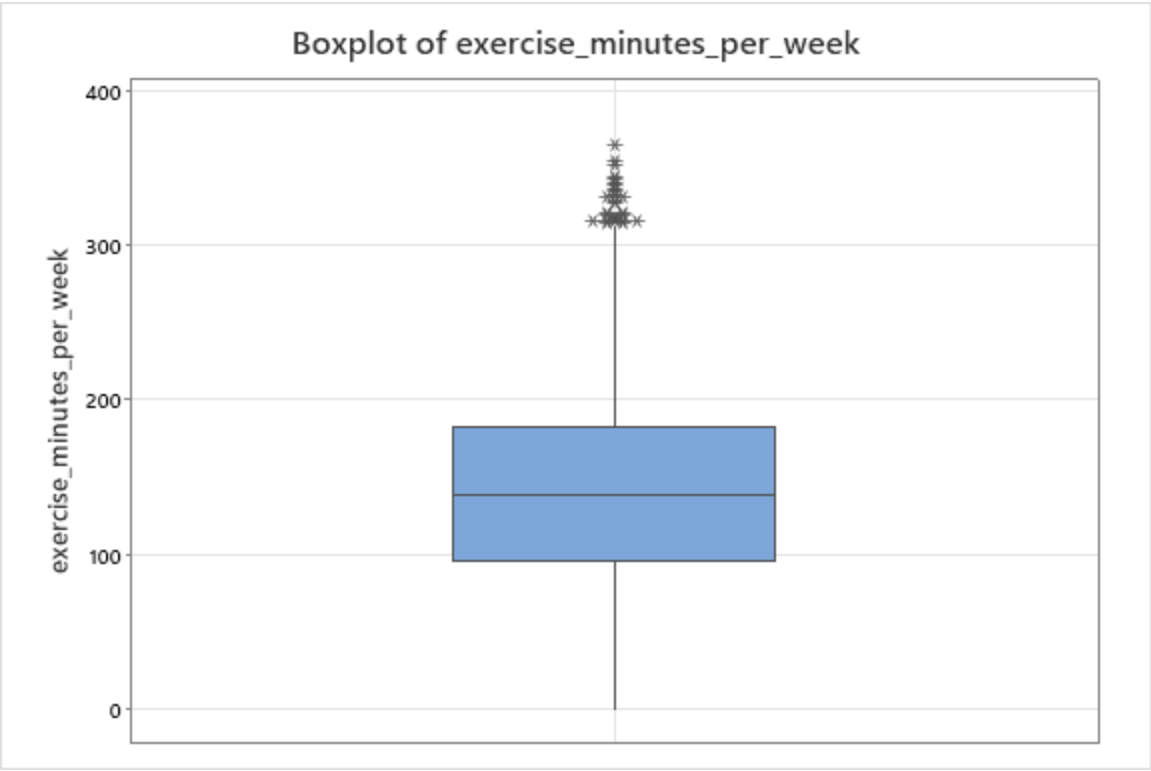


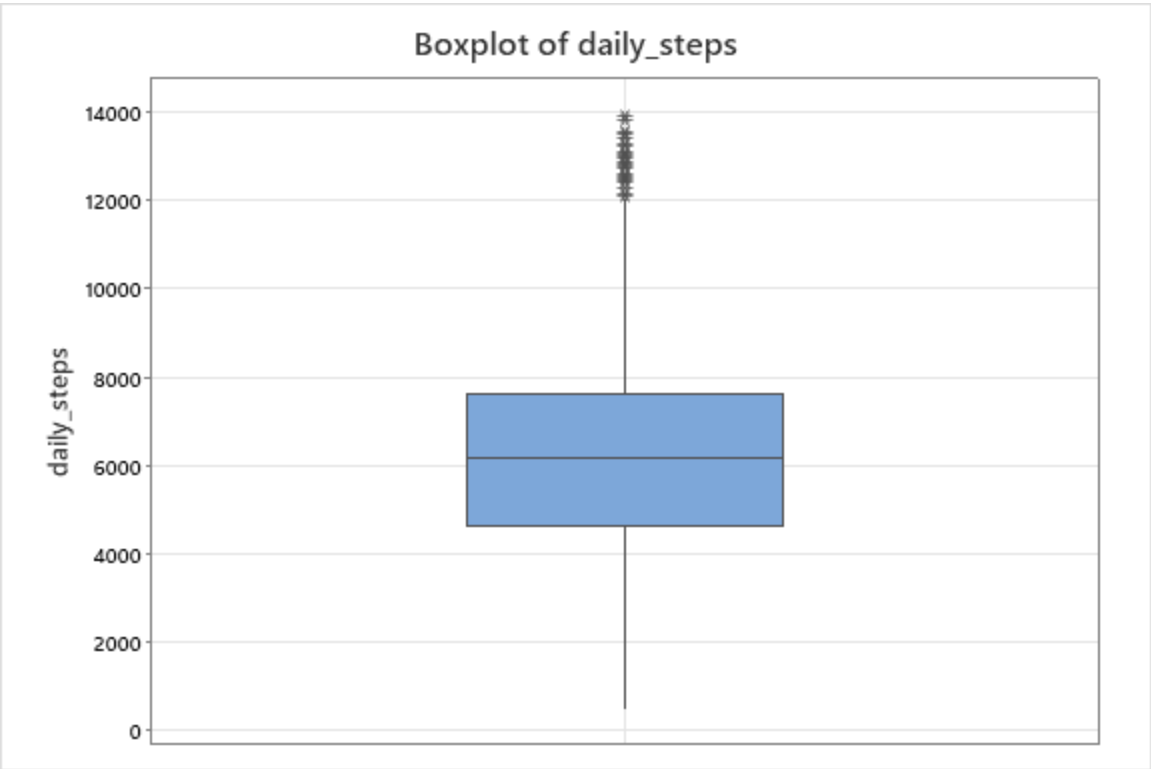
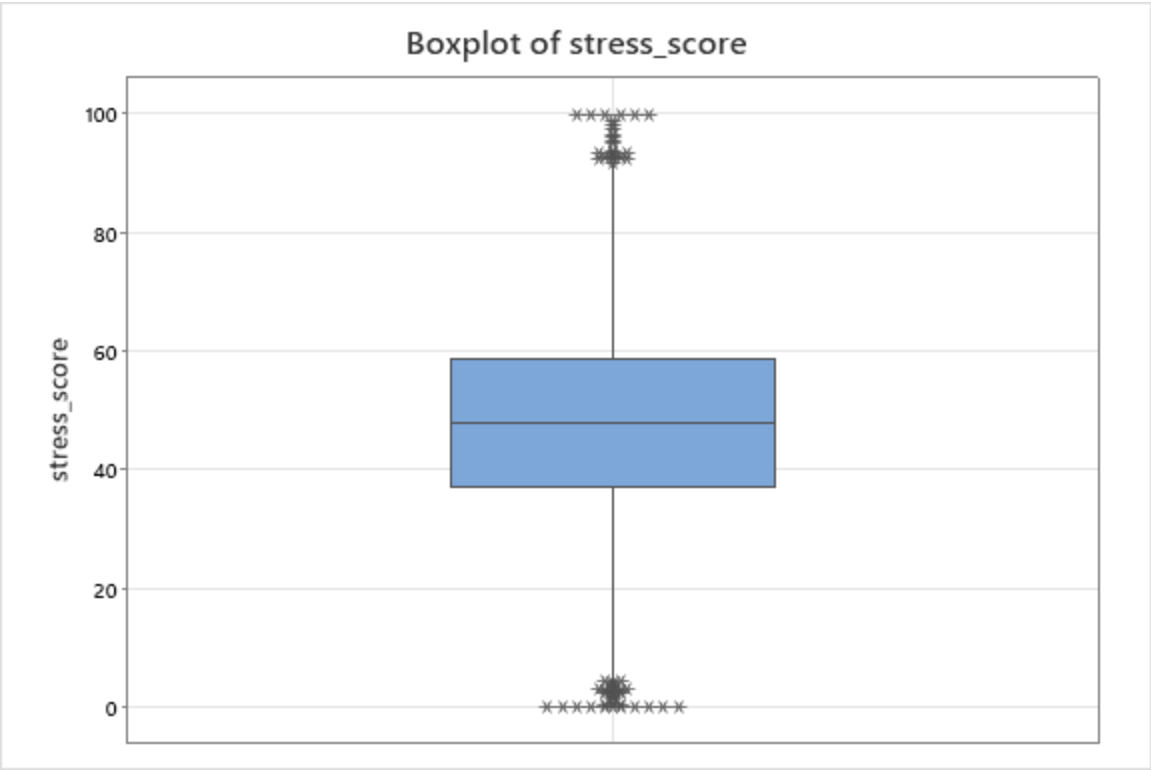


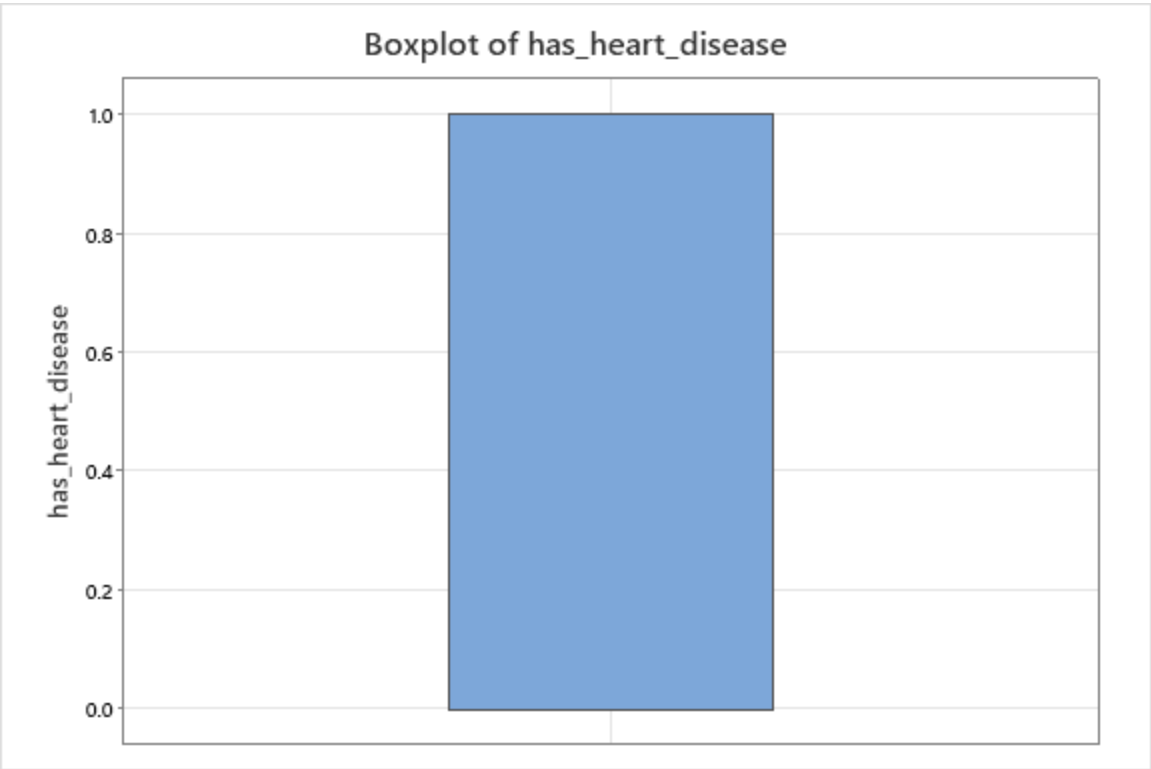
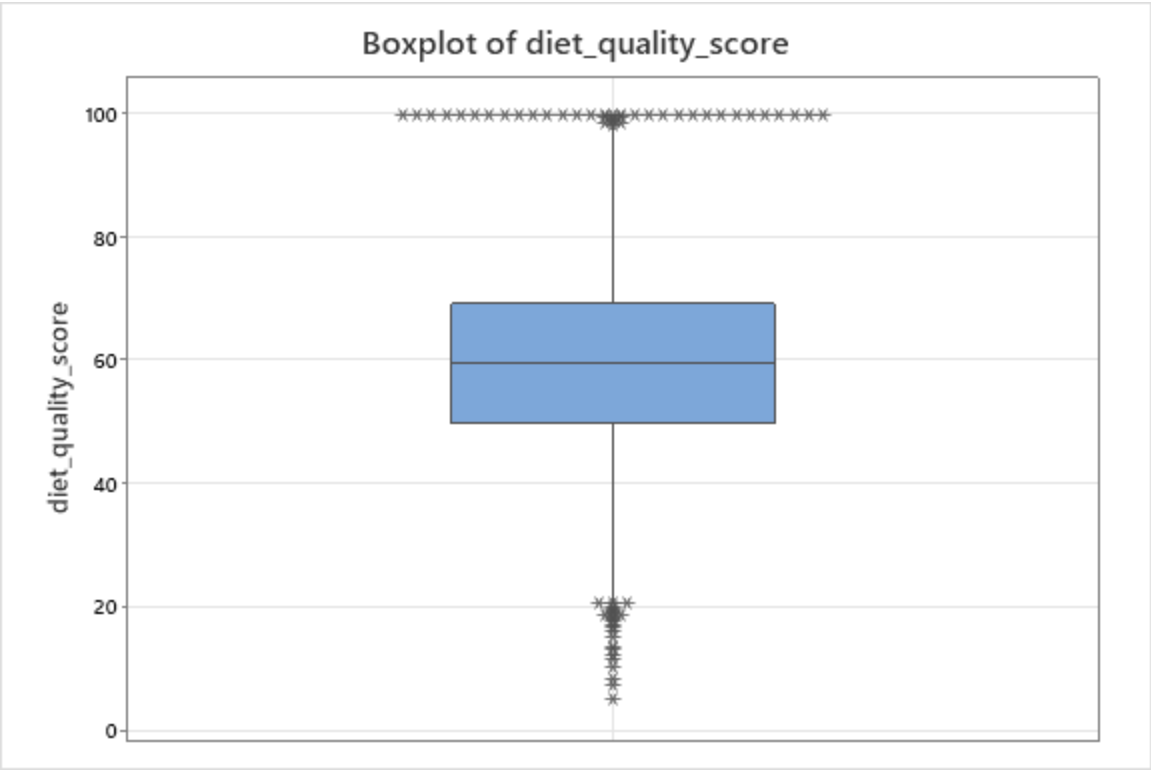












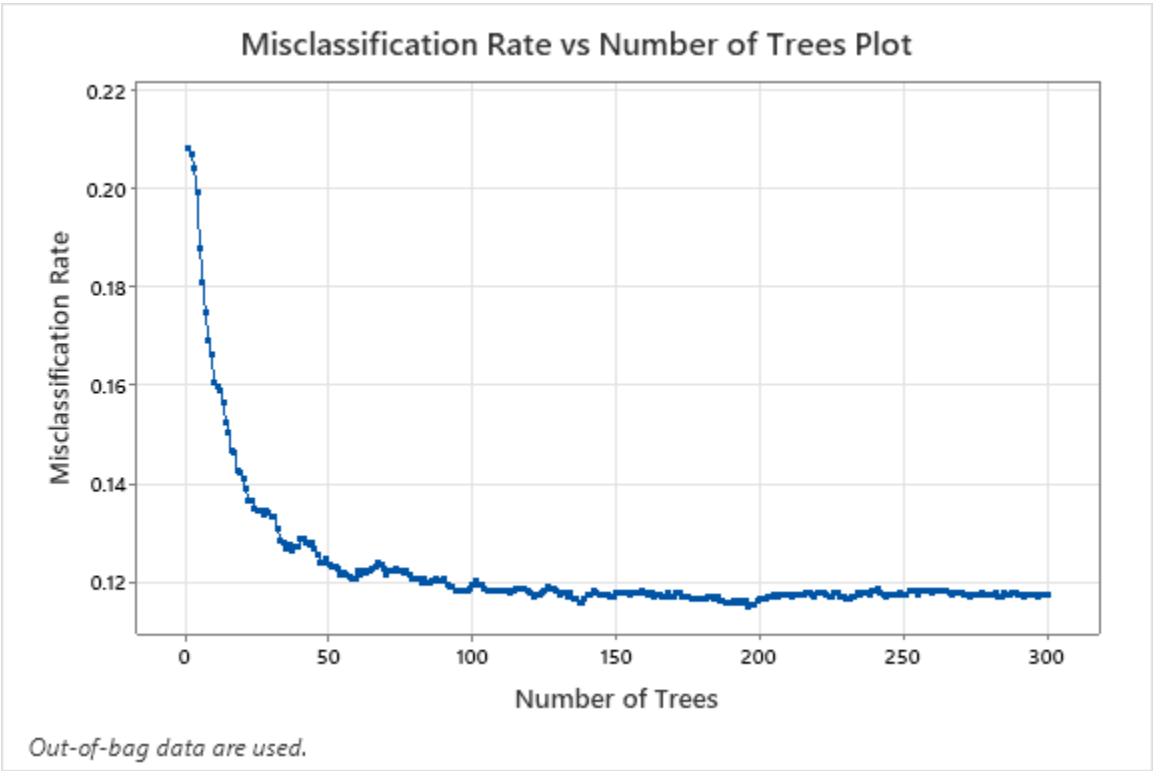
Random Forests Classifier:

Method

Model validation	Validation with out-of-bag data
Number of bootstrap samples	300
Sample size	Same as training data size of 9000
Number of predictors selected for node splitting	Square root of the total number of predictors = 5
Minimum internal node size	5
Rows used	9000

Binary Response Information

Variable	Class	Count	%
has_heart_disease	1 (Event)	2727	30.30
	0	6273	69.70
	All	9000	100.00

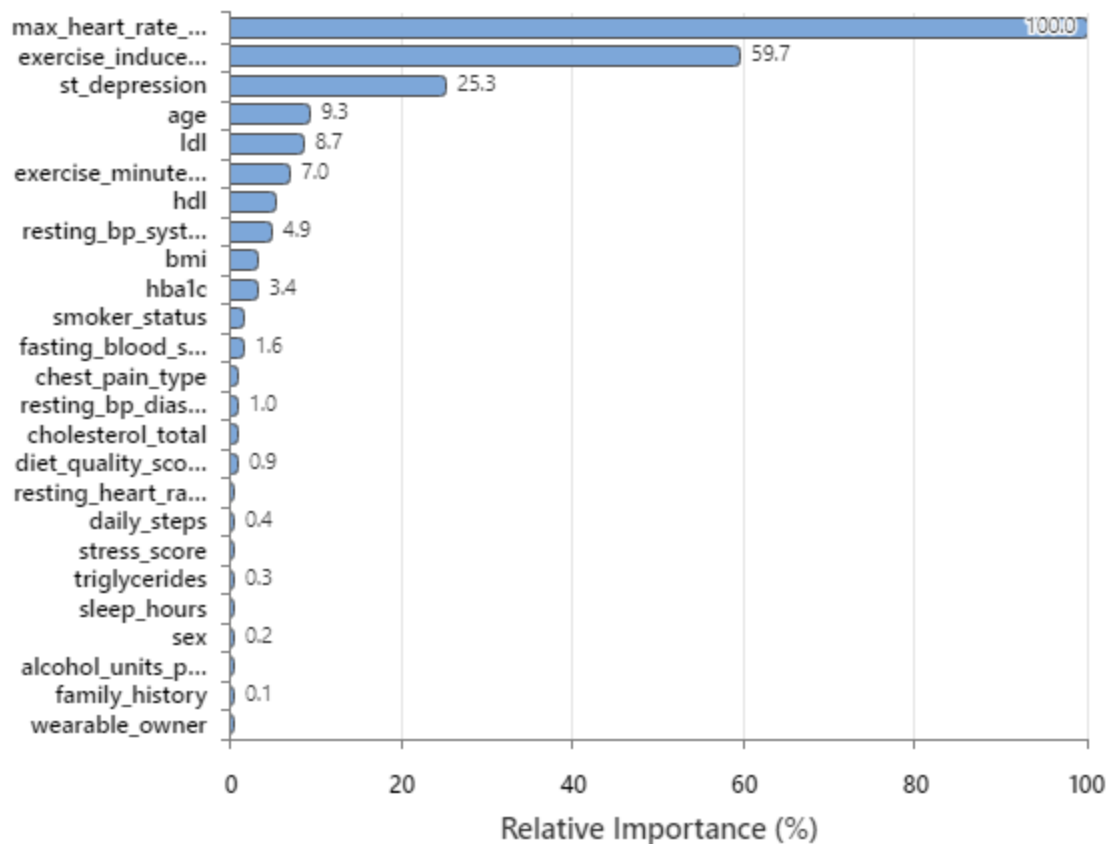


Model Summary

Total predictors	25
Important predictors	25
Statistics	Out-of-Bag
Average -loglikelihood	0.3170
Area under ROC curve	0.9392
95% CI	(0.9340, 0.9445)
Lift	3.1903
Misclassification rate	0.1172

Relative Variable Importance

Importance ranking method: Gini



Variable importance measures model improvement when splits are made on a predictor. Relative importance is defined as % improvement with respect to the top predictor.

Confusion Matrix

Actual Class	Count	Predicted Class (Out-of-Bag)		% Correct
		1	0	
1 (Event)	2727	2012	715	73.78
0	6273	340	5933	94.58
All	9000	2352	6648	88.28

Out-of-Bag	
Statistics	(%)
True positive rate (sensitivity or power)	73.78
False positive rate (type I error)	5.42
False negative rate (type II error)	26.22
True negative rate (specificity)	94.58

Misclassification

Actual Class	Count	Out-of-Bag	
		Misclassified	% Error
1 (Event)	2727	715	26.22
0	6273	340	5.42
All	9000	1055	11.72

